

GLOBAL CREDIT TRADER

The Quality Cross Over

- USD and EUR BB spreads have continued to outperform single-Bs, consistent with an ongoing ‘flight to quality’ within high yield, which has pushed the B/BB spread differential near the upper end of its historical range.
- However, this has not extended cleanly to the cross over between the low end of investment grade (BBB) and the high end of high yield (BB), where valuations remain broadly mid-range versus history. In the USD market, we think the BB/BBB ratio is fairly priced amid a somewhat more challenging growth and inflation mix. In EUR, by contrast, we see scope for more decompression given a more adverse macroeconomic backdrop and incremental technical pressures, including elevated fallen angel risk.
- While ‘generic’ rating screens (for example, BBB and BBB- debt with a Negative Outlook/Watch) can be a useful *directional* indicator of fallen angel risk, history suggests they capture only part of the story. Since 2010, we estimate that roughly half of fallen angels across both the USD and EUR markets have come from issuers that did *not* clearly signal an imminent downgrade ahead of the event. The same ‘surprise’ dynamic is even more pronounced on the upside, when tracking rising stars.
- We introduce a new framework for modeling fallen angel and rising star risk using historical rating-transition behavior, alongside a subjective overlay. On that basis, we raise our 2026 USD rising star forecast to \$75 billion (from \$40 billion previously). Elsewhere, the framework is broadly consistent with our existing 2026 forecasts: USD fallen angels of \$60 billion, and EUR rising stars and fallen angels of €20 billion and €30 billion, respectively. In EUR, that profile would imply the first year since 2020 where fallen angels outpace rising stars.
- Fallen angel risks are not meaningfully reflected in either region. A related nuance is that the market impact of fallen angels has been asymmetric. In the USD market, downgrade risk is often well telegraphed and fallen angels have tended to *outperform* post-downgrade. In EUR, they have more often *underperformed*—in part reflecting the higher prevalence of single-agency rating structures and the potential for downgrade risk to materialize more abruptly.

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Revisiting Cross Over Dynamics and Fallen-Angel Trends

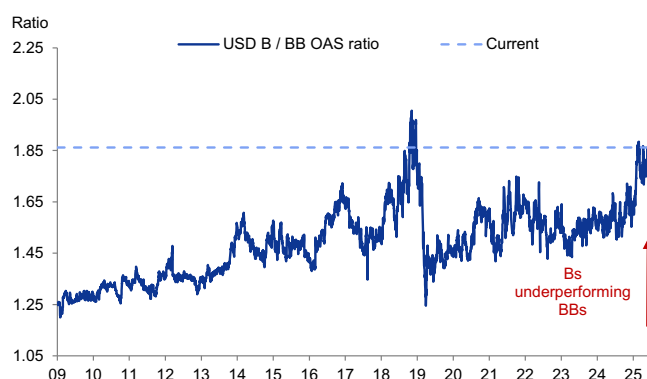
BBs benefit from a ‘flight to quality’ within HY

In the USD market, spreads of BB-rated bonds have notably outperformed B-rated bonds since late 2025 (Exhibit 1). The OAS ratio between the two rating cohorts is hovering near the widest level of the post-financial crisis era. We believe the resilience of BBs reflects, in part, a ‘flight to quality’ positioning within the USD HY market. We attribute this sentiment to the overhang of geopolitical and macroeconomic uncertainty—even as the *directional* tone has felt ‘risk on’ at the index level. The relative underperformance of B-rated spreads is also consistent with the ‘atypical beta’ behavior we highlighted in late April.

Notably, however, the ‘flight to quality’ hasn’t translated into the relative value trade off between the low end of IG (BBBs) and the high end of HY (BBs). As shown in Exhibit 2, the ratio of BB spreads vs. BBBs ranks towards the middle-to-lower end of the historical distribution—signaling that the higher rated BBB cohort has *not* outperformed BBs. While historical valuations imply some modest scope for compression in BB spreads relative to BBBs, we believe the more likely outcome is for the ratio to remain range bound around current levels, amid a more challenging growth and inflation mix.

Exhibit 1: On a spread basis, USD BB-rated bonds have meaningfully outperformed B-rated bonds since late 2025

OAS ratio of the B and BB subsets of the Bloomberg USD HY Corporate indices

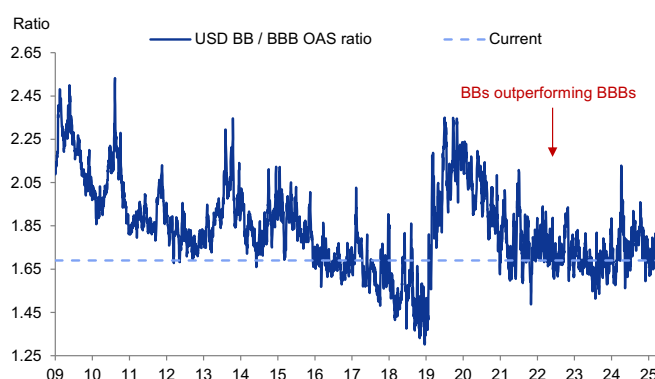


Note: As of May 26, 2026.

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 2: In contrast, USD BB vs. BBB spreads have been rangebound

OAS ratio of the BB and BBB subsets of the Bloomberg USD HY and IG Corporate indices



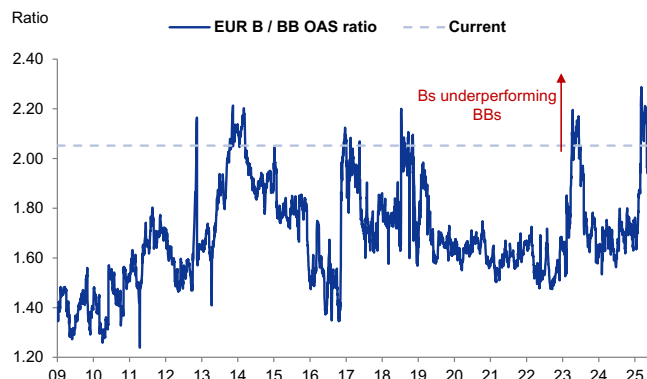
Note: As of May 26, 2026.

Source: Bloomberg, Goldman Sachs Global Investment Research

The EUR market shows a similar divergence in the ‘flight to quality’ dynamic: BBs have outperformed single-Bs (Exhibit 3), yet the BB vs. BBB spread ratio has remained relatively range bound, toward the mid-to-low end of its historical range (Exhibit 4). Unlike in the USD market, we see scope for EUR BBBs to outperform BBs over the medium term, supported by a weaker European growth backdrop due to the ongoing energy shock, as well as incremental technical headwinds for BBs from elevated fallen-angel risk.

Exhibit 3: In the EUR market, BB spreads have also outperformed the B-rated cohort

OAS ratio of the B and BB subsets of the Bloomberg Pan Euro HY Corporate indices

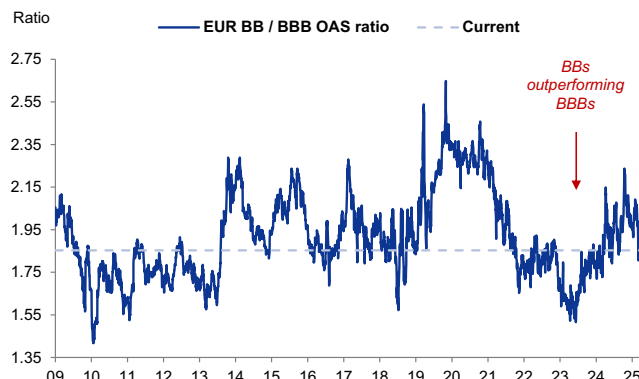


Note: As of May 27, 2026.

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 4: EUR BB vs. BBB spreads remain near the mid-to-low end of the historical range

OAS ratio of the BB and BBB subsets of the Bloomberg Pan Euro HY and IG Corporate indices



Note: As of May 27, 2026.

Source: Bloomberg, Goldman Sachs Global Investment Research

We expect a greater technical headwind from 'fallen angels' in the EUR market

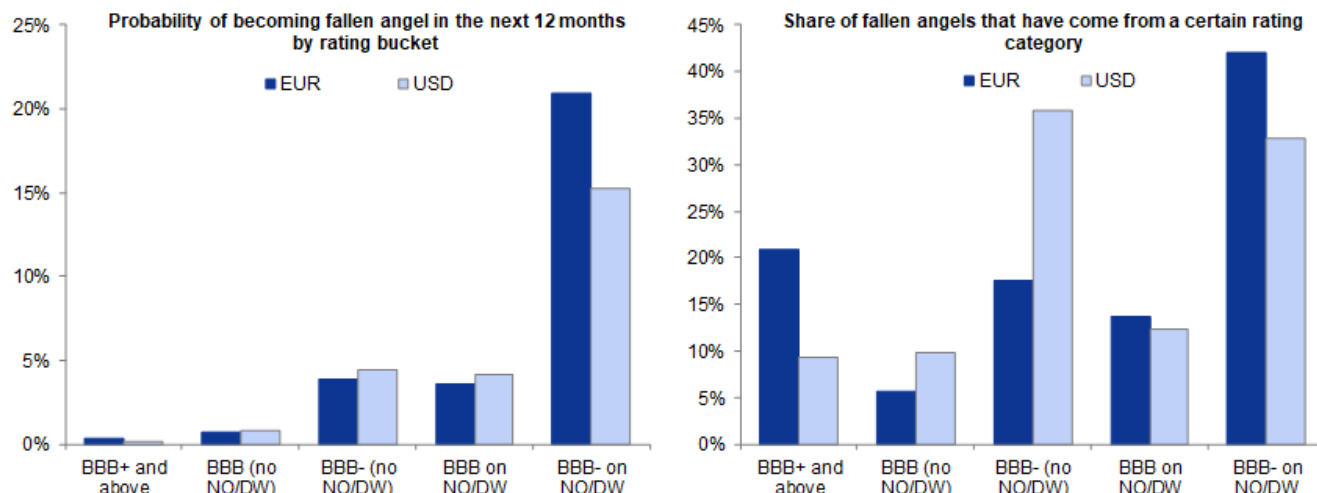
Beyond the well-telegraphed divergence in the macro backdrop between the US and Europe, we also expect technical conditions to differentiate across regions. In Europe, we see scope for a larger supply headwind into the BB segment from fallen angels, whereas USD BB technicals should benefit from more rising stars than fallen angels.

To assess fallen angel and rising star risk over the remainder of 2026, we introduce a new framework based on historical one year rating-transition behavior by rating bucket. The left panels of Exhibits 5 and 6 summarize the historical probability of becoming a fallen angel or a rising star over a one-year horizon, by rating bucket. The results are broadly intuitive: since 2010, bonds rated BBB- on Negative Outlook or Downgrade Watch have exhibited the highest propensity to become fallen angels, while BB+ credits with a Positive Outlook or Upgrade Watch have shown the highest likelihood of migrating back to investment grade (i.e., becoming rising stars).

That said, the right panels of both exhibits show that ratings and outlooks are, at best, an imperfect guide to forward-looking fallen angel and rising star risk: since 2010, issuers originally rated BBB/BBB- or higher *without* a Negative Outlook or Downgrade Watch have accounted for roughly half of all fallen angel volumes in both markets. The signal is especially noisy during abrupt macro shocks—such as COVID-19—when fundamentals can deteriorate faster than rating frameworks can adjust.

Exhibit 5: A high share of fallen angels in the EUR and USD markets were not rated BBB or BBB- on Negative outlook or Downgrade Watch prior to the event

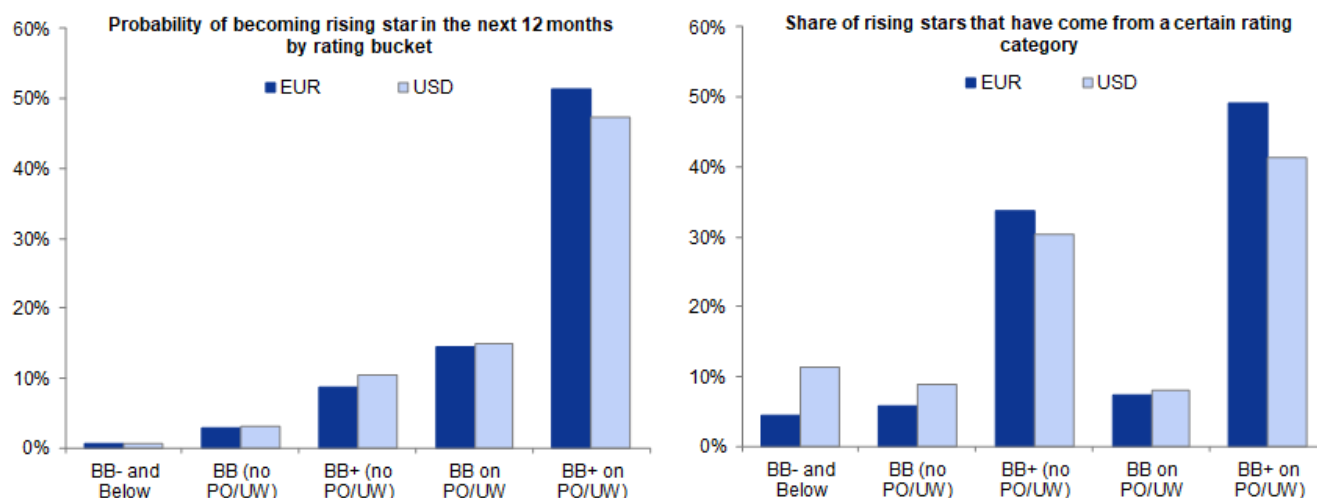
Share of bonds in each rating bucket that became fallen angels within 12 months (since 2010, left panel). Rating composition of all fallen angels prior to downgrade (right panel). The difference reflects the denominator — the left measures transition rates within each bucket, the right measures where fallen angels came from.



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 6: Similarly, a high share of rising stars in the EUR and USD markets were not BB+ or BB on Positive Outlook or Upgrade Watch prior to the event

Share of bonds in each rating bucket that became rising stars within 12 months (since 2010, left panel). Rating composition of all rising stars prior to upgrade (right panel). The difference reflects the denominator — the left measures transition rates within each bucket, the right measures where rising stars came from.



Source: Bloomberg, Goldman Sachs Global Investment Research

Accordingly, while our fallen angel and rising star forecasts are anchored in historical transition behavior, we would stress that forecast uncertainty remains elevated and, at times, warrants a subjective overlay.

For example, our model would imply roughly \$85 billion of USD rising stars in 2026; however, we are tempering that signal given the somewhat more challenging growth and inflation backdrop expected by our US economists, which may slow the pace of upgrades. As a result, we raise our official USD rising star forecast to \$75 billion (from \$40 billion previously; Exhibit 7).

Elsewhere, the framework is broadly consistent with our existing forecasts for 2026: USD

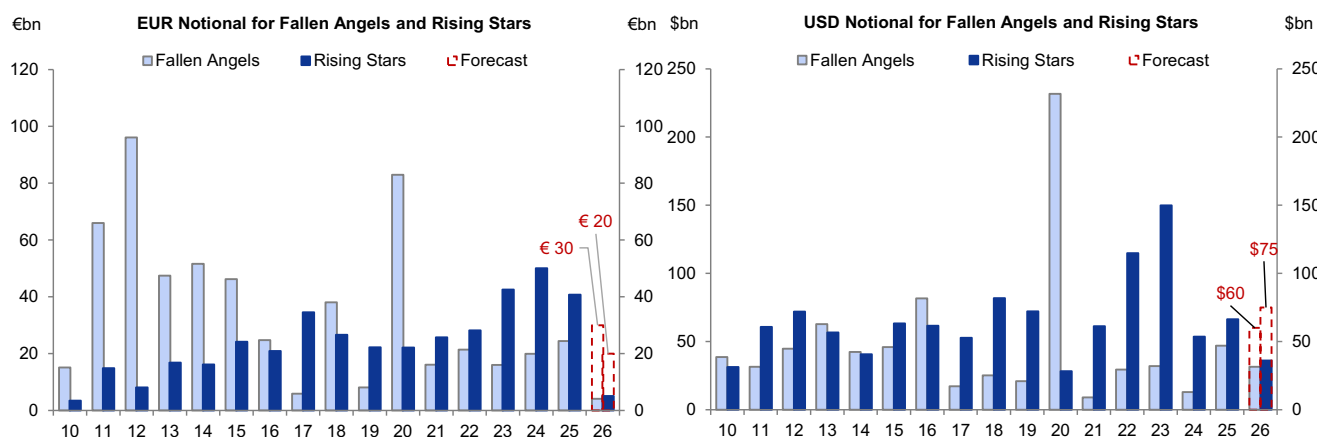
fallen angels of \$60 billion, and EUR rising stars and fallen angels of €20 billion and €30 billion, respectively. In the EUR market, that profile would imply the first year since 2020 in which fallen angels outpace rising stars, while the USD market is set up for yet another year of net rising stars.

This divergence is starting to show up in year-to-date net rising-star activity: the EUR market is roughly flat at €1 billion, versus \$5 billion in the USD market.

Notably, however, the market does not appear to be pricing substantial fallen-angel risk in either region. Exhibit 8 shows that the share of the BBB- segment of the EUR and USD IG markets (unsecured bonds only) trading at spreads wider than the average BB+ remains around 2% and 15%, respectively, near the low end of the past decade.

Exhibit 7: In the EUR market, fallen angels are set to outpace rising stars for the first time since 2020

Fallen angel and rising star notionals in the EUR market (left panel) and USD market (right panel)



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 8: Neither market is pricing significant fallen angel risk



Note: As of May 25, 2026.

Source: iBoxx, Goldman Sachs Global Investment Research

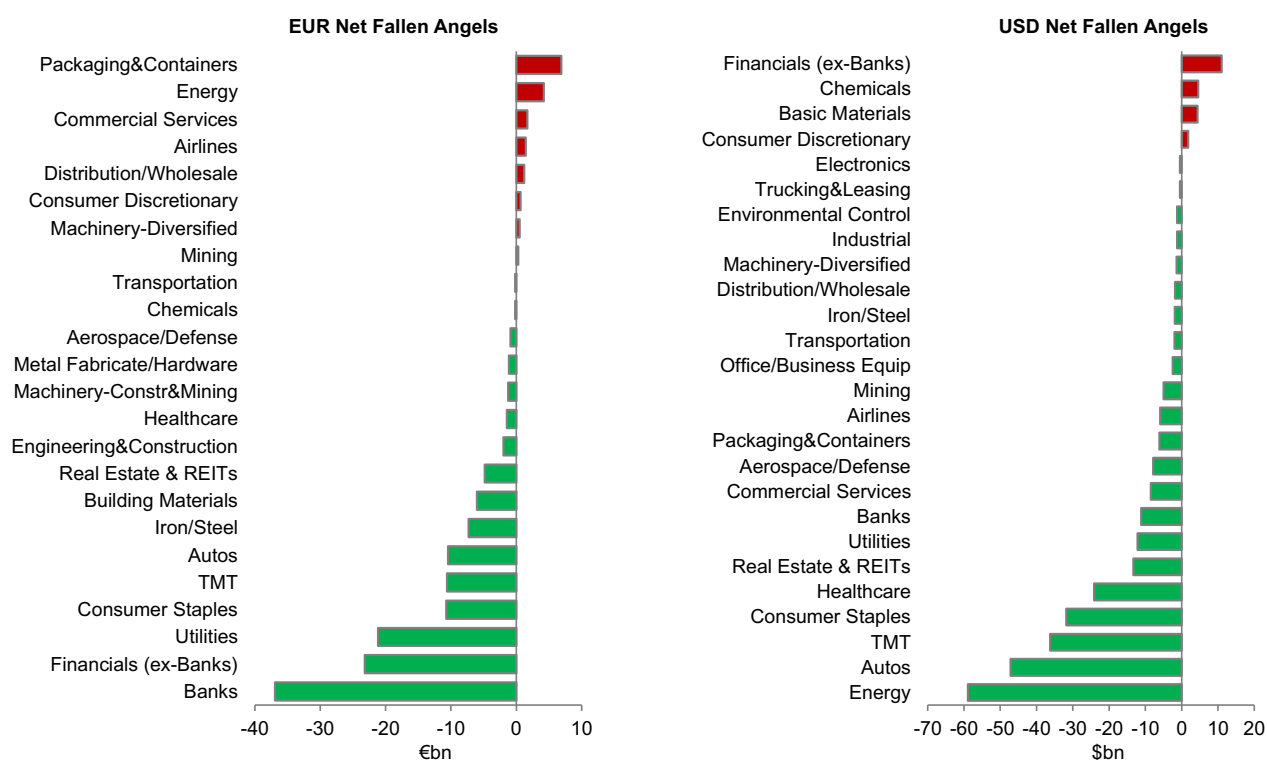
Sector drivers: Likely to be Energy intensive in EUR, Software in USD

The diverging paths between rising stars and fallen angels in the EUR and USD markets are further reinforced when we examine the composition of debt that has driven rising-star activity in recent years, alongside the segments that currently screen as most vulnerable to fallen angel risk.

First, the rising-star wave since 2021 has been more concentrated in European financials (Banks accounted for 30% of rising stars since 2021; Exhibit 9). While bank balance sheets remain strong, the sector faces incremental headwinds from higher taxes, slower growth, and, in some countries, lingering fiscal risks—all of which likely limit further upside rating momentum (at least in the near term). By contrast, USD rising stars have been more diversified across sectors, including Energy, Autos, and TMT.

Second, the current energy shock is weighing more heavily on energy intensive and interest rate sensitive sectors, which represent a large share of BBB/BBB- rated bonds currently on Negative Outlook or Downgrade Watch in the EUR market. For example, Automotives, Chemicals, REITs, and Food and Beverage account for 24% of EUR BBB and BBB- issuers on Negative Outlook or Downgrade Watch, versus 16% in the USD market (Exhibit 10). Idiosyncratic, issuer-specific, fallen angel risk is present in both markets, concentrated in Banks in the EUR market, and Software in the USD market.

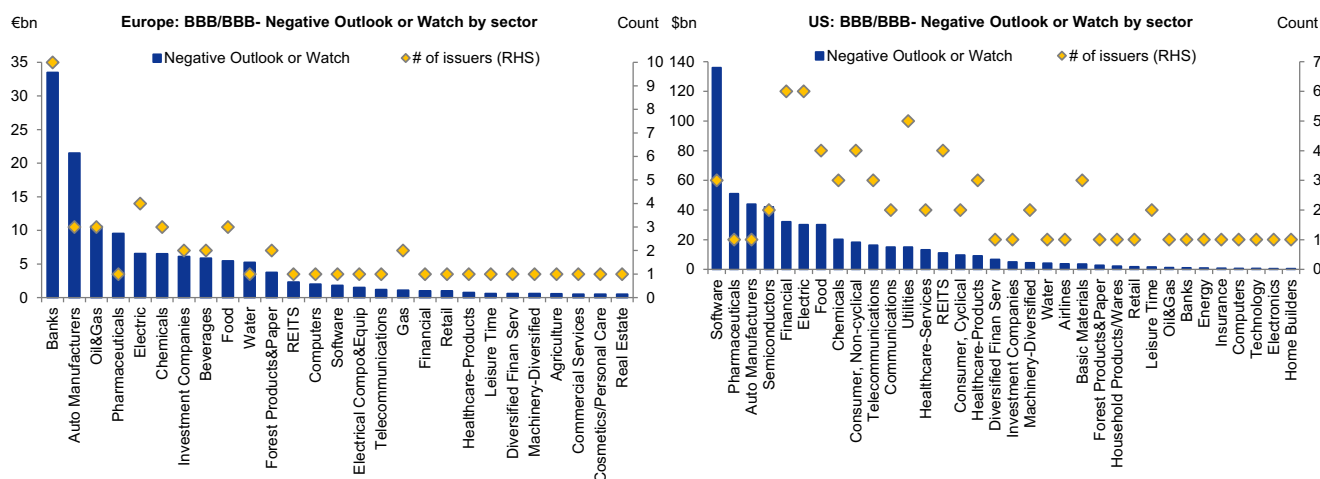
Exhibit 9: Since 2021, rising stars have been dominated by Banks and Financials in the EUR market



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 10: Energy intensive sectors screen at risk of becoming fallen angels in the EUR market, alongside Software in the USD market

Exhibit shows the sectors that are on Negative Outlook and Negative Watch



Source: Bloomberg, Goldman Sachs Global Investment Research

Fallen angels weigh more heavily on performance in the EUR market

In Exhibits 11 and 12, we capture a sample of fallen angels (since 2010) across the EUR and USD markets, to assess performance. Our sample uses iBoxx and Bloomberg data, and includes 155 fallen angels in the EUR market and 354 fallen angels for the USD market.

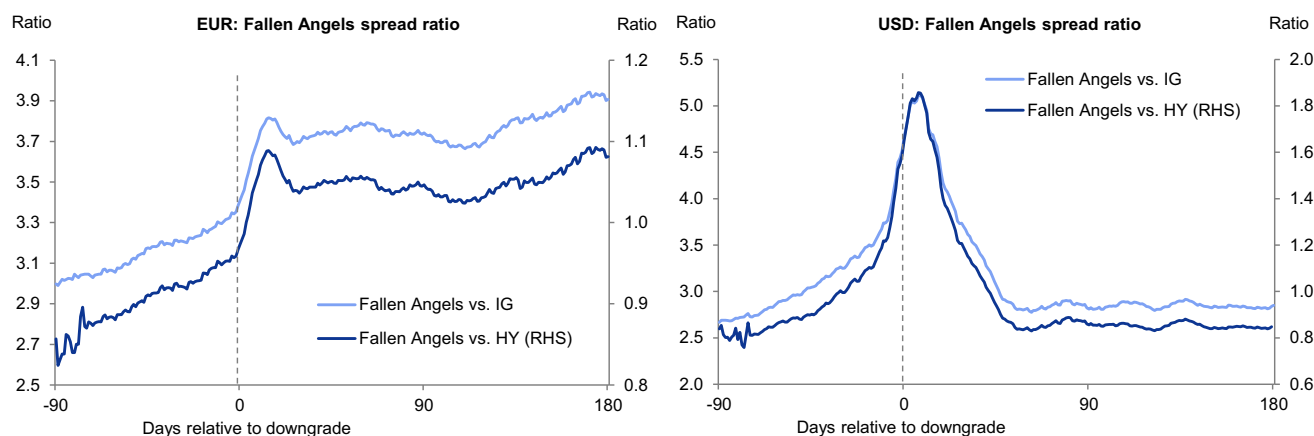
In the USD market, we find that fallen angels have tended to outperform after a downgrade (Exhibit 11). In many cases, the move to high yield is well telegraphed, and subsequent HY index inclusion can broaden the buyer base. In addition, it is common for HY investors to have a favorable view of the credit quality of recent fallen angels—especially relative to the ‘existing’ HY universe and even more so if there is a stated plan to return to IG-rated metrics.

In the EUR market, however, fallen angels have tended to underperform relatively consistently following a downgrade. One important driver appears to be the prevalence of single-rated issuers in Europe. In the EUR BBB market, 19% of notional outstanding has only one rating agency, versus less than 1% in the USD market (Exhibit 12). As a result, fallen-angel risk can materialize more abruptly and be less fully anticipated than in the USD market, where a ‘true’ fallen angel typically requires a second downgrade (per index eligibility rules).

EUR fallen angels have also tended to have more pronounced downside outcomes. Since 2010, 5% of the number of bonds that become fallen angels in the EUR have been downgraded to CCC+ or lower within one year, versus 1.5% in the USD market (Exhibit 13). That said, the market does appear to price some of this risk. BBB issuers rated by just one agency typically trade at slightly wider spreads versus their peers, particularly when also on Negative Outlook (Exhibit 14).

Exhibit 11: In the EUR market, fallen angels tend to underperform post a downgrade, unlike in the USD market

The exhibit shows the relative OAS performance before/after becoming a fallen angel since 2010, covering 155 issuers in the EUR market and 354 issuers in the USD market.

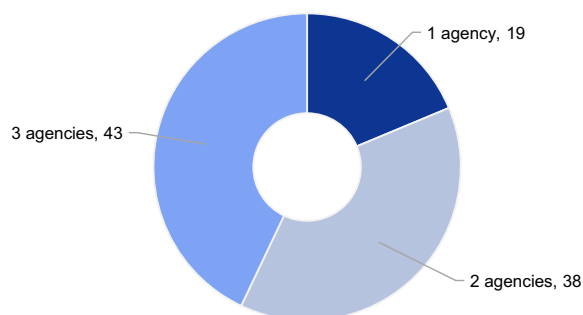


Source: iBoxx, Bloomberg, Goldman Sachs Global Investment Research

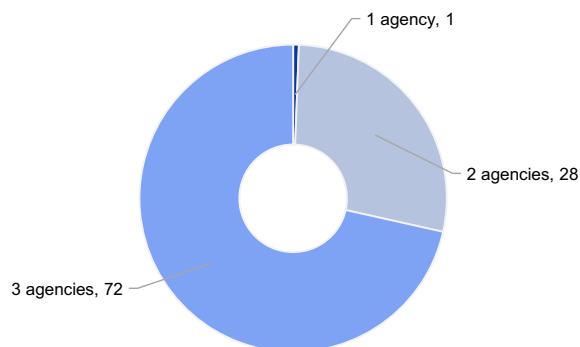
Exhibit 12: The EUR market has a high share of debt attached to just one rating agency vs. the USD market

Percentages based on notional amounts

Share of BBBs by number of rating agencies in Europe (%)



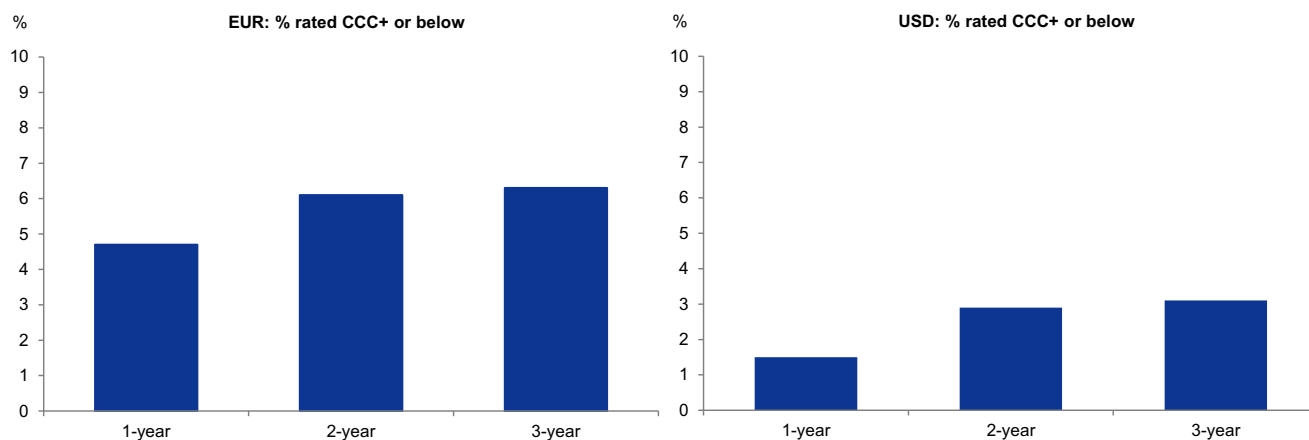
Share of BBBs by number of rating agencies in the US (%)



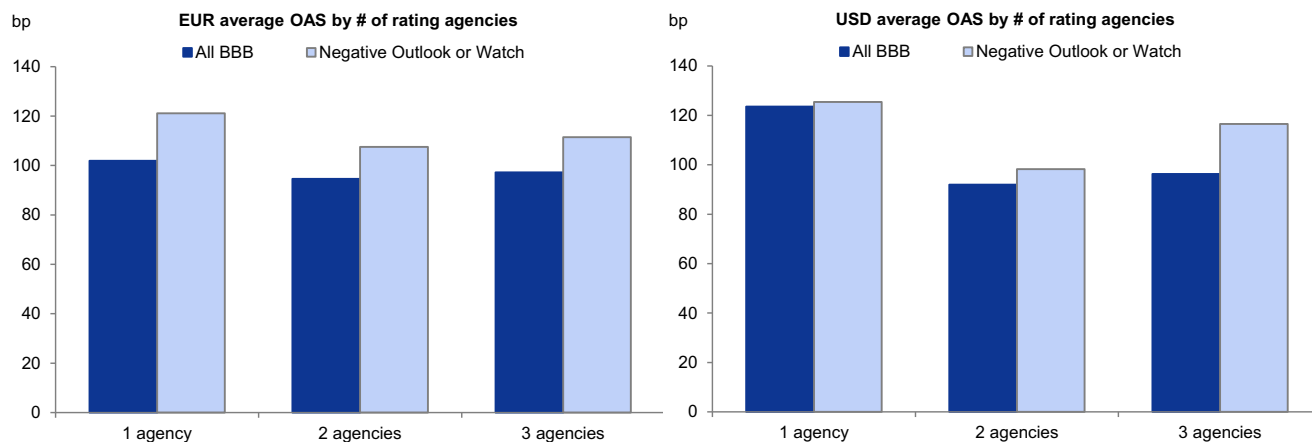
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 13: In the EUR market, fallen angels have a higher tendency to be downgraded to CCC+ or below

Exhibit shows the share of bonds that become fallen angels in the EUR and USD market which are downgraded to CCC+ or below within 1, 2 and 3 years



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 14: Issuers in the EUR and USD market with just one rating agency tend to have wider spreads

Source: iBoxx, Goldman Sachs Global Investment Research

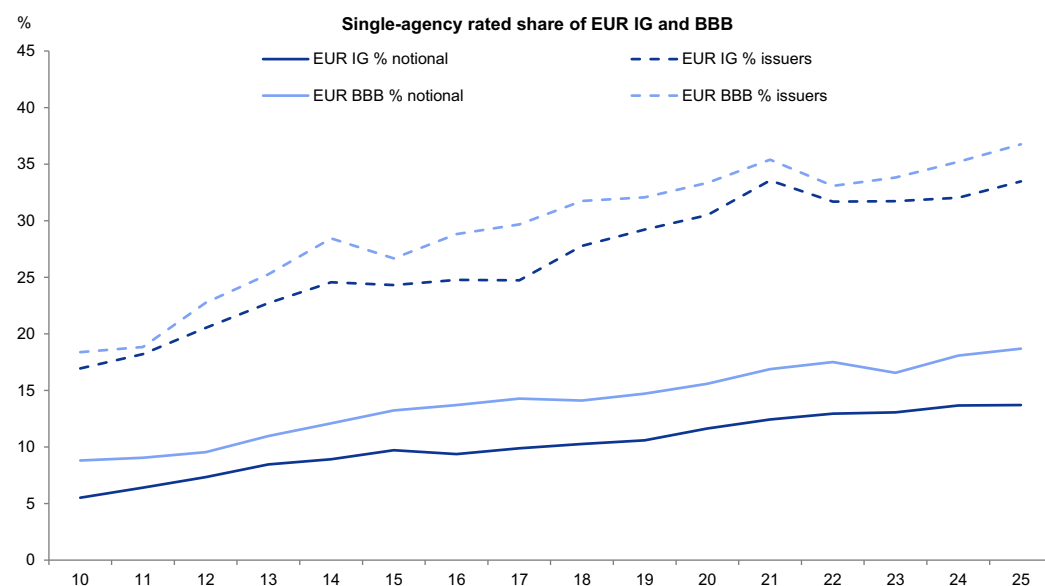
Why does Europe have such a high share of single rating-agency issuers?

Both the number of issuers and the share of issuers with only one rating in Europe have risen steadily since 2010 (Exhibit 15).

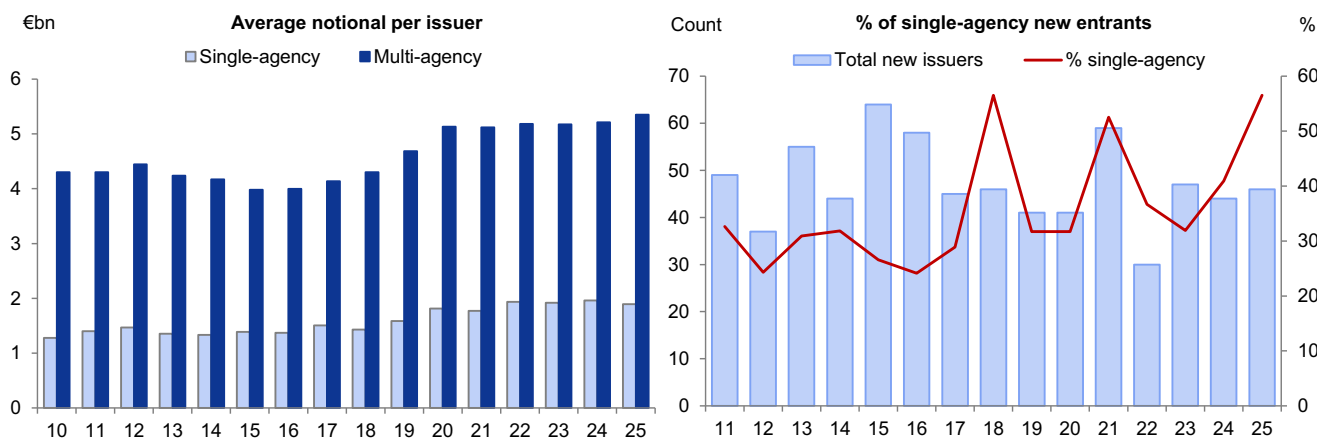
Key drivers appear to be structural and regulatory. Under the EU CRA Regulation, Article 8d, issuers (or related third parties) that intend to appoint two or more credit rating agencies are required to consider appointing at least one credit rating agency with no more than 10% total EU market share; if they do not, they must document that decision.

In practice, this can increase the cost and operational burden of maintaining a multi-rating strategy, particularly for smaller issuers. As a result, some issuers may opt to remain single-rated rather than add a second (or third) relationship that could trigger additional documentation and governance steps.

This also interacts with issuer mix and market structure. As Exhibit 16 suggests, single-rated issuers tend to be smaller (lower average notional per issuer), and a large share of new entrants to the EUR credit market begin with only one rating. Given Europe's more fragmented market, we do not expect this dynamic to fade quickly (at least absent regulatory change).

Exhibit 15: In Europe, the share of single rating agency risk has been rising

Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 16: Single rating agency risk tends to be concentrated in new and smaller issuers

Source: iBoxx, Goldman Sachs Global Investment Research

The authors would like to thank Patricia Pacheco for her contribution to this report.

Forecast dashboard: Spreads, returns, issuance, defaults, and ratings migrations

Exhibit 17: USD and EUR spread forecasts

Index	Updated as of May. 28, 2026			
	Current	2026Q2	2026Q3	2026Q4
USD spreads				
USD IG	72	95	92	90
USD Fin	76	103	98	94
USD Non-Fin	69	92	89	86
USD HY	261	335	325	315
EUR spreads				
EUR IG	89	120	117	114
EUR Fin	95	127	122	118
EUR Non-Fin	86	116	113	111
EUR HY	272	375	368	360

Source: Bloomberg, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research

Exhibit 19: Issuance forecasts

Market	2026 YTD (bn)	2026 full-year forecast (bn)
USD market		
Gross issuance		
IG	\$1,051	\$2,100
HY	\$145	\$400
Leveraged loans	\$174	--
Net issuance		
IG	\$462	\$850
HY	\$52	\$160
EUR market		
Gross issuance		
IG	€ 405	€ 850
HY	€ 47	€ 125
Net issuance		
IG	€ 96	€ 265
HY	€ 22	€ 60

Source: Dealogic, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 18: USD and EUR excess returns forecast

Index	Updated as of May. 28, 2026		
	Current duration	2026	2026 full-year forecast (%)
USD excess returns			
USD IG	6.8	0.7%	0.0%
USD HY	2.9	1.3%	1.3%
EUR excess returns			
EUR IG	4.6	0.5%	0.0%
EUR HY	2.7	0.6%	-0.4%

Source: Bloomberg, ICE-BAML, iBoxx, Goldman Sachs Global Investment Research

Exhibit 20: Ratings migration and default forecasts

Default rates are issuer-weighted and 12-month trailing

Market	Updated as of May. 28, 2026	
	2026 YTD	2026 full-year forecast
USD market		
Rising stars	\$43bn	\$75bn
Fallen angels	\$31bn	\$60bn
HY defaults (12-month trailing)	3.6%	3.0%
Leveraged loan defaults (12-month trailing)	5.1%	4.75%
EUR market		
Rising stars	€ 5bn	€20bn
Fallen angels	€ 4bn	€30bn
HY defaults (12-month trailing)	4.5%	4.7%
Leveraged loan defaults (12-month trailing)	3.3%	4.0%

Source: Bloomberg, Moody's, Goldman Sachs Global Investment Research

Forecast dashboard: Relative value views

Exhibit 21: Relative value views

Relative value views		
Corporate Credit		
Regional	Overweight USD vs. EUR	The growth, inflation and monetary policy mix is somewhat more favorable in the US, relative to Europe. As a result, and relative to the USD market, we forecast a wider peak (and a slower recovery in 2H2026) in EUR credit spreads.
	Overweight IG vs. HY	We have a slight preference for IG vs. HY in the USD market, owing to valuations and our economists' forecasts for below-potential growth.
USD market	Overweight BBBs vs. higher-rated cohorts	We prefer owning BBBs over the higher-rated segments of USD IG (As and AAs), as a way to capture additional spread. We also see scope for balance sheet deterioration at the highest end of the rating spectrum in IG.
	Overweight BBs, Neutral CCCs, Underweight Bs	Current spread levels indicate that single-Bs offer the lowest credit risk premium relative to expected losses among all HY rating buckets.
	Overweight HY bonds vs. loans	HY bonds still provide a superior trade-off between carry and duration compared to leveraged loans. In addition, loans' substantial software sector exposure is likely to present a sentiment overhang, at least in the near-term.
	Overweight CLO mezz tranches vs. leveraged loans Neutral CLO vs. USD IG Neutral CLO vs. CMOs	For floating rate investors, we prefer CLO mezzanine tranches vs. leveraged loans. Historical data points to low impairment rates for BB and BBB CLO tranches, and current valuations offer attractive carry and a cushion against moderate to elevated default outcomes. Elsewhere, we are neutral between senior CLO tranches and USD IG; we would use spread widening to add exposure in CLOs. We are also neutral between CLO AAA and CMO floaters.
	Prefer Agency MBS vs. IG	Agency MBS offers an attractive diversification benefit vs. a portfolio of IG corporate bonds.
	Overweight IG vs. HY	We prefer IG over HY given the more challenging growth, inflation, and monetary policy mix in the Euro Area.
EUR market	Overweight As vs. BBBs	In the EUR market, we prefer As over BBBs, as the weaker growth backdrop warrants more of an up-in-quality stance. That said, we would not move too far up the quality spectrum, as we expect supply technicals will be a headwind in the EUR AA market.
	Overweight HY bonds vs. loans	Similar to the US, we prefer HY bonds given the loan market's weighting toward software (and the potential negative sentiment overhang). The reaction function of the ECB may also favor floating rate credit, given the central bank's focus on inflationary pressures.
	Overweight IG 10s vs. 2s	We have a slight preference for extending duration in EUR IG to optimize carry, but have scaled back this approach.
	Overweight BBs, Neutral Bs, Underweight CCCs	At current valuations and given our expectation for heightened dispersion, we prefer moving up in quality within EUR HY.
	Underweight AT1s vs. EUR HY	Given downside tail risks and a limited valuation cushion, we recommend an underweight on AT1s vs. EUR HY.

Source: Goldman Sachs Global Investment Research

Performance dashboard: Cash and synthetic markets

Exhibit 22: Performance of USD and EUR markets

Market	Current*		Total return			Spread change (YTD,bp)	Yield change (YTD, %)
	Spread	Yield	MTD	QTD	YTD		
USD Corporate bonds							
IG	73	5.2	0.38	0.83	0.29	-5	0.37
AA	46	4.9	0.23	0.47	0.15	-3	0.31
A	61	5.0	0.27	0.76	0.36	-4	0.31
BBB	93	5.4	0.42	1.14	0.67	-8	0.29
HY	261	7.0	0.28	1.98	1.48	-5	0.50
BB	161	6.2	0.33	1.80	1.42	-8	0.26
B	295	7.7	0.47	2.30	1.91	-6	0.36
CCC	936	13.7	-0.52	2.22	-0.03	51	0.78
USD Leveraged Loans							
Leveraged loan index	437	8.2	0.00	0.09	-0.46	42	0.4
BB	257	6.3	0.00	0.00	0.70	2	-0.1
B	448	8.3	0.00	0.00	-0.93	57	0.5
CCC	1877	23.9	0.00	0.00	-5.09	314	3.3
USD Securitized products							
Agency MBS	23	5.0	-0.04	0.03	0.43	1	0.4
ABS (Autos)	57	4.5	0.19	0.52	1.19	-9	0.3
ABS (Credit cards)	28	4.2	0.12	0.42	0.93	-5	0.2
ABS (Student loans)	100	4.8	0.37	0.64	1.73	127	-0.1
USD Macro indices							
CDX IG	52	--	0.18	0.63	0.40	2	--
CDX HY	306	--	0.34	3.07	1.20	-10	--
EUR Corporate bonds							
IG	91	3.7	0.68	1.63	0.61	1	0.2
AA	65	3.6	0.61	1.46	0.51	3	0.2
A	83	3.6	0.67	1.61	0.61	0	0.2
BBB	104	3.8	0.70	1.69	0.63	2	0.2
HY	269	5.8	0.84	2.80	1.03	-1	0.2
AT1	273	5.4	1.07	3.49	2.08	-13	0.1
EUR Leveraged loans							
Leveraged loan index	497	7.2	0.00	0.18	-0.87	47	0.6
BB	282	5.0	0.00	0.00	0.75	-70	-0.6
B	471	6.9	0.00	0.00	-1.30	56	0.7
EUR Macro indices							
iTraxx Main	54	--	0.30	0.94	0.3	4	--
iTraxx Xover	268	--	1.45	4.54	1.9	23	--
iTraxx SenFin	57	--	0.34	1.13	0.6	2	--
iTraxx SubFin	93	--	0.54	2.08	1.1	0	--

*Data as of previous day close.

For USD IG and HY we use the Bloomberg indices with history back to Jun-89 and Jan-94, respectively. Corp rating level, Agency MBS and ABS we use ICE-BAML indices with history back to Dec-96, EUR IG we use the iBoxx QW5A index with history back to Jan-99, EUR HY we use the ICE-BAML index with history back to Dec-97, and USD and EUR leveraged loans we use the Morningstar LSTA LLI and ELLI indices with data going back to Jan-97 and Jan-02, respectively.

Source: Bloomberg, iBoxx, ICE-BAML, Pitchbook LCD, Goldman Sachs Global Investment Research

Performance dashboard: Sectors

Exhibit 23: Sector performance: USD IG

Sector	Metrics							Spread change				Total return			
	Index weight	Mkt val (\$bn)	OAS	Percentile rank (2010-today)	YTW	Dur	Rating	1-week	MTD	QTD	YTD	1-week	MTD	QTD	YTD
Basic Industry	2%	181.4	76		5.3	7.4	BAA1/BAA2	1	-6	-21	-13	0.6	0.4	1.2	0.8
Chemicals	1%	83.1	81		5.3	7.5	BAA1/BAA2	1	-9	-20	-17	0.6	0.6	1.3	1.2
Metals & Mining	1%	72.0	67		5.2	7.7	A3/BAA1	2	-4	-22	-11	0.6	0.3	1.3	0.5
Paper	0%	26.3	87		5.3	6.0	BAA2/BAA3	0	-5	-19	-9	0.5	0.3	1.1	0.3
Capital Goods	5%	410.3	57		5.0	6.8	A3/BAA1	0	-7	-19	-10	0.6	0.3	0.7	0.3
Aerospace & Defense	2%	150.8	64		5.2	8.3	BAA1/BAA2	1	-6	-18	-11	0.7	0.4	0.8	0.4
Building Materials	1%	39.9	71		5.1	6.5	BAA1/BAA2	0	-6	-18	-7	0.6	0.3	0.8	0.2
Diversified Manufacturing	2%	116.0	57		5.0	6.6	A3/BAA1	0	-6	-18	-7	0.6	0.2	0.6	0.3
Construction Machinery	1%	58.3	34		4.6	4.7	A1/A2	0	-8	-24	-17	0.4	0.1	0.4	0.4
Packaging	0%	14.3	72		5.0	4.5	BAA2/BAA3	0	-6	-20	-9	0.4	0.1	0.6	0.4
Environmental	0%	31.0	42		4.8	5.8	A3/A3	2	-7	-20	-14	0.5	0.1	0.3	0.1
Communications	8%	567.0	89		5.5	8.6	A3/BAA1	0	-5	-11	-5	0.8	0.4	0.5	-0.1
Cable & Satellite	1%	107.8	132		6.0	9.3	BAA1/BAA2	1	-1	-5	1	0.9	0.2	0.2	-0.1
Media & Entertainment	2%	137.1	78		5.4	8.7	A1/A2	1	2	-4	-7	0.8	0.2	0.1	-1.4
Wireless	2%	125.1	75		5.2	7.3	BAA1/BAA2	0	-8	-17	-7	0.7	0.5	0.8	0.5
Wirelines	3%	197.0	83		5.4	9.1	BAA1/BAA2	0	-8	-13	-7	0.9	0.7	0.8	0.4
Consumer Cyclical	7%	549.1	65		5.0	6.2	A2/A3	0	-9	-21	-8	0.5	0.3	0.8	0.3
Automotive	2%	170.8	80		5.0	4.3	A3/BAA1	-1	-11	-24	-9	0.4	0.4	1.1	0.7
Leisure	0%	4.8	97		5.4	6.4	BAA2/BAA2	-5	-10	-19	-5	1.0	0.7	1.3	-0.2
Gaming	0%	23.6	101		5.3	4.7	BAA2/BAA3	0	-10	-23	-9	0.4	0.4	1.3	0.7
Home Construction	0%	10.4	82		5.1	4.9	BAA1/BAA2	1	-3	-16	-2	0.5	0.1	0.7	0.3
Lodging	0%	20.0	69		5.0	4.8	BAA2/BAA3	-1	-11	-22	-7	0.5	0.3	0.8	0.3
Retailers	3%	256.5	50		5.0	7.7	A1/A2	1	-8	-19	-5	0.6	0.3	0.5	0.0
Restaurants	0%	33.3	56		5.1	7.6	BAA1/BAA2	0	-8	-21	-11	0.6	0.4	0.8	0.4
Consumer Cyc Services	0%	29.6	73		5.1	6.0	A3/BAA1	1	-7	-15	2	0.5	0.2	0.6	-0.1
Consumer Non-Cyclical	15%	1133.6	62		5.2	7.8	A2/A3	1	-7	-18	-8	0.7	0.4	0.7	0.1
Consumer Products	1%	59.4	44		4.8	6.0	A2/A3	1	-5	-20	-11	0.5	0.1	0.5	0.3
Food & Beverage	3%	235.5	67		5.2	7.6	A3/BAA1	1	-7	-19	-3	0.6	0.4	0.9	0.0
Healthcare	4%	331.2	71		5.2	7.8	A3/BAA1	1	-6	-16	-7	0.7	0.3	0.6	0.1
Pharmaceuticals	5%	408.5	53		5.1	8.5	A1/A2	1	-8	-19	-10	0.7	0.5	0.6	0.2
Supermarkets	0%	13.4	82		5.5	9.6	BAA1/BAA2	1	-6	-14	-4	0.8	0.6	0.9	0.0
Tobacco	1%	85.5	69		5.1	6.6	A3/BAA1	1	-7	-18	-11	0.5	0.3	0.8	0.3
Energy	7%	536.5	74		5.3	7.6	A3/BAA1	1	-9	-19	-18	0.6	0.5	1.1	1.2
Independent Energy	1%	94.2	74		5.3	7.5	BAA1/BAA2	1	-11	-22	-22	0.6	0.7	1.3	2.0
Integrated Energy	1%	103.3	49		5.1	8.3	AA3/A1	2	-6	-19	-16	0.6	0.3	0.7	0.6
Oil Field Services	0%	25.2	72		5.2	7.6	A3/BAA1	0	-8	-21	-25	0.7	0.4	1.2	1.5
Refining	0%	30.3	84		5.3	7.2	BAA1/BAA2	1	-7	-18	-20	0.6	0.5	1.2	1.5
Midstream	4%	283.5	82		5.3	7.5	BAA1/BAA2	1	-9	-19	-16	0.6	0.6	1.2	1.1
Technology	10%	764.5	76		5.2	7.3	A2/A3	1	-8	-16	1	0.6	0.5	0.9	0.1
Transportation	2%	150.6	66		5.3	9.1	A3/BAA1	1	-6	-18	-8	0.8	0.4	0.9	0.4
Airlines	0%	18.3	83		5.0	3.5	A3/BAA1	0	-8	-19	-8	0.3	0.2	0.8	0.7
Railroads	1%	90.4	61		5.4	10.8	A3/BAA1	1	-6	-19	-10	0.9	0.5	1.0	0.2
Transportation Services	1%	41.9	70		5.2	7.8	A3/BAA1	0	-4	-16	-6	0.7	0.3	0.7	0.5
Other Industrial	1%	38.8	59		5.3	10.3	AA3/A1	0	-6	-16	-7	0.9	0.4	0.5	0.2
Utility	10%	759.1	77		5.3	8.1	A3/BAA1	1	-7	-17	-9	0.7	0.4	0.9	0.4
Electric	9%	683.0	77		5.3	8.2	A3/BAA1	1	-7	-17	-9	0.7	0.4	0.8	0.4
Natural Gas	1%	59.3	86		5.4	7.8	A3/BAA1	1	-6	-18	-10	0.7	0.4	1.0	0.5
Other Utility	0%	16.9	66		5.2	8.1	BAA1/BAA2	1	-7	-17	-5	0.7	0.3	0.7	0.0
Financial Institutions	32%	2412.3	76		5.1	5.4	A2/A3	0	-7	-19	-2	0.5	0.3	0.9	0.2
Banking	22%	1647.8	71		5.0	4.9	A2/A3	0	-8	-18	-1	0.4	0.2	0.8	0.2
Brokerage Asset Managers Exchanges	2%	119.6	75		5.1	6.2	A3/BAA1	0	-9	-20	2	0.5	0.4	1.1	0.1
Finance Companies	1%	108.6	150		5.7	3.6	BAA2/BAA3	0	-6	-18	6	0.3	0.3	1.3	0.4
Insurance	4%	332.2	82		5.4	8.3	A3/BAA1	1	-8	-23	-8	0.7	0.6	1.4	0.3
REITS	3%	196.6	70		5.0	5.3	A3/BAA1	0	-8	-20	-8	0.5	0.2	0.8	0.3
Other Financial	0%	7.5	93		5.2	4.7	A3/BAA1	0	-3	-18	-8	0.4	0.1	1.0	0.9
US Corporate IG		7503.2	73		5.2	6.9	A3/BAA1	1	-7	-18	-6	0.6	0.4	0.8	0.3

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 24: Sector performance: USD HY

Sector	Metrics							Spread change				Total return			
	Index weight	Mkt val (\$bn)	OAS	Percentile rank (2010-today)	YTW	Dur	Rating	1-week	MTD	QTD	YTD	1-week	MTD	QTD	YTD
Basic Industry	6%	92.9	245		6.9	3.3	BA3/B1	-2	-9	-24	-16	0.6	0.6	2.5	2.4
Chemicals	3%	48.2	214		6.6	3.4	BA3/B1	-3	-12	-29	-34	0.5	0.5	2.5	3.8
Metals & Mining	3%	37.5	199		6.4	3.2	BA3/B1	-1	-11	-28	-4	0.6	0.7	2.8	1.6
Paper	0%	7.2	697		11.3	2.6	B1/B2	-1	1	-3	39	0.7	1.2	1.7	-2.8
Capital Goods	10%	150.3	215		6.6	3.0	BA3/B1	-1	-6	-17	1	0.5	0.2	1.6	-0.3
Aerospace & Defense	2%	30.9	108		5.7	3.0	BA3/B1	0	-2	-22	8	0.3	0.2	1.3	1.1
Building Materials	2%	36.1	353		8.0	3.3	BA3/B1	-1	-6	-16	24	0.7	-0.1	1.7	-1.8
Diversified Manufacturing	2%	27.8	156		5.9	2.5	BA2/BA3	-2	-7	-21	49	0.3	-0.1	1.0	0.8
Construction Machinery	1%	21.6	119		5.7	3.0	BA3/B1	-1	-8	-35	-9	0.4	0.2	1.9	1.6
Packaging	2%	25.2	321		7.6	3.3	B1/B2	-3	-7	-5	-23	0.9	1.0	2.2	-2.7
Environmental	1%	8.8	144		5.8	3.1	BA3/B1	0	-2	-16	-3	0.4	0.2	1.2	1.2
Communications	15%	218.8	402		8.4	3.7	B1/B2	1	5	-7	16	0.2	-0.4	1.4	1.4
Cable & Satellite	5%	81.0	546		9.7	2.9	B1/B2	1	9	1	36	0.3	-1.2	-0.5	0.0
Media & Entertainment	5%	80.8	309		7.6	4.2	BA3/B1	1	2	-13	-8	-0.1	0.0	2.8	1.6
Wireless	1%	19.8	264		7.1	4.1	BA3/B1	2	8	-17	5	0.3	-0.6	2.1	0.7
Wirelines	3%	37.2	363		8.1	4.2	B1/B2	0	-1	-12	16	0.4	0.4	2.6	3.7
Consumer Cyclical	18%	272.1	234		6.7	3.1	BA3/B1	0	-2	-22	-7	0.5	0.4	2.1	1.3
Automotive	3%	46.0	250		6.8	3.0	BA3/B1	-2	-1	-28	-9	0.6	0.4	2.5	1.5
Leisure	2%	29.2	189		6.3	3.4	BA3/B1	-3	-5	-18	5	0.5	0.5	1.6	1.1
Gaming	3%	45.4	259		7.1	3.0	BA3/B1	4	0	-16	7	0.3	0.3	2.0	1.2
Home Construction	1%	18.9	194		6.3	2.9	BA3/B1	-2	-5	-23	-4	0.5	0.3	2.0	1.4
Lodging	1%	16.5	119		5.6	3.8	BA2/BA3	-2	-10	-28	-10	0.5	0.4	1.8	1.0
Retailers	4%	61.0	238		6.6	3.3	BA3/B1	0	-4	-29	-28	0.5	0.4	2.4	1.5
Restaurants	1%	12.5	158		5.9	3.8	BA3/B1	0	3	-15	-3	0.3	0.4	1.6	1.3
Consumer Cyc Services	3%	42.5	298		7.4	2.6	B1/B2	1	-1	-15	8	0.2	0.3	2.2	1.2
Consumer Non-Cyclical	11%	158.4	242		6.8	3.2	B1/B2	-2	-3	-14	-2	0.3	0.1	1.7	1.7
Consumer Products	2%	22.8	332		7.6	4.0	B1/B2	0	7	-7	2	0.2	-1.1	0.8	-0.2
Food & Beverage	2%	28.9	171		6.2	3.5	B1/B2	-1	1	-6	0	0.5	0.1	1.2	1.3
Healthcare	5%	71.3	205		6.5	2.9	B1/B2	-5	-11	-18	-1	0.3	0.5	1.8	2.0
Pharmaceuticals	2%	25.3	381		8.3	3.0	B2/B3	0	2	-18	-13	0.3	-0.1	3.5	3.4
Supermarkets	1%	9.8	153		5.9	3.6	BA2/BA3	2	-1	-8	121	0.3	-0.1	0.6	0.5
Tobacco	0%	0.3	190		6.5	3.3	BA3/BA3	4	-16	-23	11	0.1	1.2	2.2	0.3
Energy	11%	164.6	153		6.1	3.0	BA2/BA3	11	1	-23	-44	0.2	0.2	1.6	4.2
Independent Energy	3%	45.1	167		6.2	2.7	BA3/B1	4	9	-18	-33	0.2	0.0	1.6	4.3
Oil Field Services	2%	26.6	172		6.3	2.7	BA3/B1	1	21	-17	-43	0.2	0.1	1.4	4.0
Refining	0%	6.1	224		7.0	2.4	BA3/B1	3	-3	-10	-38	0.1	0.4	1.2	5.5
Midstream	6%	86.8	134		5.8	3.3	BA2/BA3	22	-10	-29	-50	0.3	0.3	1.7	4.0
Technology	9%	138.9	331		7.7	3.1	BA3/B1	1	-11	-19	5	0.6	1.4	2.6	0.9
Transportation	2%	31.5	439		8.8	2.7	BA3/B1	-3	1	-13	21	0.8	0.1	1.9	-0.5
Airlines	1%	13.3	362		7.9	2.7	BA3/B1	-6	-6	-19	31	1.0	0.6	2.4	0.0
Railroads	0%	1.9	1243		17.1	2.6	B2/B3	1	8	13	29	0.4	0.0	-6.9	-7.2
Transportation Services	1%	16.2	410		8.6	2.7	B1/B2	-1	5	-14	17	0.6	-0.3	2.6	-0.1
Other Industrial	1%	16.0	212		6.6	3.3	BA3/B1	3	-1	-18	-18	0.4	0.0	2.0	1.1
Utility	3%	44.8	161		6.1	4.0	BA2/BA3	-3	-4	-21	10	0.6	0.3	1.9	2.0
Financial Institutions	13%	196.5	238		6.8	3.2	BA3/B1	0	-4	-26	7	0.4	0.3	2.6	0.8
Banking	0%	3.1	183		6.1	4.3	BA1/BA2	0	-3	-33	-15	0.4	0.3	3.0	0.1
Brokerage Asset Managers Exchanges	1%	18.1	176		6.3	2.9	BA3/B1	0	-10	-34	-7	0.3	0.6	3.0	1.8
Finance Companies	4%	63.4	258		7.0	3.1	BA2/BA3	-1	-6	-25	24	0.5	0.5	2.9	-0.1
Insurance	4%	63.0	222		6.7	3.6	BA3/B1	1	1	-24	16	0.4	0.0	1.9	0.3
REITS	2%	29.6	242		6.7	2.8	B1/B2	-2	-5	-27	-11	0.4	0.5	3.0	2.2
Other Financial	1%	19.3	287		7.5	2.9	BA3/B1	-1	-7	-23	-9	0.4	0.6	2.7	1.9
US Corp High Yield		1484.9	261		7.0	3.2	BA3/B1	0	-3	-18	-2	0.4	0.3	2.0	1.5

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 25: Sector performance: EUR IG

Sector	Metrics							Spread Change				Total Return			
	%Mkt	MktVal (€bn)	OAS	Percentile	YTM	Dur	Rating	1-Wk	MTD	QTD	YTD	1-Wk	MTD	QTD	YTD
Consumer Goods	12.5	€ 388.9	84.9		3.7	4.6	A	-1	-5	-18	1	0.63	0.74	1.69	0.59
Automobiles & Parts	5.1	€ 158.4	101.7		3.7	3.7	A	-2	-7	-25	0	0.52	0.74	1.74	0.63
Beverages	2.9	€ 91.1	80.9		3.7	5.9	A	-1	-4	-14	0	0.79	0.80	1.80	0.64
Food Producers	1.9	€ 60.1	69.9		3.5	5.2	A	0	-1	-13	0	0.67	0.70	1.56	0.52
Household Goods	0.2	€ 7.3	51.5		3.5	5.4	AA	-1	-4	-16	0	0.68	0.65	1.50	0.50
Personal Goods	1.7	€ 52.0	66.0		3.4	4.4	A	-1	-2	-17	3	0.58	0.63	1.52	0.36
Tobacco	0.6	€ 20.0	110.9		3.9	4.8	BBB	-2	-4	-17	-1	0.69	0.83	1.75	0.77
Consumer Services	4.6	€ 142.9	87.4		3.8	5.2	A	-1	-3	-12	5	0.67	0.73	1.52	0.46
Food & Drug Retailers	0.7	€ 20.3	85.7		3.6	4.3	BBB	-1	-3	-17	0	0.57	0.68	1.60	0.62
General Retailers	0.6	€ 18.8	91.1		4.0	6.9	AA	-1	-2	-11	12	0.90	0.92	1.62	0.51
Media	1.2	€ 36.0	92.1		3.7	4.0	BBB	-1	-3	-18	8	0.53	0.63	1.58	0.41
Travel & Leisure	2.2	€ 67.8	84.7		3.8	5.6	A	0	-2	-10	3	0.72	0.75	1.44	0.42
Energy	3.2	€ 98.0	95.5		3.8	4.9	A	-1	-3	-16	-2	0.63	0.73	1.70	0.92
Oil & Gas Producers	2.8	€ 87.7	96.0		3.8	4.8	A	-1	-4	-17	-3	0.63	0.74	1.71	0.91
Oil Equipment / Services & Distribution	0.2	€ 7.8	91.9		3.9	5.6	A	0	1	3	15	0.68	0.69	1.68	1.03
Renewable Energy	0.1	€ 2.5	92.5		3.7	5.0	BBB	-1	-2	-14	-9	0.69	0.77	1.61	1.19
Industrials	9.7	€ 299.7	81.1		3.6	4.5	A	-1	-2	-17	0	0.60	0.64	1.59	0.53
Aerospace & Defense	0.4	€ 13.6	70.0		3.5	4.4	A	0	-2	-15	6	0.56	0.57	1.48	0.33
Construction & Materials	1.6	€ 50.2	83.8		3.6	4.2	BBB	-1	-2	-18	0	0.57	0.60	1.62	0.59
Electronic & Electrical Equipment	0.9	€ 28.4	70.2		3.5	4.8	A	-1	-2	-17	0	0.66	0.69	1.64	0.53
General Industrials	1.3	€ 39.4	72.2		3.6	5.3	A	-1	-2	-20	-2	0.69	0.69	1.75	0.56
Industrial Engineering	1.1	€ 34.9	77.3		3.5	3.4	A	-1	-5	-20	0	0.45	0.61	1.51	0.60
Industrial Transportation	3.3	€ 103.1	85.5		3.6	4.8	A	-1	-2	-15	1	0.62	0.65	1.54	0.46
Support Services	1.0	€ 30.2	95.7		3.7	4.0	BBB	-2	-3	-21	1	0.56	0.64	1.64	0.68
Health Care	5.5	€ 170.3	78.4		3.7	5.8	A	-1	-2	-15	-1	0.74	0.70	1.67	0.69
Health Care Equipment & Services	2.4	€ 75.3	82.5		3.8	5.7	A	-1	-4	-16	-1	0.73	0.71	1.65	0.63
Pharmaceuticals & Biotechnology	3.1	€ 95.0	75.3		3.7	5.9	A	0	-1	-15	-1	0.75	0.69	1.67	0.73
Basic Materials	2.7	€ 84.6	83.9		3.6	4.8	A	-1	-3	-20	-2	0.66	0.74	1.78	0.72
Chemicals	2.1	€ 63.7	82.8		3.6	5.0	A	-1	-3	-19	-1	0.68	0.75	1.78	0.69
Forestry & Paper	0.2	€ 5.6	90.1		3.6	4.2	BBB	-2	-5	-26	3	0.60	0.80	1.91	0.49
Industrial Metals	0.1	€ 3.7	89.5		3.6	5.1	BBB	-1	-1	-20	-3	0.68	0.64	1.83	0.85
Mining	0.4	€ 11.6	86.1		3.6	4.1	BBB	-1	-4	-23	-10	0.57	0.68	1.71	0.90
Technology	2.5	€ 77.0	94.8		3.9	5.3	A	0	1	-15	9	0.65	0.58	1.64	0.25
Software & Computer Services	2.0	€ 63.5	98.8		3.9	5.7	A	0	1	-14	10	0.70	0.58	1.65	0.24
Technology Hardware & Equipment	0.4	€ 13.4	64.7		3.3	3.5	A	0	-3	-24	5	0.43	0.60	1.58	0.31
Telecommunications	5.5	€ 170.9	91.9		3.8	5.5	BBB	-1	-2	-16	0	0.74	0.75	1.70	0.63
Fixed Line Telecommunications	4.4	€ 136.6	90.3		3.8	5.6	BBB	-1	-3	-16	0	0.74	0.76	1.68	0.64
Mobile Telecommunications	1.1	€ 34.3	98.4		3.8	5.3	BBB	-1	1	-16	3	0.72	0.71	1.81	0.63
Financials	42.4	€ 1,317.3	94.9		3.7	4.0	A	-1	-3	-21	0	0.53	0.66	1.65	0.63
Banks Sen	26.5	€ 823.6	83.9		3.5	3.8	A	-1	-3	-19	0	0.49	0.64	1.52	0.61
Banks Sub	4.1	€ 126.6	126.2		3.9	3.5	BBB	-1	-7	-28	0	0.48	0.74	1.83	0.83
Financial Services Sen	2.9	€ 89.4	83.3		3.6	4.6	A	-2	-1	-17	0	0.64	0.67	1.66	0.72
Financial Services Sub	0.0	€ 1.0	113.8		3.7	2.1	A	-9	-14	-35	-25	0.41	0.73	1.44	1.30
Insurance Sen	1.5	€ 45.2	77.7		3.5	4.1	A	-1	-1	-22	2	0.54	0.57	1.66	0.46
Insurance Sub	2.6	€ 81.2	135.0		4.1	4.8	BBB	-1	-4	-21	1	0.64	0.75	1.88	0.69
Real Estate Sen	4.8	€ 149.7	113.5		3.9	4.7	BBB	-1	-2	-26	5	0.62	0.65	2.09	0.51
Real Estate Sub	0.0	€ 0.5	172.3		4.5	5.7	A	-2	-8	-34	0	0.84	0.89	2.99	0.00
Utilities	11.5	€ 355.6	96.9		3.8	5.2	BBB	-1	-1	-12	3	0.69	0.68	1.56	0.64
Electricity	6.9	€ 213.5	99.7		3.9	5.3	BBB	-1	-2	-13	3	0.70	0.70	1.59	0.66
Gas / Water & Multiutilities	4.6	€ 142.1	92.8		3.8	5.2	BBB	0	0	-11	2	0.67	0.63	1.53	0.62
Index	100.0	€ 3,105.2	90.6		3.7	4.6	A	-1	-3	-18	1	0.60	0.68	1.64	0.62

Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 26: Sector performance: EUR HY

Sector	Metrics							Spread Change				Total Return			
	%Mkt	MktVal (€bn)	OAS	Percentile	YTM	Dur	Rating	1-Wk	MTD	QTD	YTD	1-Wk	MTD	QTD	YTD
Consumers	27.9	€ 122.7	307		5.7	2.7	BB	0	-12	-50	32	0.40	0.85	2.68	0.93
Consumer Goods	15.5	€ 68.1	276		5.4	2.8	BB	0	-12	-45	39	0.36	0.72	2.63	1.04
Consumer Services	12.4	€ 54.6	347		6.1	2.7	BB	0	-8	-46	28	0.46	1.01	2.75	0.80
Energy	10.0	€ 44.1	252		5.2	3.1	BB	0	-14	-37	-6	0.40	0.66	1.80	0.92
Energy	2.3	€ 10.3	252		5.1	2.7	BB	0	-20	-58	-49	0.30	0.99	2.41	2.50
Utilities	7.7	€ 33.8	252		5.2	3.2	BB	0	-13	-32	5	0.43	0.55	1.62	0.47
Industrials	30.4	€ 133.5	333		6.0	2.7	BB	-1	-14	-61	-12	0.41	1.15	3.03	1.76
Basic Materials	5.9	€ 25.7	412		6.7	2.7	B	-3	-46	-161	-151	0.46	2.21	6.13	5.97
Construction & Materials	1.3	€ 5.5	667		9.3	2.6	BB	8	-22	-47	59	0.30	2.07	3.50	0.03
Health Care	8.6	€ 37.6	293		5.6	2.5	B	-1	-5	-48	27	0.34	0.69	2.29	1.33
Industrials	14.7	€ 64.6	296		5.6	2.8	BB	-2	-11	-44	10	0.45	0.94	2.30	0.68
TMT	20.1	€ 88.1	322		5.9	3.4	BB	-4	-6	-44	36	0.58	0.91	2.65	0.58
Technology	3.9	€ 17.1	427		7.0	3.7	BB	-7	-31	-59	48	0.66	1.25	2.85	-1.25
Telecommunications	16.2	€ 71.0	293		5.6	3.3	BB	-4	-2	-42	21	0.57	0.83	2.60	0.98
Financials	11.6	€ 50.9	368		6.3	3.1	BB	-2	5	-44	41	0.49	0.30	2.20	0.05
Core Financials	4.3	€ 18.9	193		4.5	3.1	BB	-2	-13	-46	14	0.42	0.74	2.09	0.60
Financial Services	1.6	€ 7.1	898		11.6	2.0	B	5	186	217	296	0.30	-2.33	-1.50	-2.86
Real Estate	5.7	€ 24.9	400		6.6	3.4	BB	-4	-1	-65	22	0.60	0.72	3.36	0.51
Index	100.0	€ 439.2	319		5.8	2.9	BB	-1	-10	-50	17	0.45	0.87	2.64	1.02

0 50 100

Source: iBoxx, Goldman Sachs Global Investment Research

Performance dashboard: USD Credit ETFs

Exhibit 27: USD ETF performance

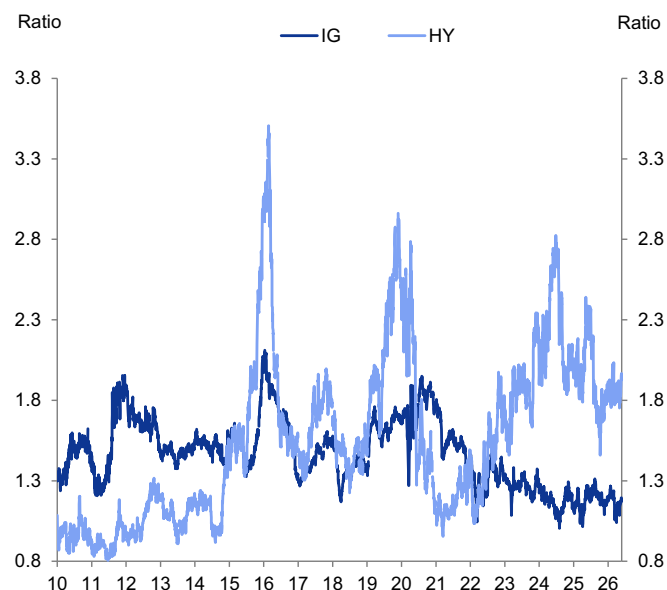
Sector	ETF Ticker	AUM	Weekly Flow (\$Mn.)	YTD cum flow (% of AUM)	Shares Outstanding (Mn.)	Short Interest (%)	Price	% NAV Premium	YTD Total Return	QTD Total Return	MTD Total Return	Benchmark Index
Investment Grade	LQD	\$29.61B	282	0.5%	272	19.34	109.19	0.05%	0.6	1.0	0.7	iBoxx USD Liquid Investment Grade Index
	VCSH	\$43.08B	32	23.0%	545	0.91	79.12	0.04%	0.7	0.5	0.2	Bloomberg Barclays US Corporate 1-5 Yrs Total Return Index Value Unhedged
	VCIT	\$66.1B	248	33.1%	801	2.27	82.73	0.06%	0.3	0.8	0.2	Bloomberg Barclays US Credit Corporate 5-10 Yrs Total Return Index Value Unhedged
	FLOT	\$9.51B	112	25.2%	187	--	51.04	0.10%	1.7	0.9	0.4	Bloomberg Barclays US FRN < 5 yr Total Return Index Value Unhedged
	IGSB	\$22.B	-32	4.2%	420	0.73	52.48	0.01%	0.7	0.6	0.2	ICE BofAML 1-5 Year US Corporate Bond Index
	IGIB	\$18.09B	-3	20.0%	341	1.88	53.21	0.02%	0.4	0.8	0.3	Bloomberg Barclays U.S. Interim
	SPSB	\$10.22B	51	22.7%	340	1.70	30.04	0.02%	0.9	0.6	0.2	Bloomberg Barclays US Corporate 1-3 Yrs Total Return Index Value Unhedged
	SPIB	\$11.65B	307	16.3%	348	3.35	33.50	0.06%	0.5	0.6	0.2	Bloomberg Barclays Intermediate Corporate Total Return Index Value Unhedged
	VCLT	\$8.5B	-16	-37.8%	114	23.86	75.30	0.25%	1.1	1.7	1.4	Bloomberg Barclays Long U.S. Corporate Total Return Index Value Unhedged
High Yield	HYG	\$15.88B	-495	6.3%	198	42.22	80.21	0.12%	1.5	1.8	0.3	iBoxx USD Liquid High Yield Index
	JNK	\$7.68B	0	-0.1%	80	11.06	96.64	0.10%	1.6	2.1	0.3	Bloomberg Barclays VLI: High Yield Total Return Index Value Unhedged USD
	SJNK	\$4.76B	15	2.5%	190	2.36	25.12	0.12%	1.5	1.7	0.2	Bloomberg Barclays US High Yield 350M Cash 0-5yr 2% Cap Total Return Index
	SHYG	\$7.57B	0	22.9%	178	0.64	42.55	0.08%	1.6	1.7	0.3	Markit iBoxx USD Liquid High Yield 0-5 Total Return Index
	HYS	\$1.68B	0	28.4%	18	0.04	93.76	-0.02%	1.3	1.7	0.4	ICE BofAML 0-5 Year US High Yield Constrained Index
	HYLS	\$1.64B	2	0.9%	40	0.22	40.86	-0.01%	0.4	1.8	0.3	ICE BofAML US High Yield Constrained Index
	ANGL	\$3.08B	-7	-0.7%	106	0.94	29.16	0.06%	1.4	2.7	0.2	ICE BofAML US Fallen Angel High Yield Index
	BKLN	\$7.16B	-90	-20.3%	348	24.59	20.48	-0.17%	0.1	1.4	0.0	S&P/LSTA U.S. Leveraged Loan 100 Index
	SRLN	\$5.25B	0	-30.7%	129	6.62	40.58	-0.09%	0.7	2.3	0.5	Markit iBoxx USD Liquid Leverage Loans Total Return Index
Loans	FTSL	\$2.3B	0	6.8%	51	0.16	44.97	0.07%	0.6	1.4	0.3	S&P/LSTA US Leveraged Loan 100 Market Value Index
	EMB	\$14.09B	67	-4.6%	147	11.55	96.21	0.36%	1.7	3.4	0.9	J.P. Morgan EMBI Global Core Index
	EMLC	\$4.85B	0	78.0%	191	1.34	25.57	-0.05%	1.0	2.9	0.6	J.P. Morgan Government Bond Index Emerging Markets Global Core Index
CLOs	JAAA	\$27.32B	109	64.7%	539	17.34	50.71	0.04%	1.8	1.1	0.4	
	PAAA	\$10.11B	538	494.4%	196	--	51.50	0.02%	2.0	1.0	0.4	
	JBBB	\$1.2B	19	-11.7%	25	--	47.55	0.25%	1.7	2.5	0.4	
	CLOI	\$1.34B	11	68.9%	25	11.22	52.95	0.00%	1.9	1.3	0.6	
	CLOA	\$2.15B	16	229.1%	41	--	51.98	-0.02%	2.0	1.0	0.4	

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Performance dashboard: Bond-level dispersion

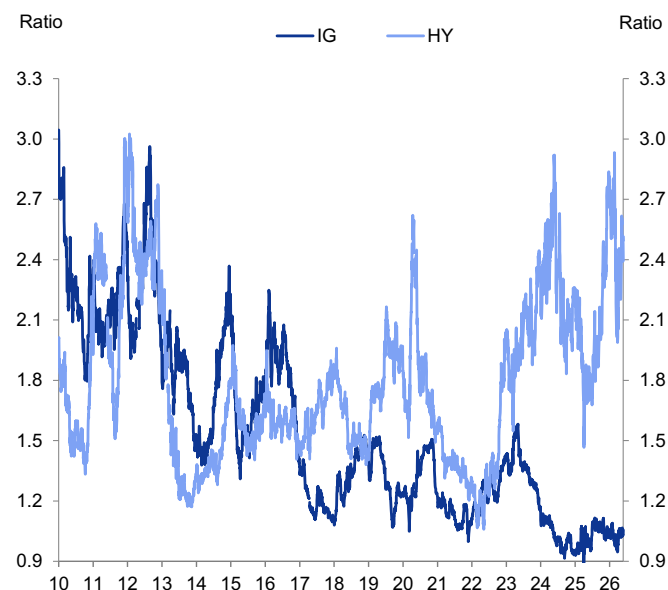
Note: Spread dispersion is the inter-decile range divided by the median across the bond index constituents and return dispersion is the inter-decile range of 10-day returns divided by the median across the bond index constituents.

Exhibit 28: USD spread dispersion



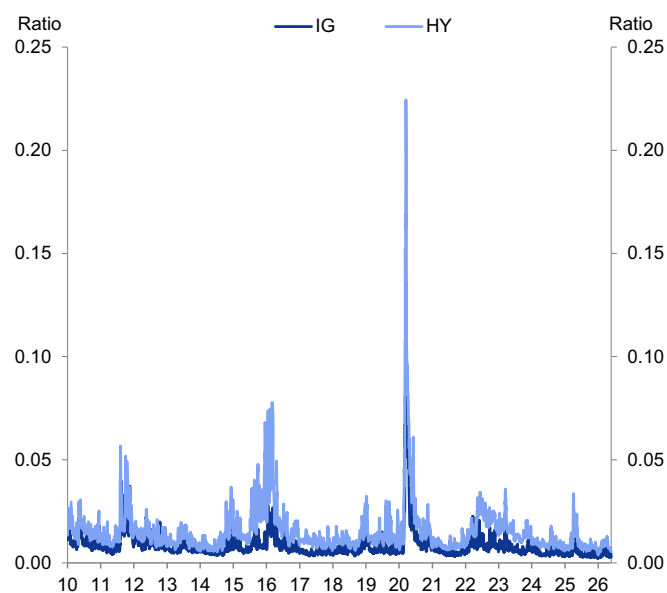
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 29: EUR spread dispersion



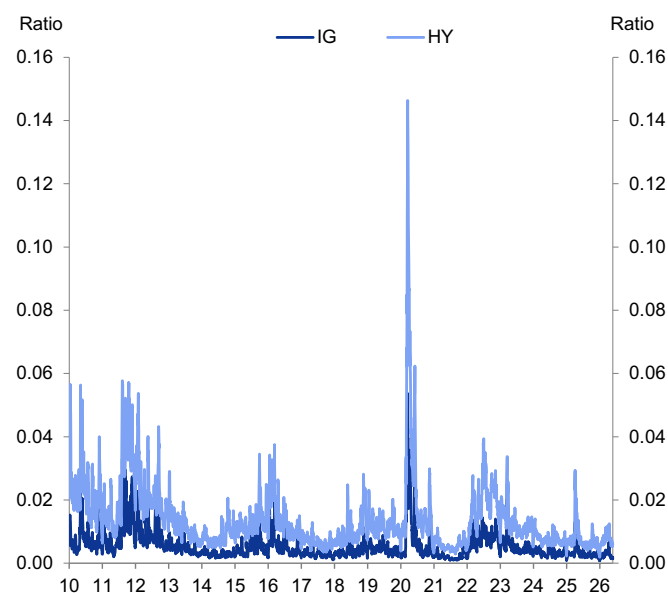
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 30: USD return dispersion



Source: iBoxx, Goldman Sachs Global Investment Research

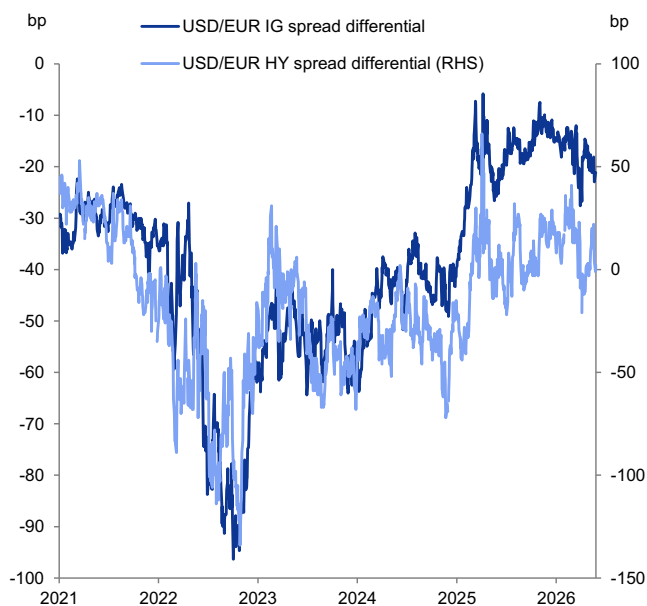
Exhibit 31: EUR return dispersion



Source: iBoxx, Goldman Sachs Global Investment Research

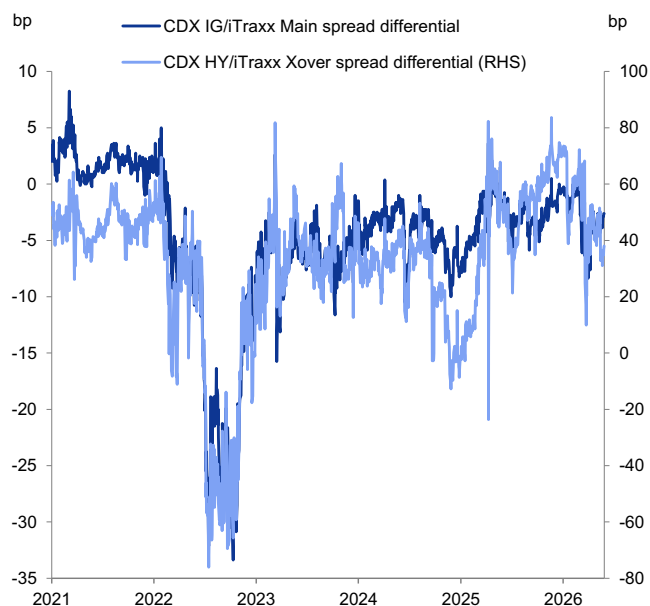
Relative value dashboard: Excess spreads in the USD and EUR markets

Exhibit 32: USD/EUR IG and HY spread differential



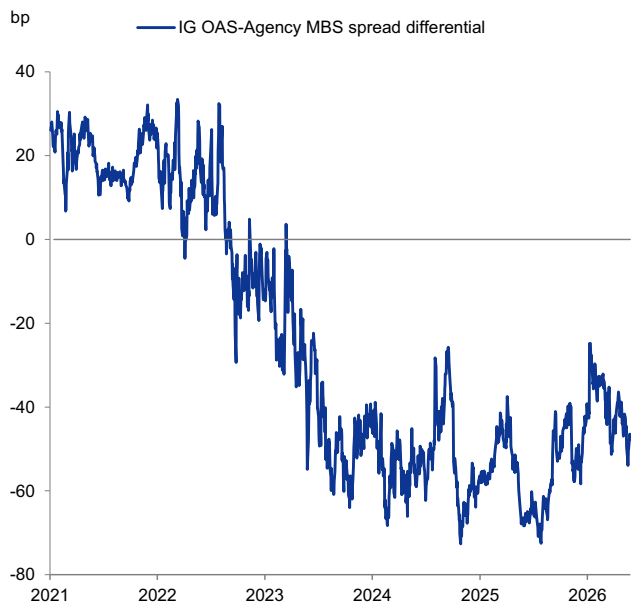
Source: iBoxx, Goldman Sachs Global Investment Research

Exhibit 33: CDX IG/iTraxx Main and CDX HY/iTraxx Xover spread differential



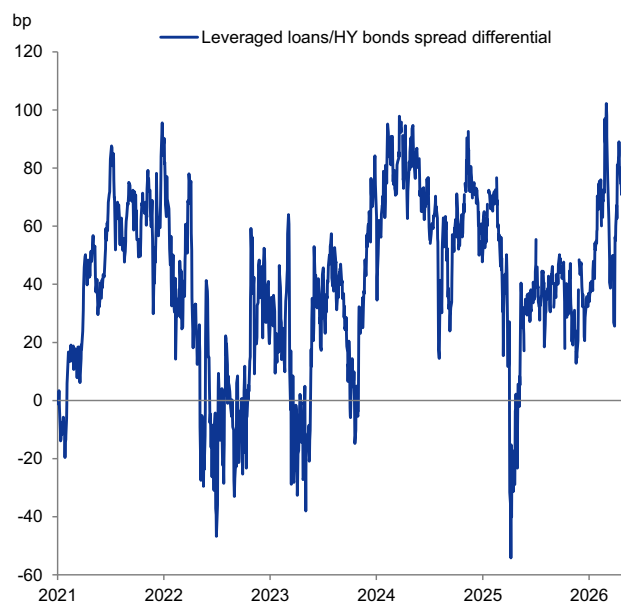
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 34: IG OAS/Agency MBS spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 35: Leveraged loans/HY bonds spread differential



Source: Bloomberg, Morningstar LCD, Goldman Sachs Global Investment Research

Performance dashboard: Biggest movers in the USD bond market

Exhibit 36: Top and bottom 20 performers in the USD IG market

Issuer	Sector	Security	Amount outstanding (\$mn)	Ratings (Moody's/S&P/Fitch)	1-week price return (%)	1-week price change (pts)	Current price
Top 20 performers							
Arcelormittal	Iron/Steel	MTNA 7 10/15/39	686	Baa2/BBB/--	3.6%	4.0	113.54
Intel Corp	Semiconductors	INTC 3.1 02/15/60	1,000	Baa2/BBB/BBB	3.4%	2.0	56.63
Toledo Hospital/The	Healthcare-Services	TOLHOS 5 3/4 11/15/38	500	A1/AA/--	3.3%	3.3	97.86
Oracle Corp	Software	ORCL 6 1/2 04/15/38	1,250	Baa2/BBB/BBB	3.3%	3.3	100.84
Comcast Corp	Media	CMCSA 5.65 06/01/54	1,200	A3/A-/A-	3.1%	2.8	90.79
Ally Financial Inc	Diversified Finan Serv	ALLY 6.184 07/26/35	750	Baa3/BBB-/BBB-	2.8%	2.8	101.15
At&T Inc	Telecommunications	T 6 04/30/56	1,650	Baa2/BBB/BBB+	2.7%	2.7	96.82
Ibm Corp	Computers	IBM 5.8 02/03/56	750	A3/A-/A-	2.7%	2.6	95.66
Vale Overseas Limited	Iron/Steel	VALEBZ 8 1/4 01/17/34	335	--/--/--	2.6%	3.0	116.64
Las Vegas Sands Corp	Lodging	LVS 6.2 08/15/34	500	Baa3/BBB-/BBB	2.6%	2.6	102.16
American Honda Finance	Auto Manufacturers	HNDA 4.9 01/10/34	750	A3/BBB+/A-	2.5%	2.5	97.67
Micron Technology Inc	Semiconductors	MU 6.05 11/01/35	221	Baa2/BBB/BBB+	2.5%	2.6	103.07
Comcast Corp	Media	CMCSA 7.05 03/15/33	750	A3/A-/A-	2.3%	2.6	111.07
Equitable Holdings Inc	Insurance	EQH 6.7 03/28/55	500	Baa2/BBB/--	2.3%	2.4	103.04
Amazon.Com Inc	Internet	AMZN 2 1/2 06/03/50	2,500	A1/AA/AA-	2.3%	1.3	57.57
Oneok Inc	Pipelines	OKE 5.05 11/01/34	1,600	Baa2/BBB/BBB	2.3%	2.3	97.60
American Electric Power	Electric	AEP 5.8 03/15/56	1,500	Baa3/BBB-/BB+	2.3%	2.3	99.52
Sempra	Electric	SRE 6 3/8 04/01/56	800	Baa3/BBB-/BBB-	2.3%	2.3	100.25
Apollo Global Management	Diversified Finan Serv	APO 5.15 08/12/35	850	A2/A/A	2.1%	2.0	97.23
Salesforce Inc	Software	CRM 6.7 03/15/66	1,000	A2/A+/--	2.1%	2.1	100.96
Bottom 20 performers							
Concentrix Corp	Software	CNXC 6.85 08/02/33	550	Baa3/BBB-/BBB	-2.4%	-2.2	92.79
Dow Chemical Co/The	Chemicals	DOW 9.4 05/15/39	503	Baa3/BBB-/BBB	-1.4%	-1.9	131.13
Meta Platforms Inc	Internet	META 5.6 05/15/53	2,500	Aa3/AA-/--	-1.4%	-1.3	93.06
Pfizer Inc	Pharmaceuticals	PFE 7.2 03/15/39	2,500	A2/A/--	-1.3%	-1.5	118.14
Duke Energy Florida Llc	Electric	DUK 4.2 12/01/30	500	A1/A/--	-1.2%	-1.2	99.55
Barclays Plc	Banks	BACR 5.367 02/25/31	1,750	Baa1/BBB+/A	-1.2%	-1.2	102.70
Ford Motor Credit Co Llc	Auto Manufacturers	F 5.73 09/05/30	1,250	Ba1/BBB-/BBB-	-1.2%	-1.2	102.06
Oracle Corp	Software	ORCL 5 1/2 08/03/35	1,750	Baa2/BBB/BBB	-1.1%	-1.0	97.22
Jefferies Fin Group Inc	Diversified Finan Serv	JEF 4.15 01/23/30	1,000	Baa2/BBB/BBB+	-0.9%	-0.9	97.67
American Honda Finance	Auto Manufacturers	HNDA 5 1/8 07/07/28	800	A3/BBB+/A-	-0.9%	-0.9	101.91
American Express Co	Diversified Finan Serv	AXP 4.918 07/20/33	1,750	A2/A-/A	-0.9%	-0.9	100.79
Chevron Usa Inc	Oil&Gas	CVX 4 1/2 10/15/32	1,250	Aa2/AA-/--	-0.9%	-0.9	100.80
Chevron Usa Inc	Oil&Gas	CVX 4.98 04/15/35	750	Aa2/AA-/--	-0.8%	-0.8	101.73
Mcdonald'S Corp	Retail	MCD 5.15 09/09/52	750	Baa1/BBB+/--	-0.8%	-0.7	91.79
Charles Schwab Corp	Diversified Finan Serv	SCHW 4.914 11/14/36	1,000	A2/A-/A	-0.8%	-0.7	97.70
Salesforce Inc	Software	CRM 5.55 03/15/36	4,500	A2/A+/--	-0.7%	-0.7	100.75
Johnson & Johnson	Pharmaceuticals	JNJ 4 3/8 12/05/33	850	Aaa/AAA/--	-0.7%	-0.7	100.20
Us Bancorp	Banks	USB 4.967 07/22/33	1,300	A3/A-/A-	-0.7%	-0.7	99.61
Natwest Group Plc	Banks	NWG 3.032 11/28/35	850	Baa1/BBB+/A-	-0.7%	-0.6	91.72
Citigroup Inc	Banks	C 4 1/8 07/25/28	2,100	Baa2/BBB/BBB+	-0.7%	-0.7	99.85

Note: Only Bloomberg index-eligible bonds.

Source: Bloomberg, TRACE, data compiled by Goldman Sachs Global Investment Research

Exhibit 37: Top and bottom 20 performers in the USD HY market

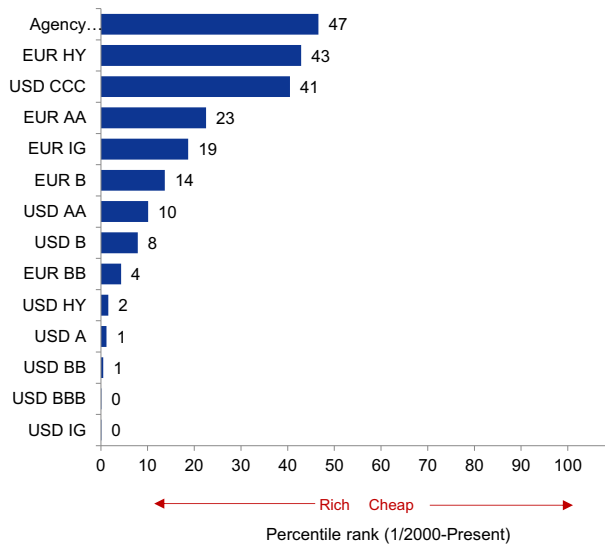
Issuer	Sector	Security	Amount outstanding (\$mn)	Ratings (Moody's/S&P/Fitch)	1-week price return (%)	1-week price change (pts)	Current price
Top 20 performers							
Cd&R Smokey / Radio Sys	Housewares	PETSAF 9 1/2 10/15/29	775	B3/B/--	17.9%	9.1	51.00
Jeldwen Holding	Building Materials	JELD 7 09/01/32	350	Caa2/CCC/--	9.2%	5.3	57.25
Newfold Digital Hldgs Gr	Internet	EIGI 11 3/4 04/30/29	319	--/--/BB-	8.6%	6.5	75.50
Macy'S Retail Hldgs Llc	Retail	M 5 1/8 01/15/42	250	Ba2/BB+/BBB-	3.6%	2.8	75.38
Getty Images Inc	Internet	ABEGET 11 1/4 02/21/30	540	B3/B+/--	3.1%	2.5	81.50
Cnt Prnt/Cdk Glo li/Fin	Software	CDK 8 06/15/29	755	B3/B/--	2.9%	1.5	50.54
S&S Holdings Llc	Distribution/Wholesale	SSACTI 8 3/8 10/01/31	600	B3/B/--	2.7%	2.5	93.27
Unisys Corp	Computers	UIS 10 5/8 01/15/31	700	B2/B+/--	2.6%	2.4	94.10
Fmc Corp	Chemicals	FMC 6 3/8 05/18/53	500	Ba2/BB+/BB+	2.2%	1.7	76.50
Alta Equipment Group	Commercial Services	ALTG 9 06/01/29	500	Caa1/B/--	2.1%	2.0	92.79
Harrow Inc	Pharmaceuticals	HROW 8 5/8 09/15/30	300	B3/--/B	1.9%	1.9	99.82
Fmc Corp	Chemicals	FMC 5.65 05/18/33	500	Ba2/BB+/BB+	1.9%	1.7	89.25
Cleveland-Cliffs Inc	Iron/Steel	CLF 4 7/8 03/01/31	325	B2/--/B+	1.9%	1.8	91.70
Embecka Corp	Healthcare-Products	EMBC 6 3/4 02/15/30	200	B3/B/--	1.8%	1.5	81.25
Fmc Corp	Chemicals	FMC 4 1/2 10/01/49	500	Ba2/BB+/BB+	1.8%	1.1	61.39
Masterbrand Inc	Building Materials	MASBRA 7 07/15/32	700	B1/BB/BB+	1.7%	1.7	97.33
Specialty Building Produ	Retail	SBPLLC 7 3/4 10/15/29	510	Caa1/B/--	1.6%	1.5	90.50
Bausch Health Cos Inc	Pharmaceuticals	BHCCN 6 1/4 02/15/29	821	Ca/CCC+/-	1.6%	1.2	71.09
Cable One Inc	Media	CABO 4 11/15/30	548	B3/B/--	1.6%	1.0	63.50
Uwm Holdings Llc	Diversified Finan Serv	UWMC 6 1/4 03/15/31	1,000	Ba3/--/BB-	1.6%	1.4	89.87
Bottom 20 performers							
Cntrl Parent/Cdk Glb Inc	Software	CDK 7 1/4 06/15/29	750	B3/B/--	-27.2%	-19.0	70.00
Embarq Llc	Telecommunications	EMBARQ 7.995 06/01/36	1,215	Ca/CCC/--	-10.2%	-3.1	30.08
Liberty Interactive Llc	Retail	LINTA 8 1/4 02/01/30	504	WR/D/--	-4.4%	-0.3	5.81
Csc Holdings Llc	Media	CSCHLD 5 3/8 02/01/28	1,000	Caa2/CCC+/-	-3.0%	-2.0	66.00
Csc Holdings Llc	Media	CSCHLD 11 3/4 01/31/29	2,050	Caa2/CCC+/-	-2.7%	-1.8	64.75
Lcpr Sr Secured Fin Dac	Media	LILAPR 5 1/8 07/15/29	820	Caa3/CCC+/CCC+	-2.4%	-1.5	64.00
Gray Media Inc	Media	GTN 5 3/8 11/15/31	1,220	Caa1/CCC/--	-2.4%	-1.8	74.25
Cerdia Finanz GmbH	Chemicals	PLATIN 9 3/8 10/03/31	900	B2/B/--	-2.2%	-2.0	93.25
Paramount Global	Media	PARA 5.9 10/15/40	300	Baa3/BB+/BB+	-2.0%	-1.5	77.50
Whirlpool Corp	Home Furnishings	WHR 5.15 03/01/43	250	B1/BB-/BB-	-1.9%	-1.3	68.65
Whirlpool Corp	Home Furnishings	WHR 6 1/2 06/15/33	600	B1/BB-/BB-	-1.8%	-1.6	91.31
Whirlpool Corp	Home Furnishings	WHR 4.6 05/15/50	500	B1/BB-/BB-	-1.6%	-1.0	64.11
Zi Tech Llc/Zi Fin Corp	Software	ZMINFO 3 7/8 02/01/29	650	B1/B+/--	-1.6%	-1.3	79.25
Whirlpool Corp	Home Furnishings	WHR 6 1/8 06/15/30	600	B1/BB-/BB-	-1.6%	-1.5	95.25
Lcpr Sr Secured Fin Dac	Media	LILAPR 6 3/4 10/15/27	1,161	Caa3/CCC+/CCC+	-1.6%	-1.0	66.02
Ew Scripps Co	Media	SSP 9 7/8 08/15/30	750	Caa2/CCC+/CCC-	-1.5%	-1.5	98.00
Scripps Escrow li Inc	Media	SSP 5 3/8 01/15/31	392	Caa3/CCC-/CC	-1.4%	-1.0	72.40
Whirlpool Corp	Home Furnishings	WHR 5 1/2 03/01/33	300	B1/BB-/BB-	-1.1%	-1.0	84.47
Gray Media Inc	Media	GTN 4 3/4 10/15/30	790	Caa1/CCC/--	-1.1%	-0.8	76.32
Whirlpool Corp	Home Furnishings	WHR 4.7 05/14/32	300	B1/BB-/BB-	-1.1%	-0.9	83.75

Note: Only Bloomberg index-eligible bonds.

Source: Bloomberg, TRACE, data compiled by Goldman Sachs Global Investment Research

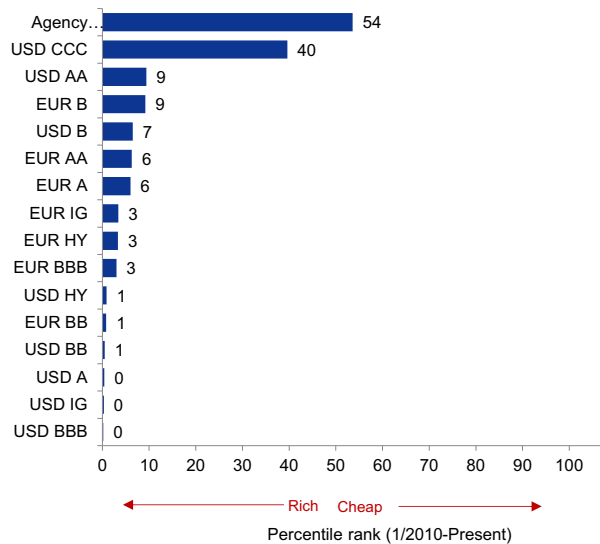
Valuation dashboard: Percentile ranks for spreads and yields

Exhibit 38: Percentile ranks for spreads across markets (2000-current)



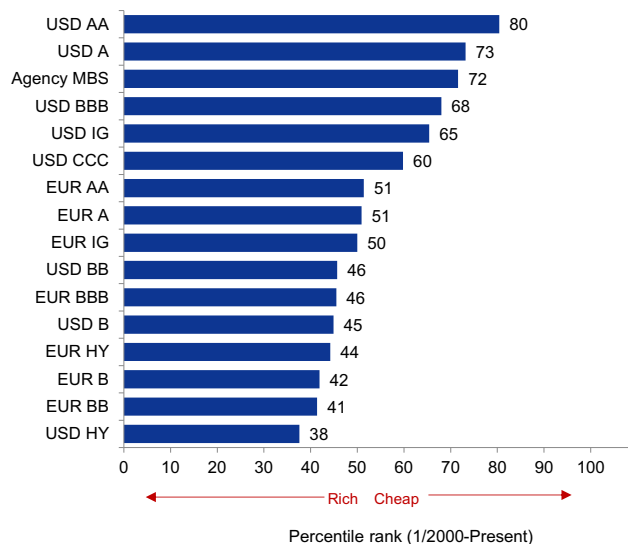
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 39: Percentile ranks for spreads across markets (2010-current)



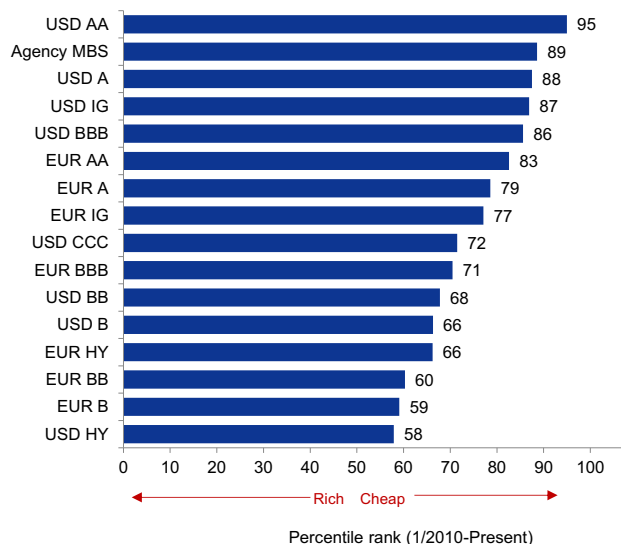
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 40: Percentile ranks for yields across markets (2000-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

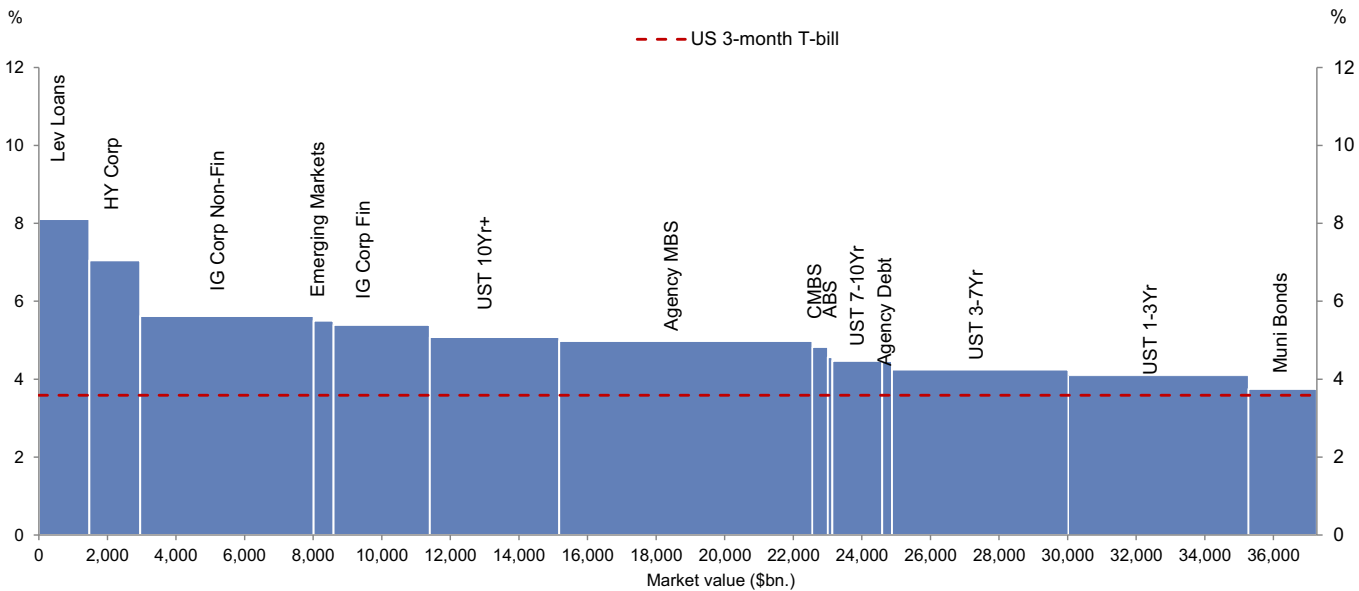
Exhibit 41: Percentile ranks for yields across markets (2010-current)



Source: Bloomberg, Goldman Sachs Global Investment Research

Valuation dashboard: Sizing the carry in USD and EUR fixed income markets

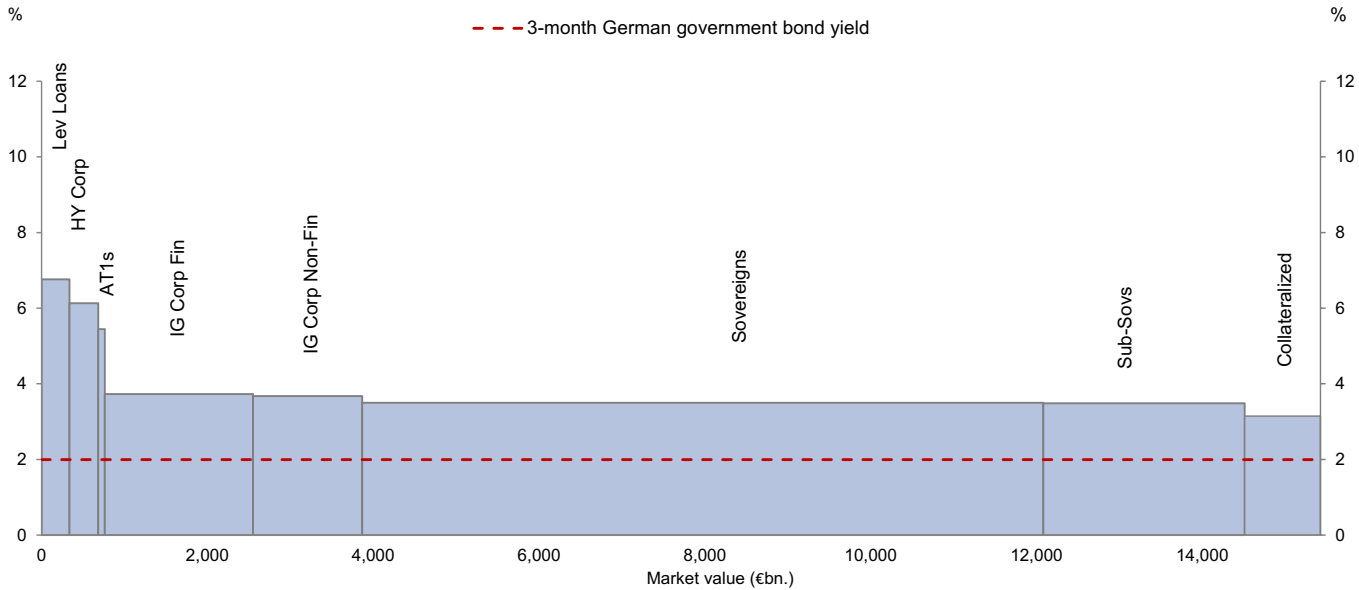
Exhibit 42: Average yield vs. total market value outstanding across the USD fixed income complex



Note: We use the iBoxx indices for the IG corporate credit and US Treasury markets, the Bloomberg indices for the HY, MBS, CMBS and ABS markets, and the Morningstar LSTA index for the leveraged loan market.

Source: iBoxx, Bloomberg, Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 43: Average yield vs. total market value outstanding for the EUR fixed income markets

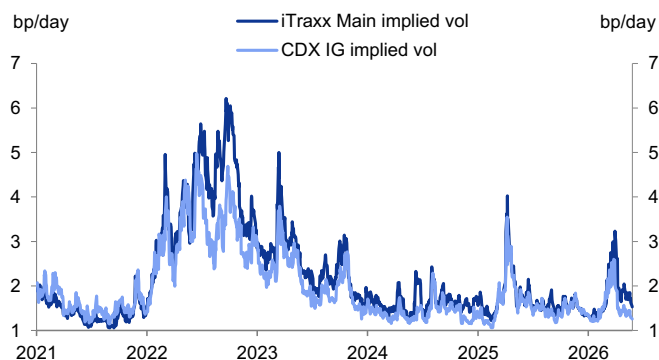


Source: iBoxx, Goldman Sachs Global Investment Research

Valuation dashboard: Macro index volatility

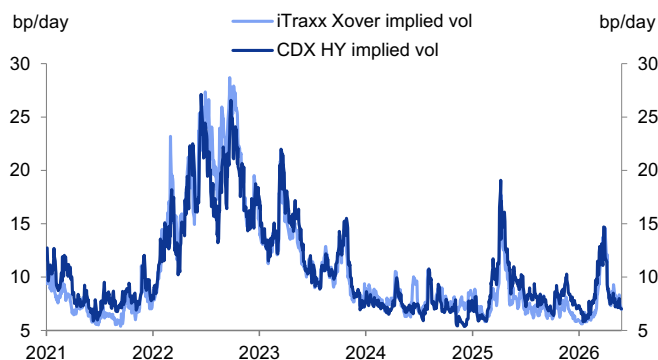
Note: Skew is calculated using difference in 3m implied volatility between 25-delta payers and receivers, divided by 3m ATM implied volatility.

Exhibit 44: CDX IG and iTraxx Main vol 3m ATM implied volatility



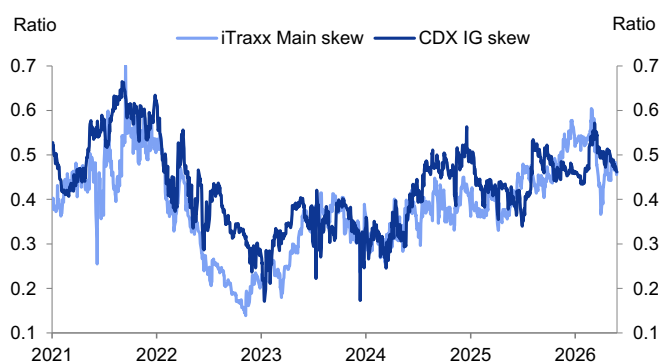
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 45: CDX HY and iTraxx Xover vol 3m ATM implied volatility



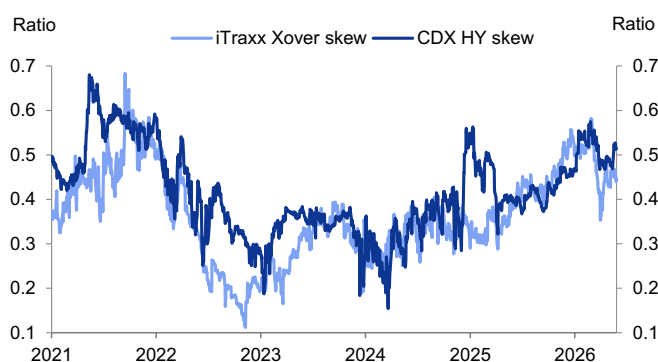
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 46: CDX IG and iTraxx Main skew



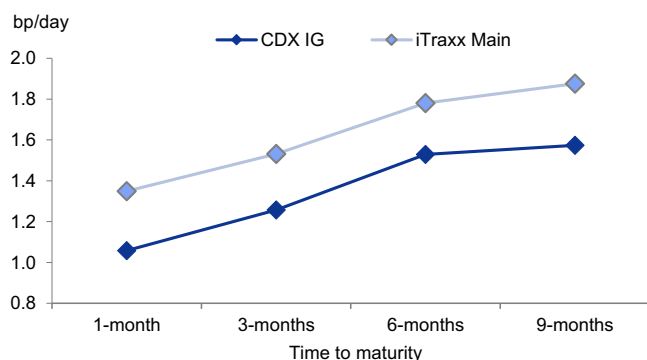
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 47: CDX HY and iTraxx Xover skew



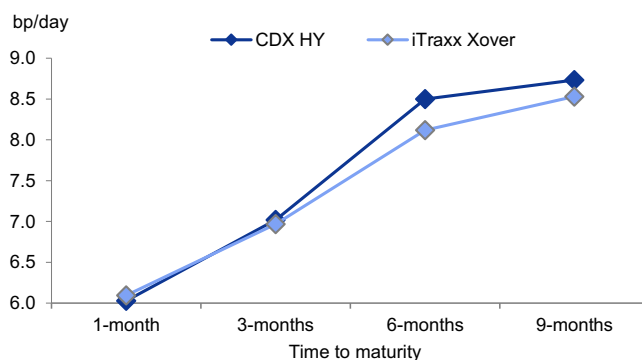
Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 48: CDX IG and iTraxx Main ATM implied volatility term structure



Source: Goldman Sachs, Goldman Sachs Global Investment Research

Exhibit 49: CDX HY and iTraxx Xover ATM implied volatility term structure



Source: Goldman Sachs, Goldman Sachs Global Investment Research

Valuation dashboard: CLOs

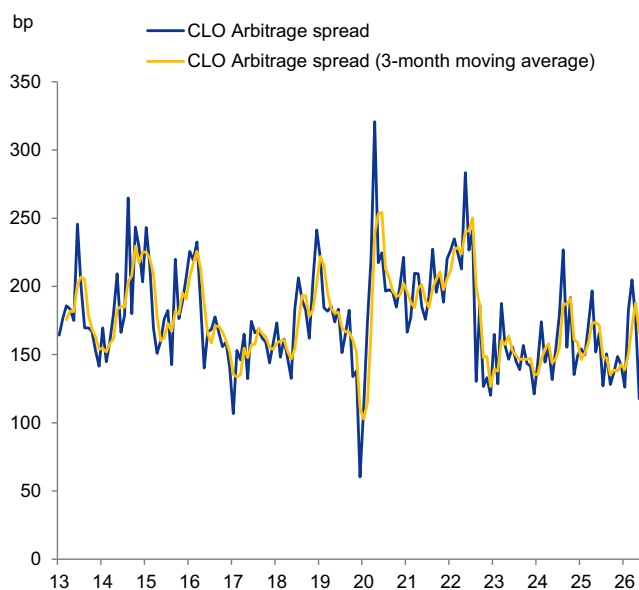
Exhibit 50: CLO spread percentile ranks

USD	Current spread (bp)	Percentile rank		
		3-year	5-year	10-year
AAA	104	8%	5%	2%
AA	143	3%	2%	1%
A	163	0%	0%	0%
BBB	284	32%	20%	14%
BB	762	62%	59%	64%

Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 52: CLO new issue arbitrage spread

New issue CLO arbitrage spread, i.e., the difference between new issue loan spreads and new issue CLO spreads (across the capital stack)



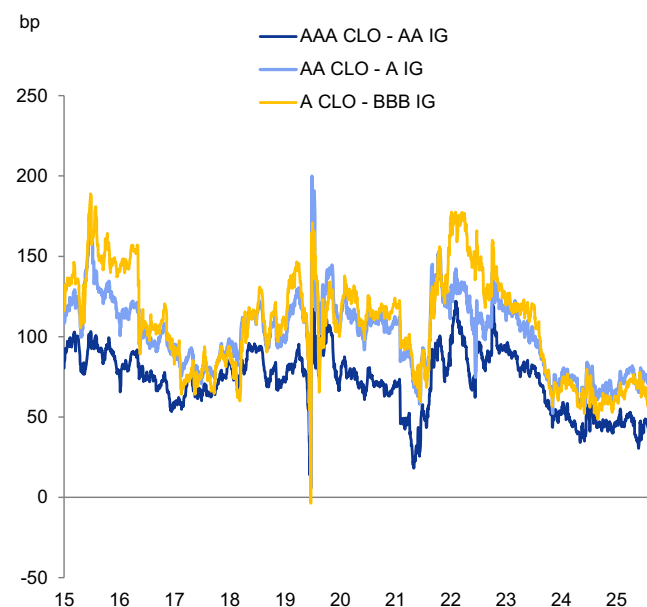
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 51: CLO equity vs. CLO BB yield

Source: Intex, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 53: CLO vs. USD IG spread differential

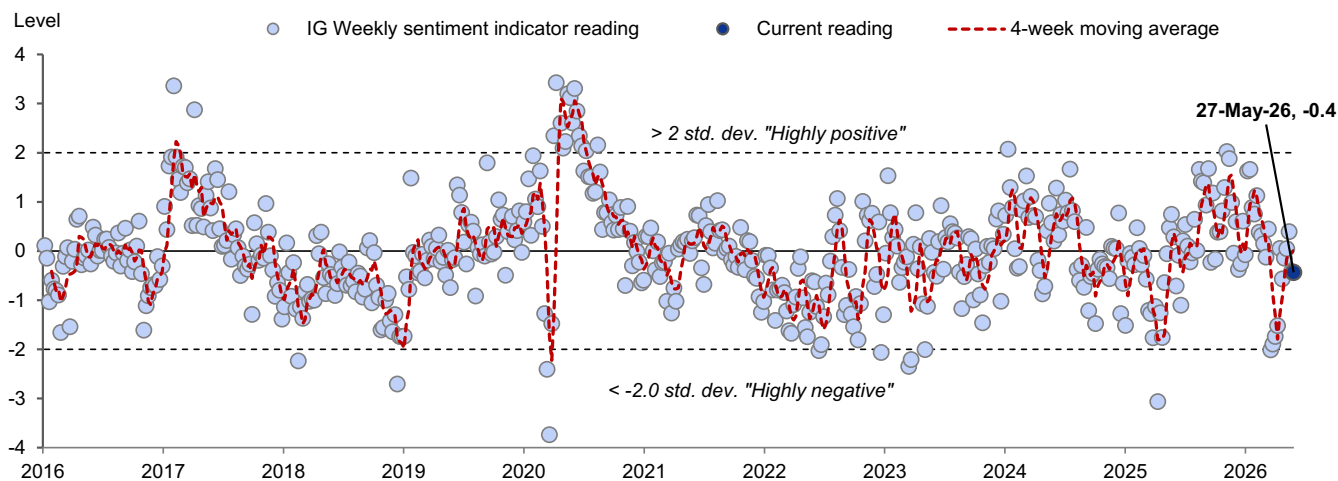
CLO by rating and USD IG bond by rating spread differential



Source: Bloomberg, Goldman Sachs Global Investment Research

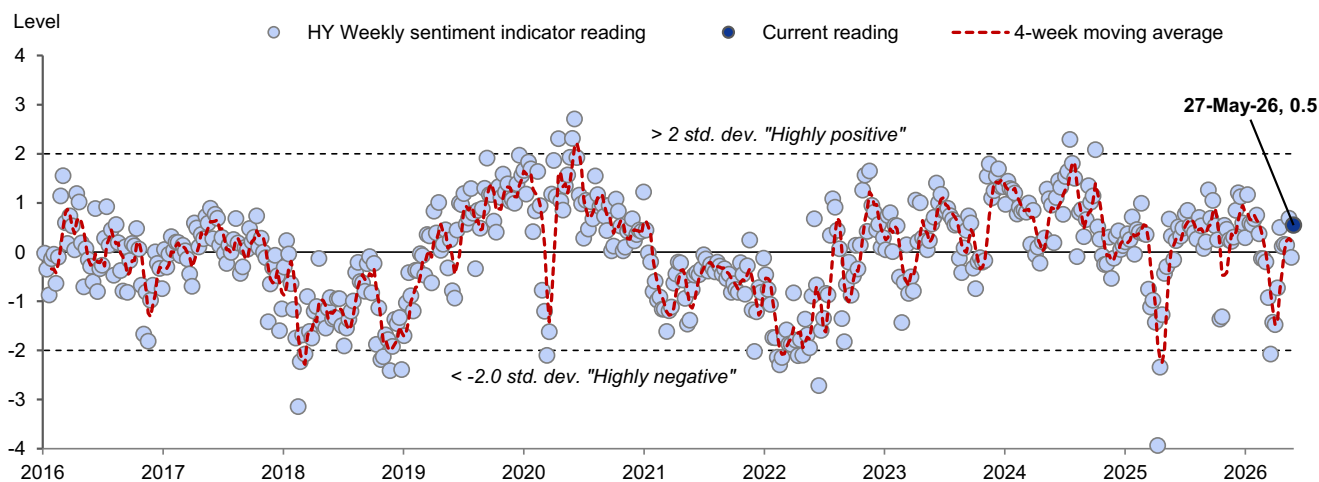
USD IG and HY Sentiment indicators

Exhibit 54: IG Weekly sentiment indicator



Source: See table below.

Exhibit 55: HY Weekly sentiment indicator



Source: See table below.

Exhibit 56: Variables included in the USD IG and HY weekly sentiment indicators

Measure	Frequency	Data Source
Synthetic index Positioning	Weekly	DTCC
Net Dealer Positioning	Weekly	FRBNY
Net Dealer Lift	Weekly	TRACE
Active US Credit Fund flows	Weekly	EPFR
Passive US Credit Fund flows	Weekly	EPFR
US Money Market Fund flows	Weekly	EPFR
Short interest on the largest credit ETFs	Semimonthly	GS FICC & Equities
Credit Mutual Fund cash positions	Monthly	ICI

Source: Goldman Sachs Global Investment Research

Microstructure dashboard: Top 20 most traded capital structures in the USD IG and HY markets

Exhibit 57: Top 20 most liquid issuers by weekly turnover in the USD IG market

Ticker	Issuer	Notional bonds outstanding (\$bn)	Notional volume traded on the week (\$mn)	Weekly turnover ratio (Wed. to Wed. %)	Median rating	Average age of bonds outstanding (Years)	Average duration of bonds outstanding (Years)
CRM	Salesforce Inc	33.5	466.0	1.4	A	1.5	7.3
UBER	Uber Technologies Inc	7.8	95.9	1.2	BBB+	1.9	6.5
ZTS	Zoetis Inc	6.8	81.8	1.2	BBB+	6.0	6.7
DVN	Devon Energy Corp	6.2	68.5	1.1	BBB+	9.6	9.1
BCRED	Blackstone Private Credit Fund	6.5	70.7	1.1	BBB-	1.2	4.1
APH	Amphenol Corp	11.8	125.3	1.1	A-	1.7	6.6
FANG	Diamondback Energy Inc	9.3	97.9	1.1	BBB	2.9	8.3
GOOGL	Alphabet Inc	51.0	527.9	1.0	AA	1.4	10.0
GIS	General Mills Inc	6.3	63.4	1.0	BBB	4.4	5.1
HUM	Humana Inc	10.9	109.7	1.0	BBB	3.9	7.2
SUZANO	Suzano Austria GmbH	7.8	78.1	1.0	BBB-	5.8	5.2
BBVASM	Banco Bilbao Vizcaya Argentaria SA	6.1	61.0	1.0	A-	1.7	4.3
ETN	Eaton Corp	13.9	138.1	1.0	A-	2.7	6.2
SJM	The JM Smucker Co	5.8	56.6	1.0	BBB	5.1	7.6
EXR	Extra Space Storage LP	5.3	51.9	1.0	BBB	2.5	4.6
EQIX	Equinix Inc	10.7	103.5	1.0	BBB+	3.9	5.8
KHC	Kraft Heinz Foods Co	13.7	130.2	0.9	BBB	8.5	9.5
DOW	The Dow Chemical Company	12.7	117.7	0.9	BBB-	7.5	9.4
SW	WRKCo Inc	7.4	67.0	0.9	BBB	3.3	6.1
AZO	AutoZone Inc	6.1	54.2	0.9	BBB	3.9	4.1

Source: TRACE, iBoxx, data compiled by Goldman Sachs Global Investment Research

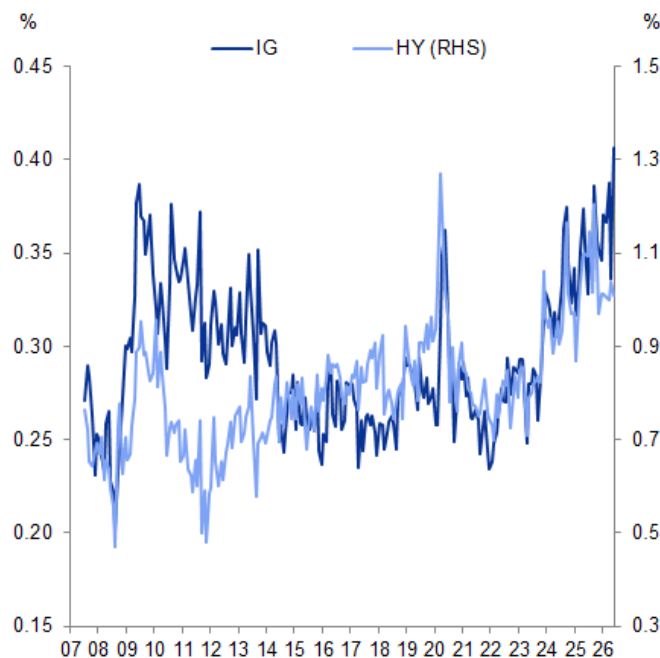
Exhibit 58: Top 20 most liquid issuers by weekly turnover in the USD HY market

Ticker	Issuer	Notional bonds outstanding (\$bn)	Notional volume traded on the week (\$mn)	Weekly turnover ratio (Wed. to Wed. %)	Median rating	Average age of bonds outstanding (Years)	Average duration of bonds outstanding (Years)
KIOXIA	Kioxia Holdings Corp	2.2	176.5	8.0	BBB-	0.8	3.3
CAR	Avis Budget Car Rental LLC / Avis Budget Finance Inc	2.9	60.8	2.1	B	3.0	2.4
AAP	Advance Auto Parts Inc	2.5	45.4	1.9	BB-	1.8	3.3
GSYCN	goeasy Ltd	2.4	42.6	1.8	B-	1.7	3.1
IHRT	iHeartCommunications Inc	2.1	30.6	1.5	CCC+	1.4	3.1
GT	The Goodyear Tire & Rubber Company	3.0	42.2	1.4	B+	3.8	4.0
MIK	Michaels Cos Inc/The	2.8	39.3	1.4	B-	0.2	5.2
JBLU	JetBlue Airways Corp / JetBlue Loyalty LP	2.0	28.5	1.4	B	1.7	4.1
APLD	APLD ComputeCo LLC	2.4	30.7	1.3	BB-	0.4	2.2
CRWV	CoreWeave Inc	6.5	83.3	1.3	B+	0.5	3.2
SEE	Sword Purchaser LLC	2.4	26.6	1.1	B+	0.1	4.0
FUN	Six Flags Entertainment Corp/Canada's Wonderland Co/Millen	3.1	31.2	1.0	CCC+	2.3	3.4
NAVI	Navient Corp	2.8	26.4	1.0	BB-	3.3	2.9
GTN	Gray Media Inc	4.8	46.0	1.0	CCC	2.8	3.6
CC	The Chemours Company	2.5	23.9	0.9	B+	2.8	3.2
MERIDI	Meridian Arc Holdco LLC	5.7	53.9	0.9	BB	0.0	3.5
NCLH	NCL Corp Ltd	5.0	45.1	0.9	B-	1.8	4.1
MLTPLN	MPH Acquisition Holdings LLC	2.1	19.1	0.9	CCC+	1.2	3.9
SABHLD	Sabre GLBL Inc	3.2	28.6	0.9	B-	0.7	2.7
VSTJET	VistaJet Malta Finance PLC / XO Management Holding Inc	2.0	17.5	0.9	B-	3.9	1.9

Source: TRACE, iBoxx, data compiled by Goldman Sachs Global Investment Research

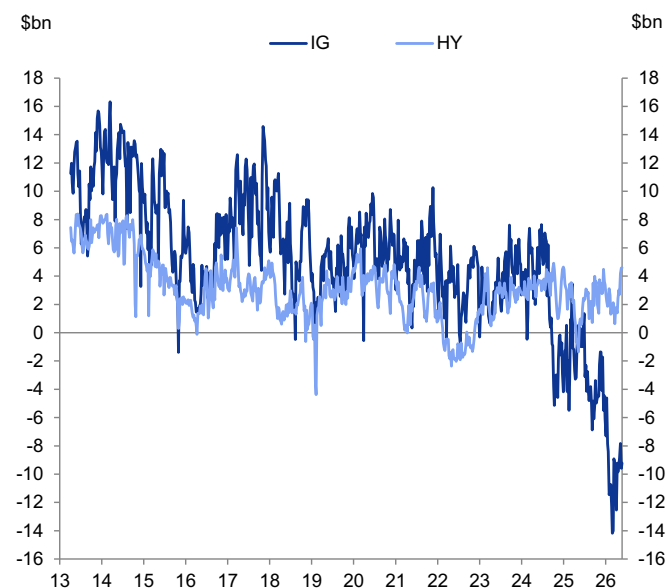
Microstructure dashboard: Liquidity conditions in the USD secondary bond market

Exhibit 59: Seasonally adjusted daily average turnover (trading volume/amount outstanding)



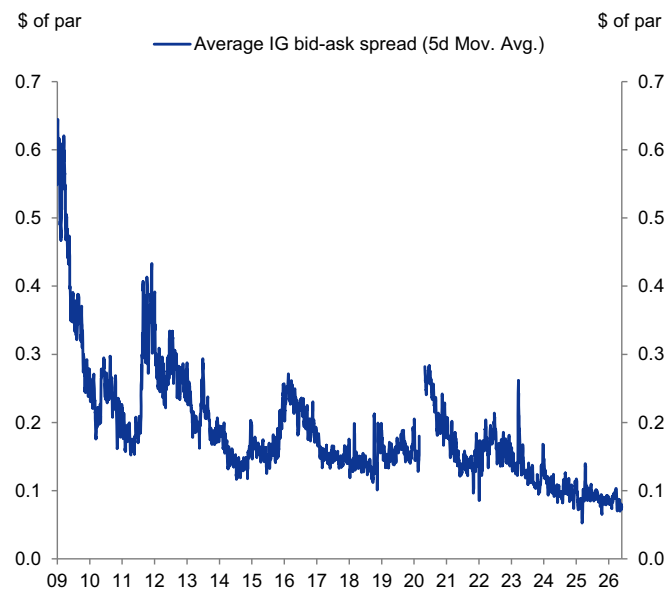
Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 60: Primary dealer net positioning in USD corporate bonds



Source: FRBNY, Goldman Sachs Global Investment Research

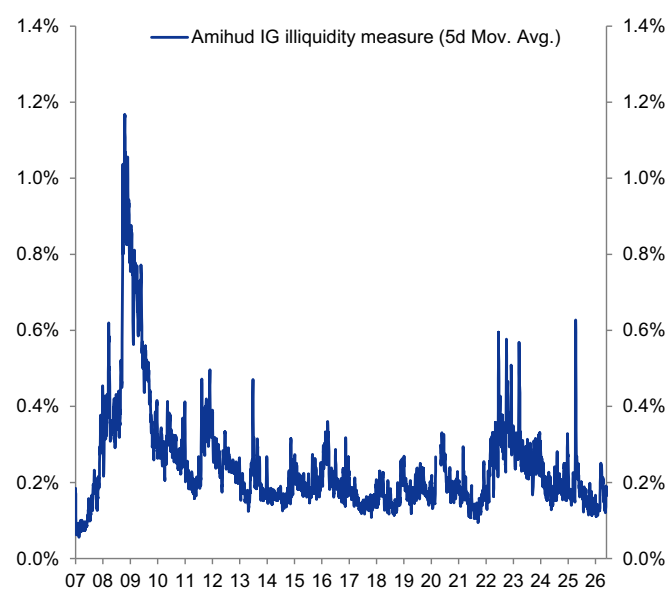
Exhibit 61: Average IG bid-ask spread 5-day moving average



Note: March and April 2020 are omitted.

Source: Trace, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 62: Amihud IG illiquidity measure 5-day moving average

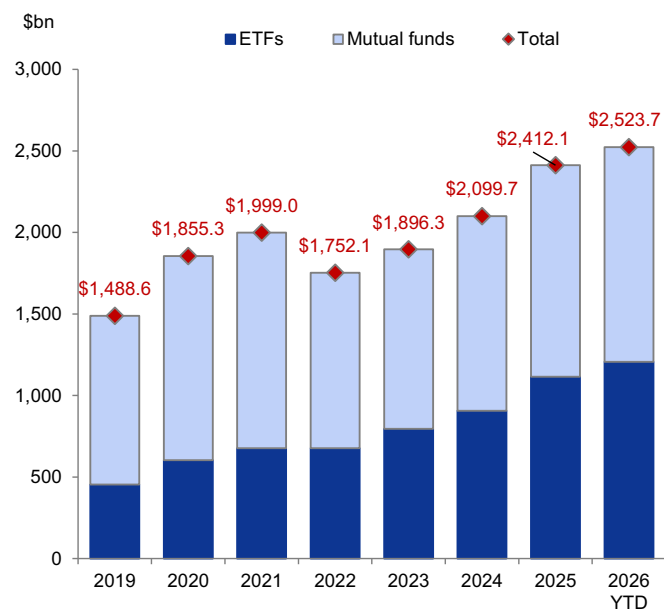


Note: March and April 2020 are omitted.

Source: TRACE, Bloomberg, Goldman Sachs Global Investment Research

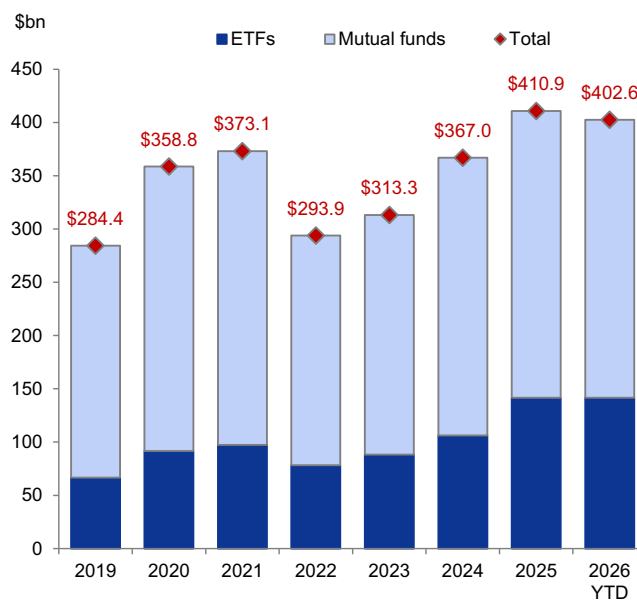
Flow dashboard: Mutual fund and ETF AUM

Exhibit 63: AUM in USD IG-focused Mutual Funds and ETFs



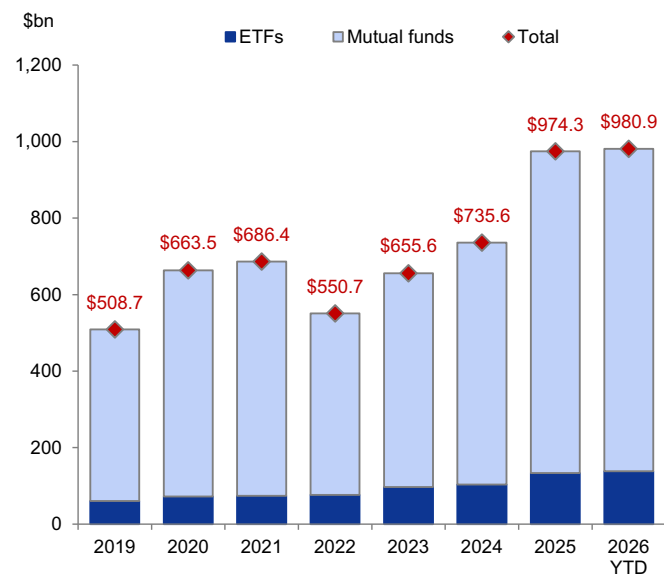
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 64: AUM in USD HY-focused Mutual Funds and ETFs



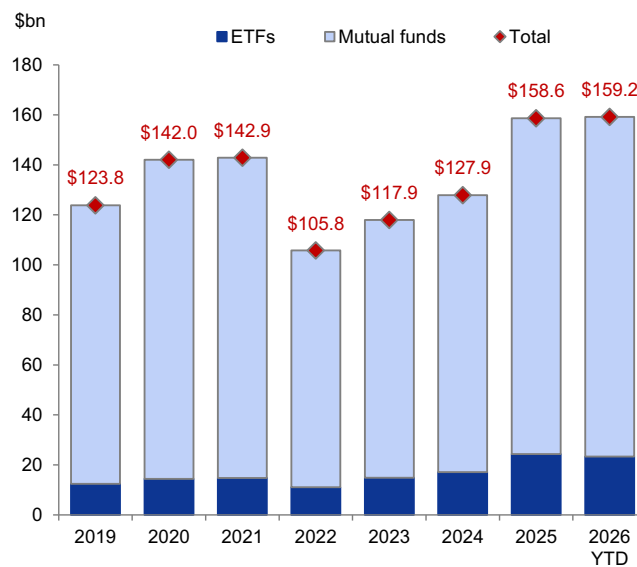
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 65: AUM in EUR IG-focused Mutual Funds and ETFs



Source: EPFR, Goldman Sachs Global Investment Research

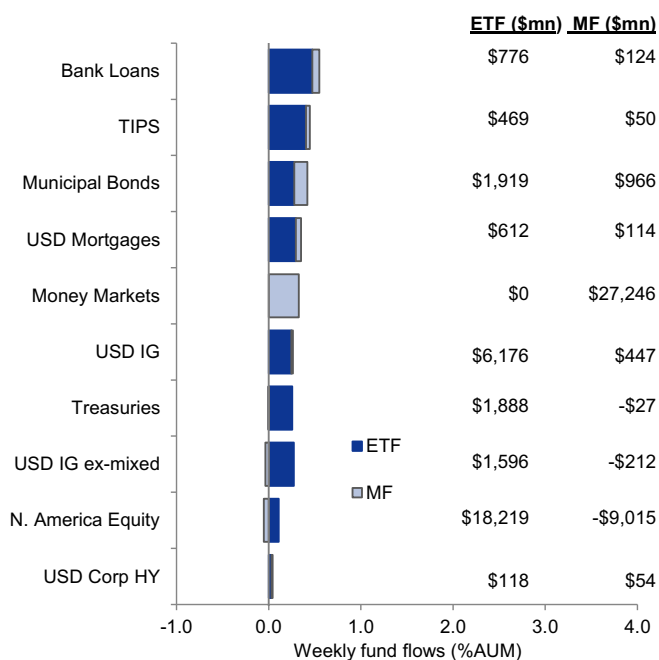
Exhibit 66: AUM in EUR HY-focused Mutual Funds and ETFs



Source: EPFR, Goldman Sachs Global Investment Research

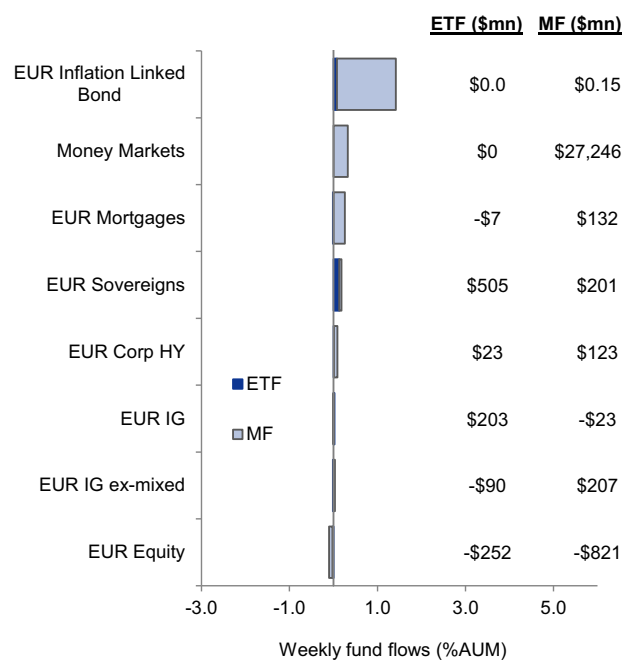
Flow dashboard: Mutual fund and ETF flows

Exhibit 67: Weekly mutual fund flows into US fixed income and equity markets



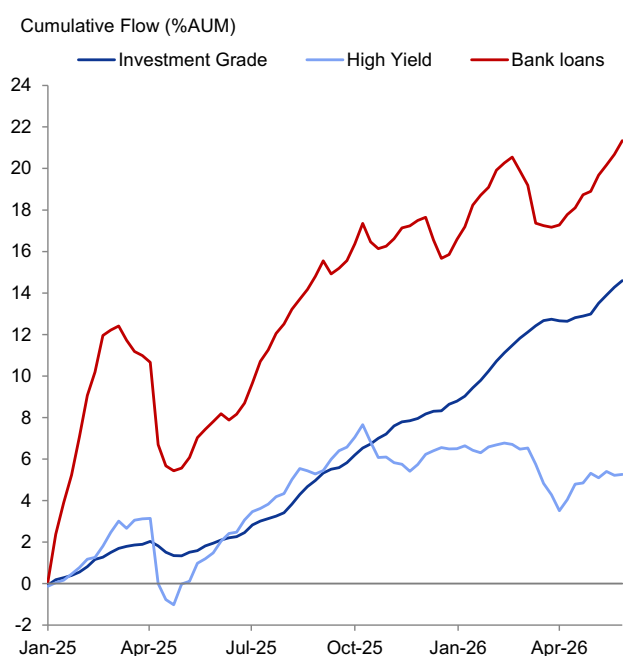
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 68: Weekly mutual fund flows into European fixed income and equity markets



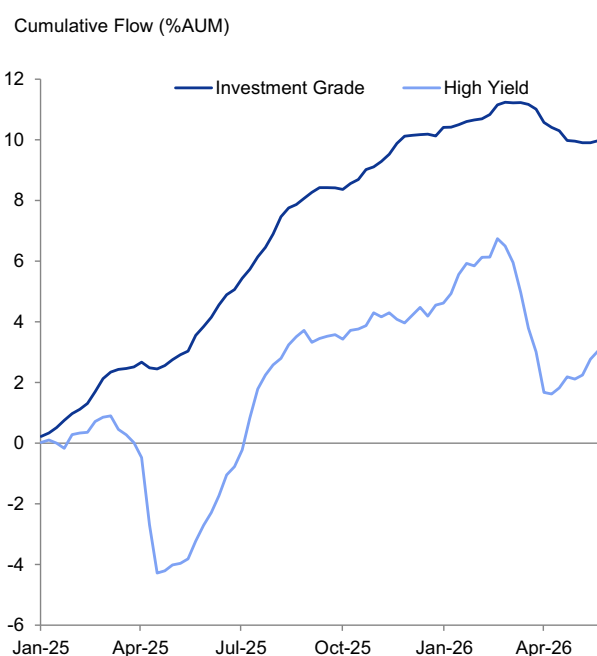
Source: EPFR, Goldman Sachs Global Investment Research

Exhibit 69: Year-to-date cumulative flows as a percentage of beginning of the year AUM for US credit funds



Source: EPFR, Goldman Sachs Global Investment Research

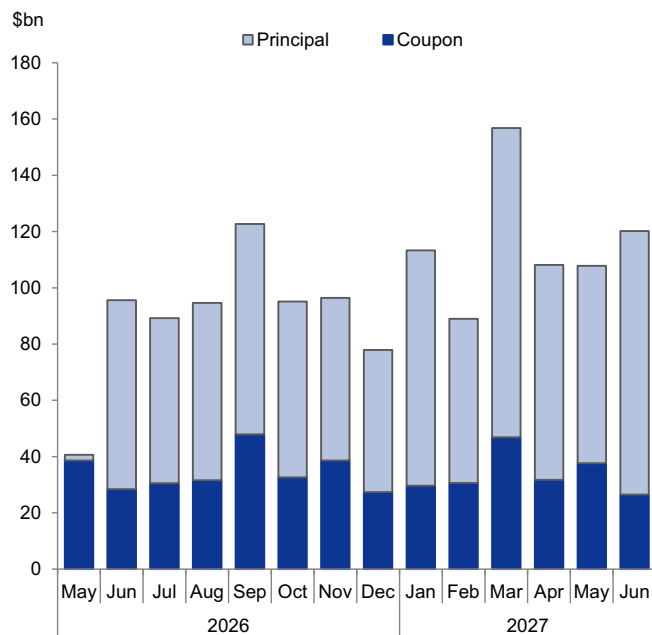
Exhibit 70: Year-to-date cumulative flows as a percentage of beginning of the year AUM for European credit funds



Source: EPFR, Goldman Sachs Global Investment Research

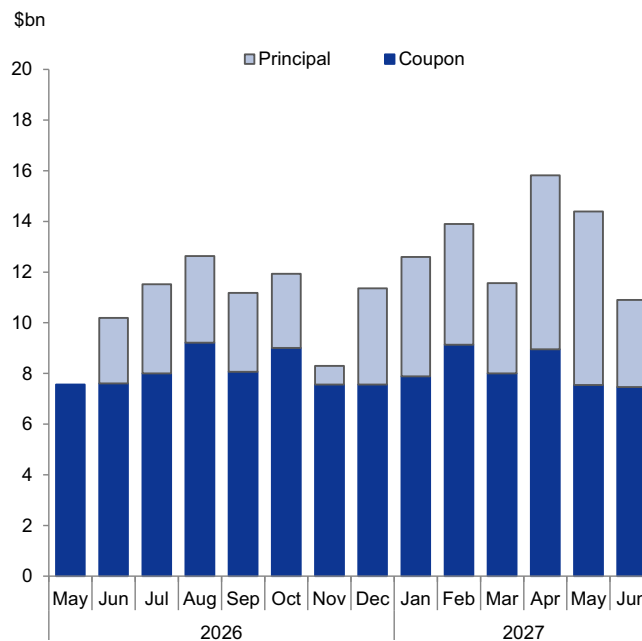
Flow dashboard: Upcoming coupon and principal payments

Exhibit 71: Our estimates of upcoming coupon and principal payments in the USD IG market



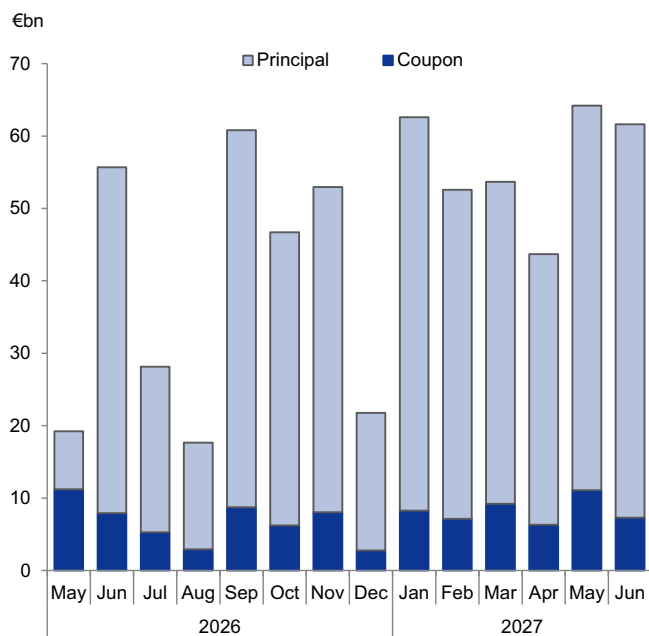
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 72: Our estimates of upcoming coupon and principal payments in the USD HY market



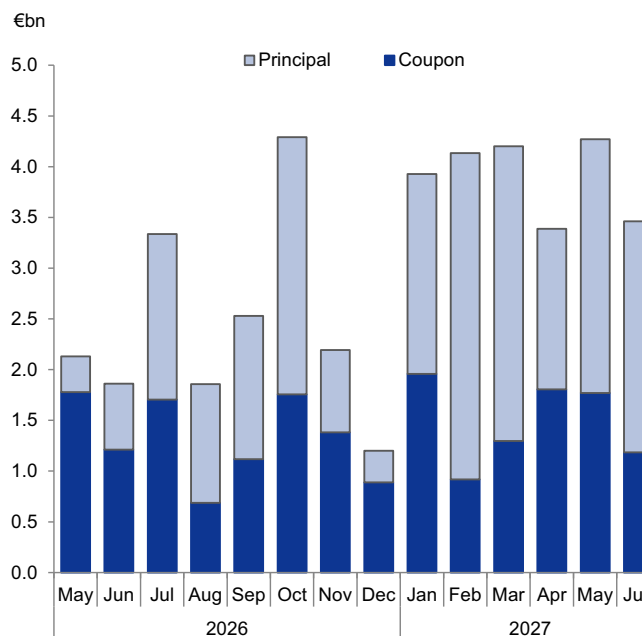
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 73: Our estimates of upcoming coupon and principal payments in the EUR IG market



Source: Bloomberg, Goldman Sachs Global Investment Research

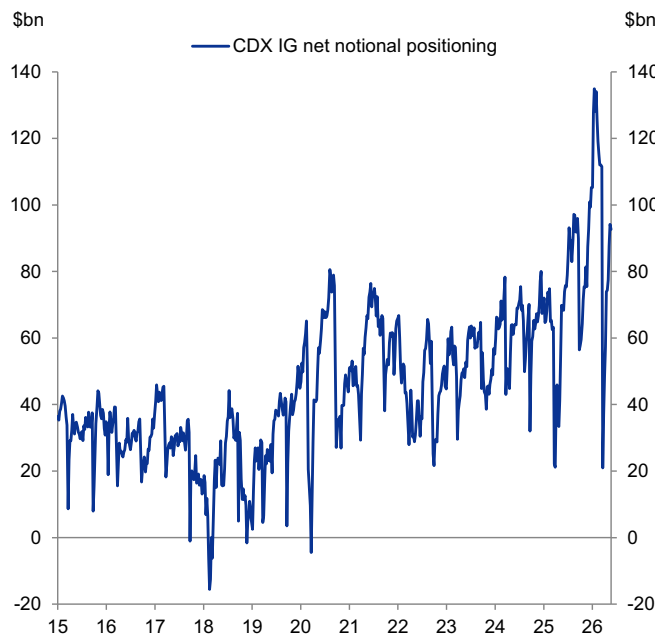
Exhibit 74: Our estimates of upcoming coupon and principal payments in the EUR HY market



Source: Bloomberg, Goldman Sachs Global Investment Research

Flow dashboard: Positioning in macro indices

Exhibit 75: Net notional positioning in the CDX IG 5-year on-the-run contract



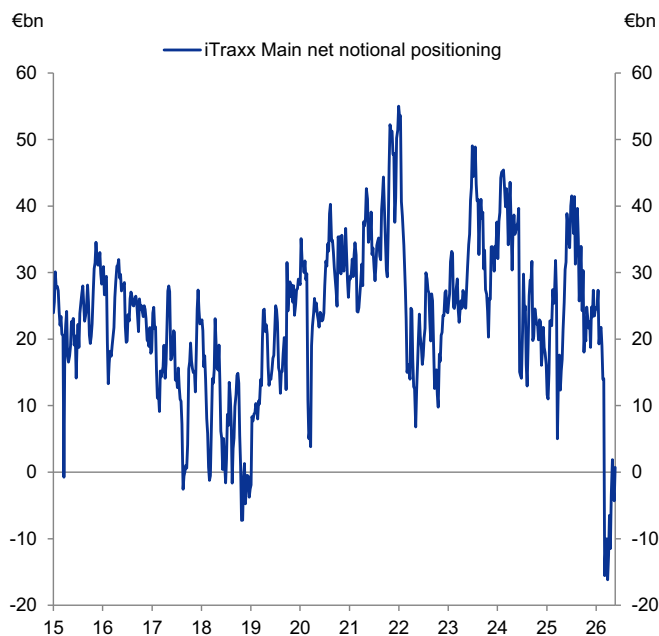
Source: DTCC, Goldman Sachs Global Investment Research

Exhibit 76: Net notional positioning in the CDX HY 5-year on-the-run contract



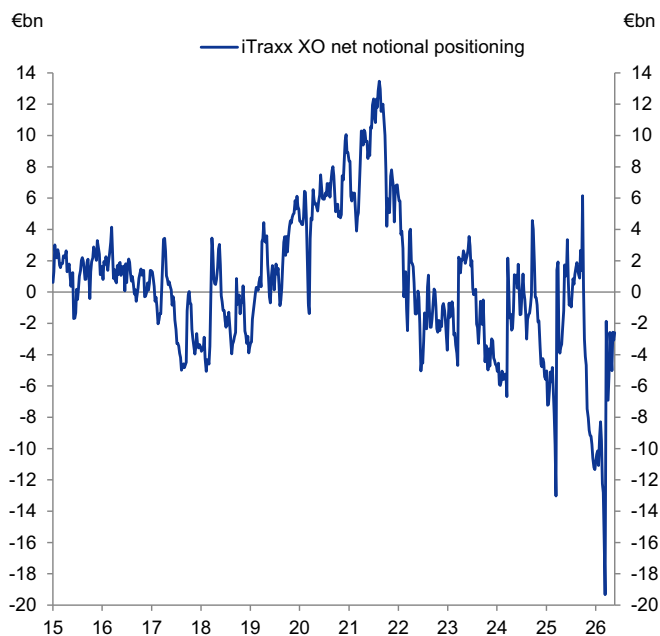
Source: DTCC, Goldman Sachs Global Investment Research

Exhibit 77: Net notional positioning in the iTraxx Main 5-year on-the-run contract



Source: DTCC, Goldman Sachs Global Investment Research

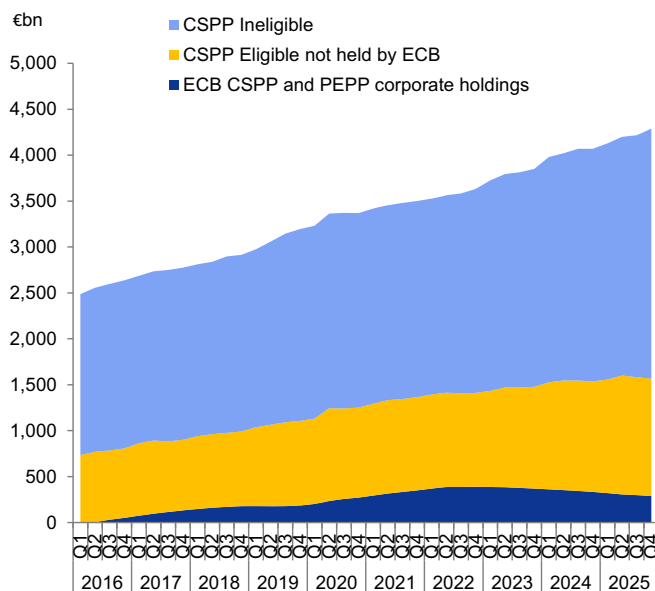
Exhibit 78: Net notional positioning in iTraxx over 5-year on-the-run contract



Source: DTCC, Goldman Sachs Global Investment Research

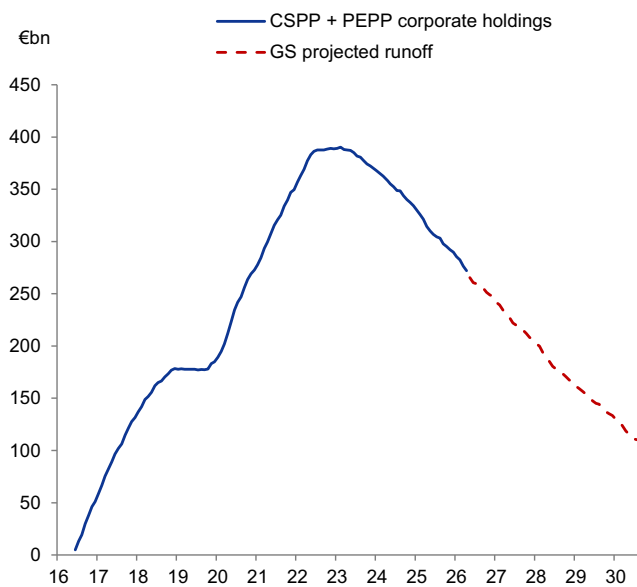
Flow dashboard: ECB QT

Exhibit 79: Ownership share of the ECB in the EUR IG market



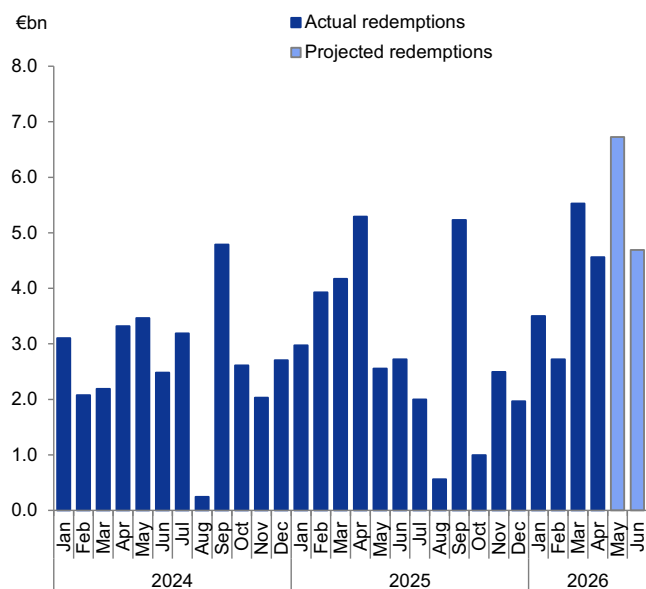
Source: ECB, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 80: Run-off projections for the ECB corporate bond holdings



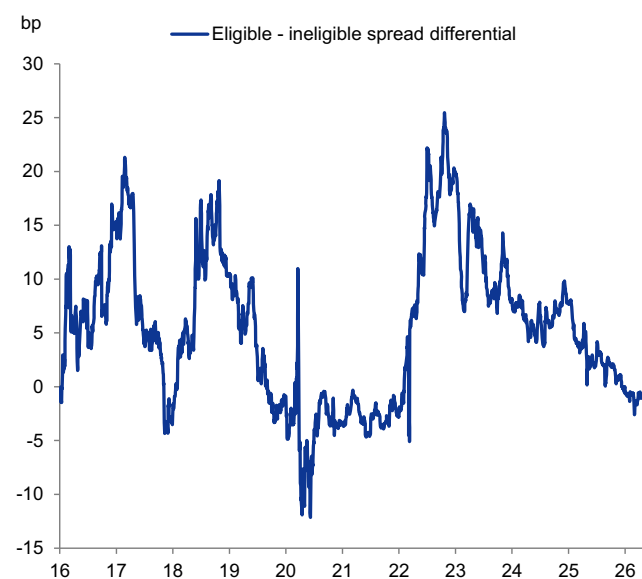
Source: ECB, Bloomberg, Goldman Sachs Global Investment Research

Exhibit 81: CSPP monthly run-off pace



Source: ECB, Goldman Sachs Global Investment Research

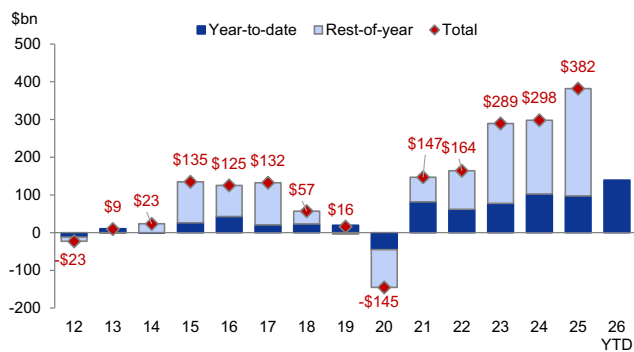
Exhibit 82: Performance of CSPP-eligible bonds vs. non-eligible bonds



Source: iBoxx, Goldman Sachs Global Investment Research

Foreign demand and hedging cost

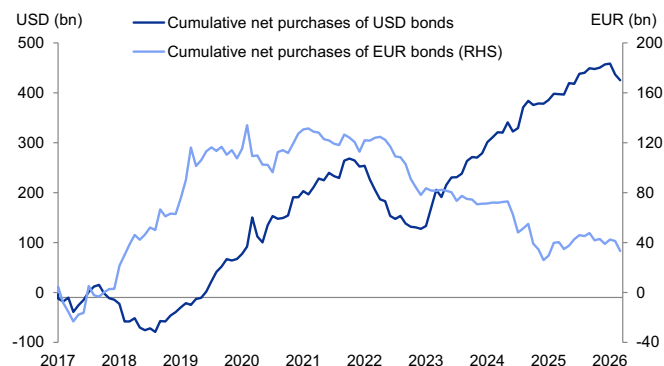
Exhibit 83: Net purchases of USD corporate bonds by foreign investors



Note: As of March 2026.

Source: Department of Treasury, Goldman Sachs Global Investment Research

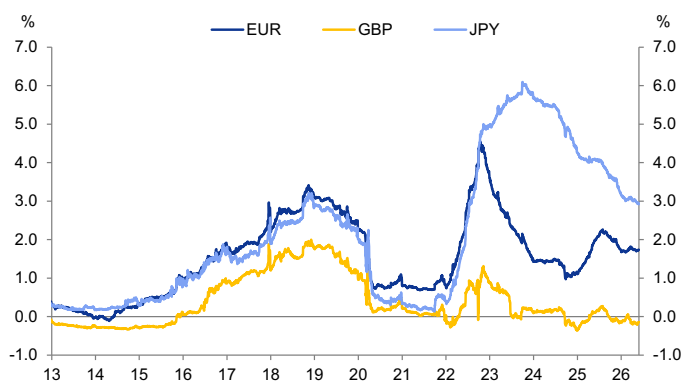
Exhibit 84: Cumulative net foreign purchases of USD and EUR bonds by Japanese investors



Note: As of March 2026.

Source: Bank of Japan, Haver Analytics, Goldman Sachs Global Investment Research

Exhibit 85: Annualized 3-month rolling cost of USD hedging against EUR, GBP and JPY

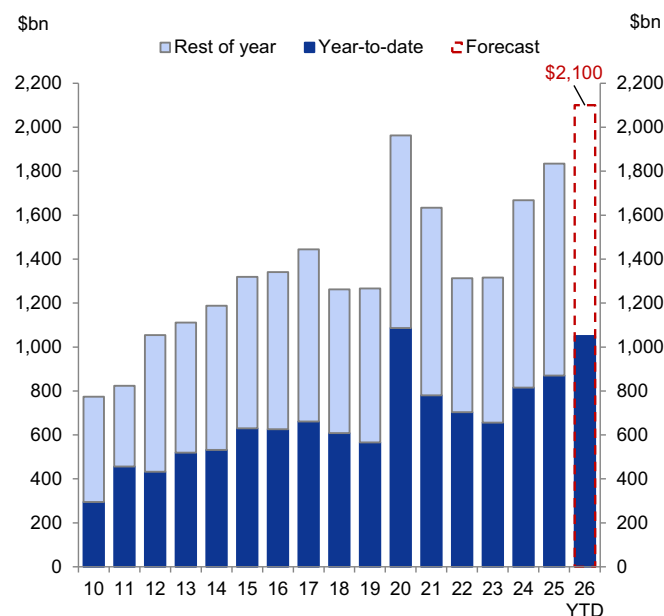


Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Primary market dashboard: USD gross issuance

Exhibit 86: USD IG gross issuance

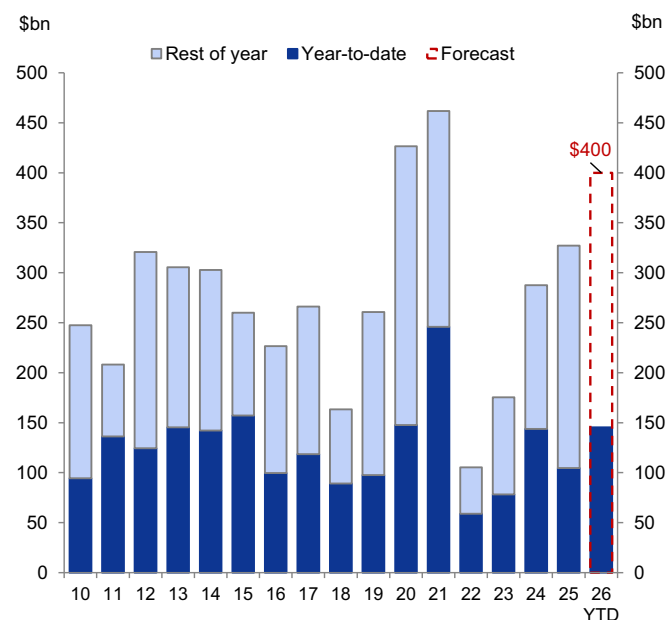
Historical investment grade issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 87: USD HY gross issuance

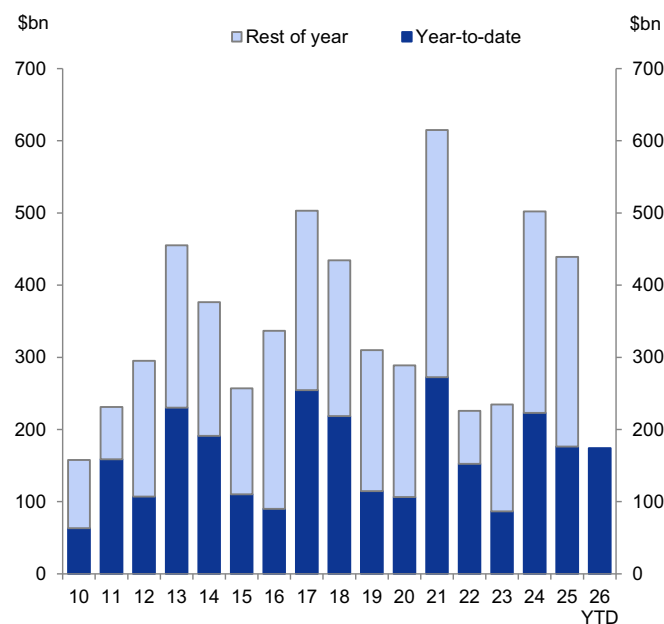
Historical USD high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 88: USD institutional leveraged loan gross issuance

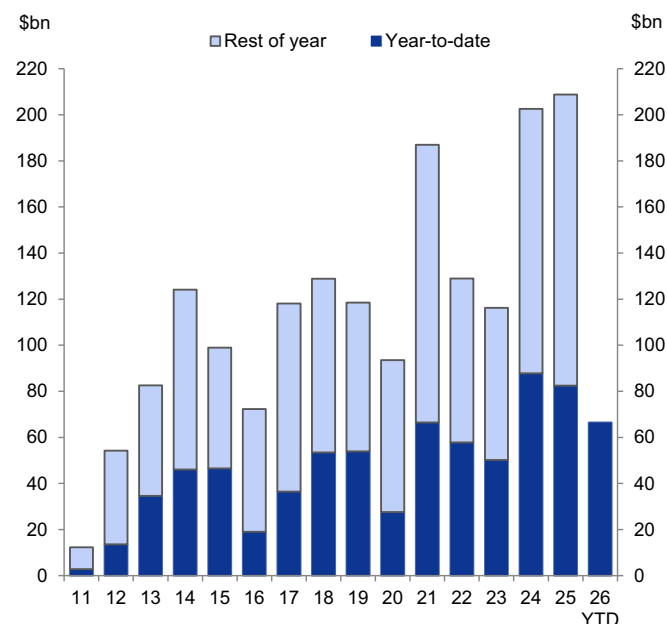
Historical leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 89: USD CLO creation

Historical USD CLO creation: Same period year to date vs. rest of the year

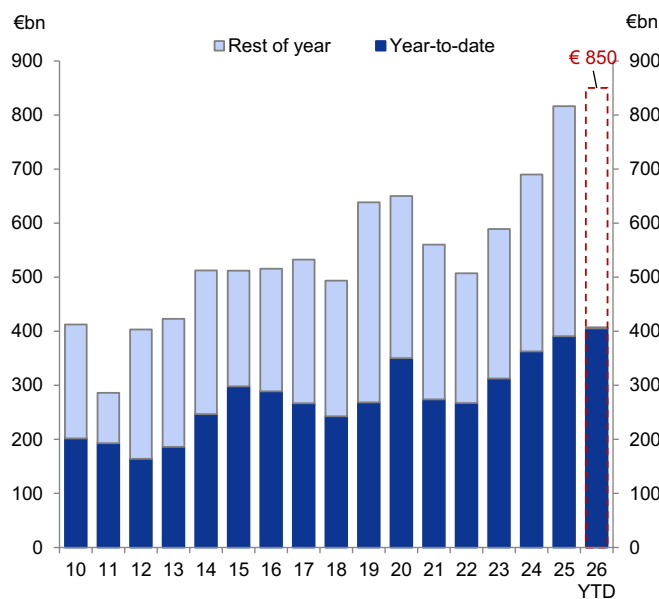


Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Primary market dashboard: EUR gross issuance

Exhibit 90: EUR IG gross issuance

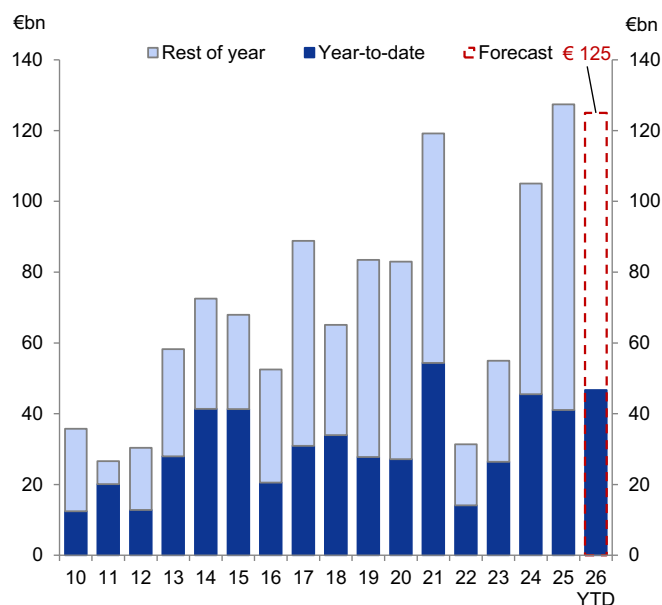
Historical EUR investment grade issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 91: EUR HY gross issuance

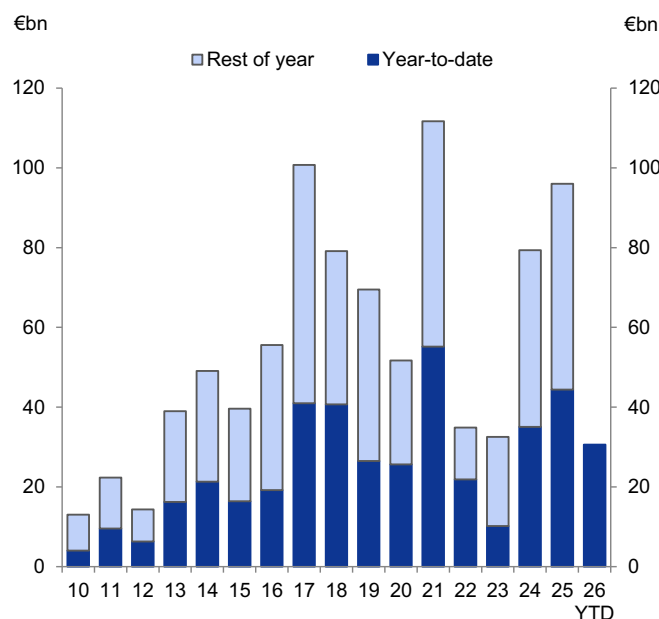
Historical EUR high yield issuance: Same period year to date vs. rest of the year



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 92: EUR Leveraged loan gross issuance

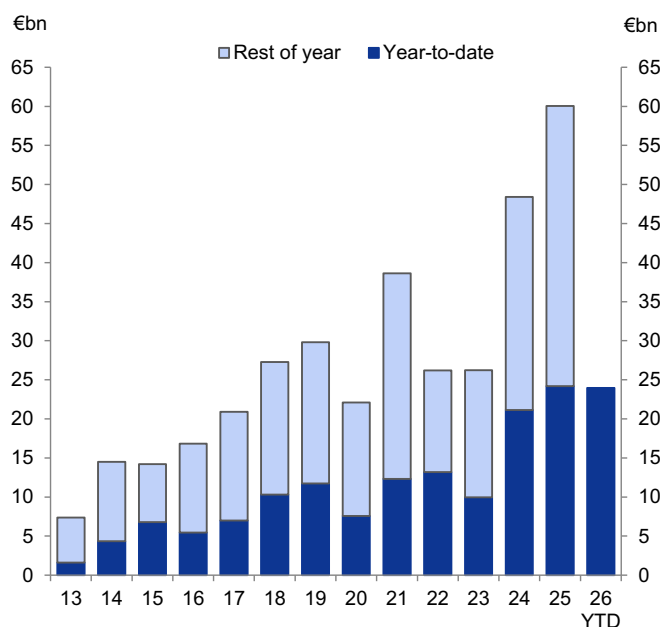
Historical EUR leveraged loan issuance: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 93: EUR CLO creation

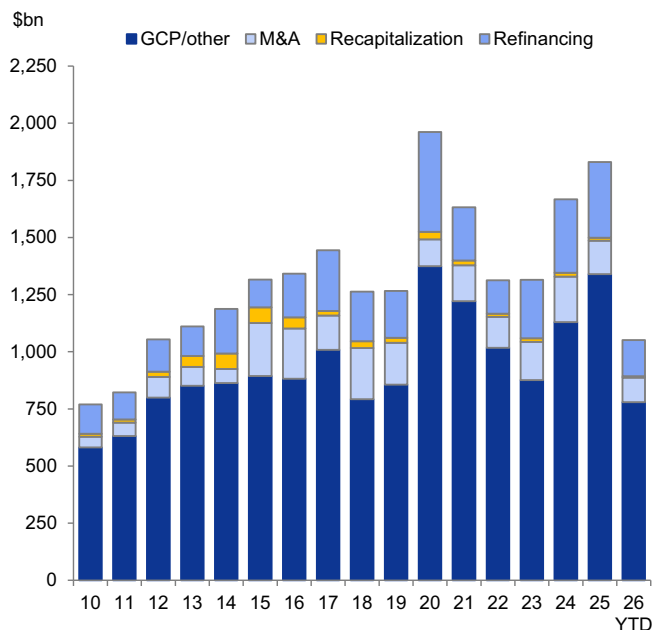
Historical EUR CLO creation broken: Same period year to date vs. rest of the year



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

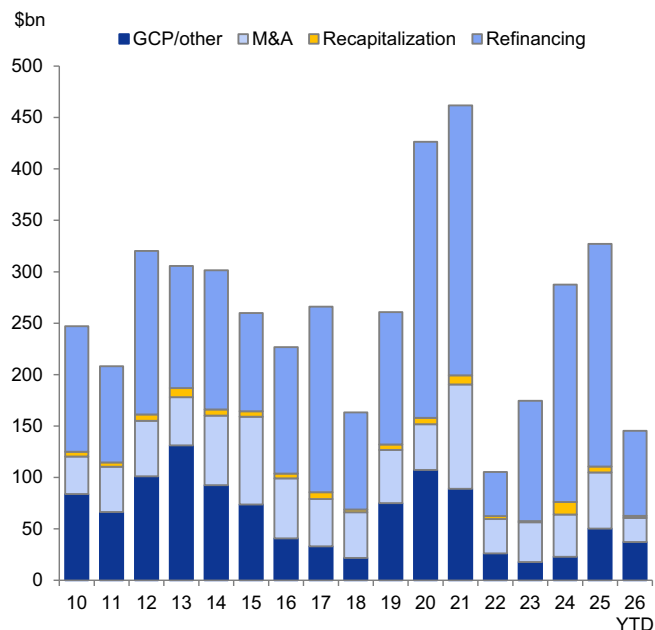
Primary market dashboard: Use of proceeds

Exhibit 94: Annual USD IG issuance volumes by use of proceeds



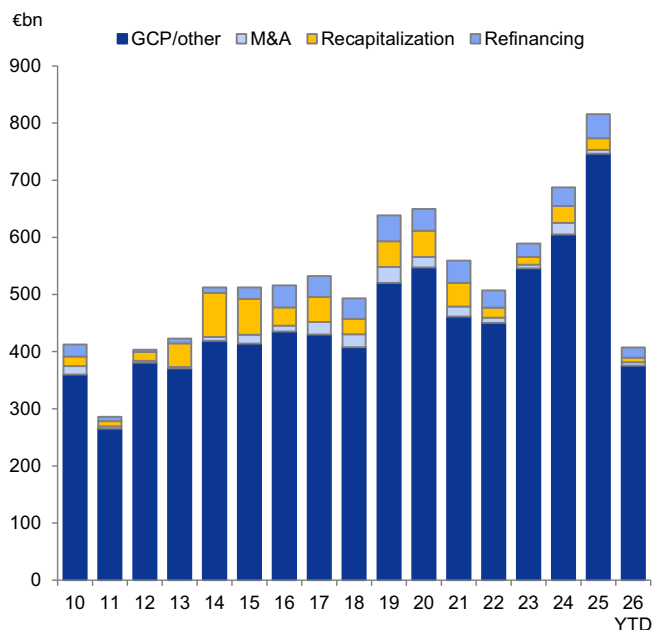
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 95: USD HY issuance volumes by use of proceeds



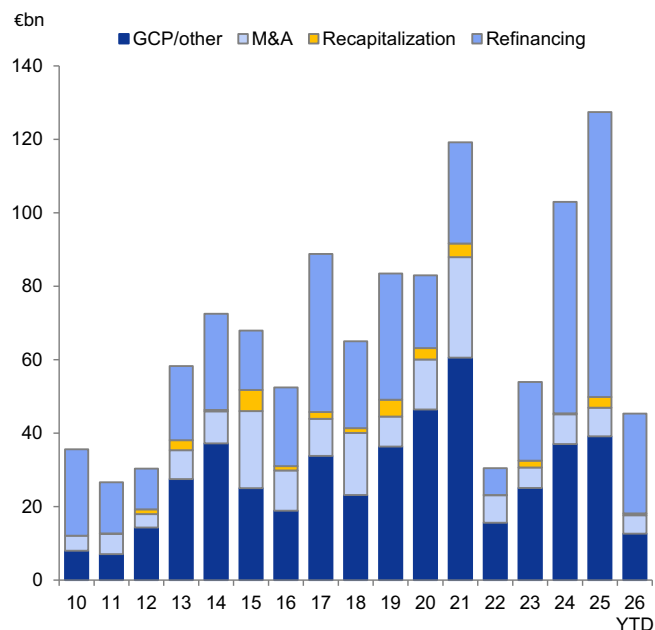
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 96: EUR IG issuance volumes by use of proceeds



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 97: EUR HY issuance volumes by use of proceeds

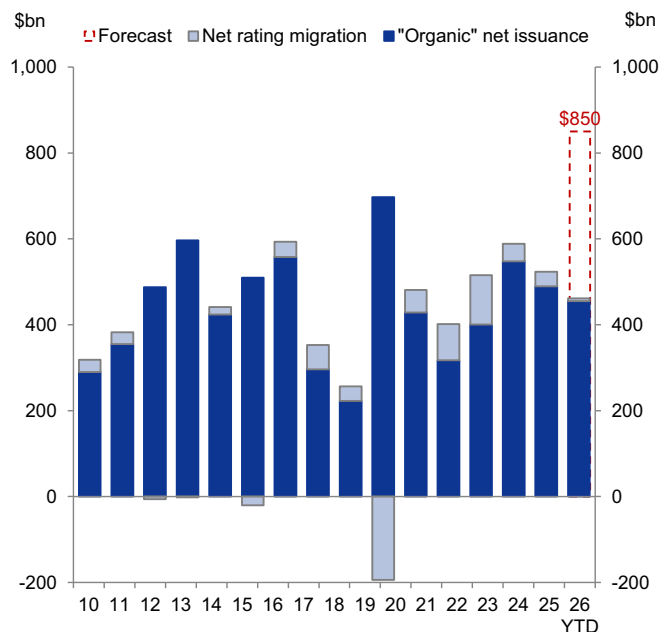


Source: Dealogic, Goldman Sachs Global Investment Research

Primary market dashboard: Net issuance

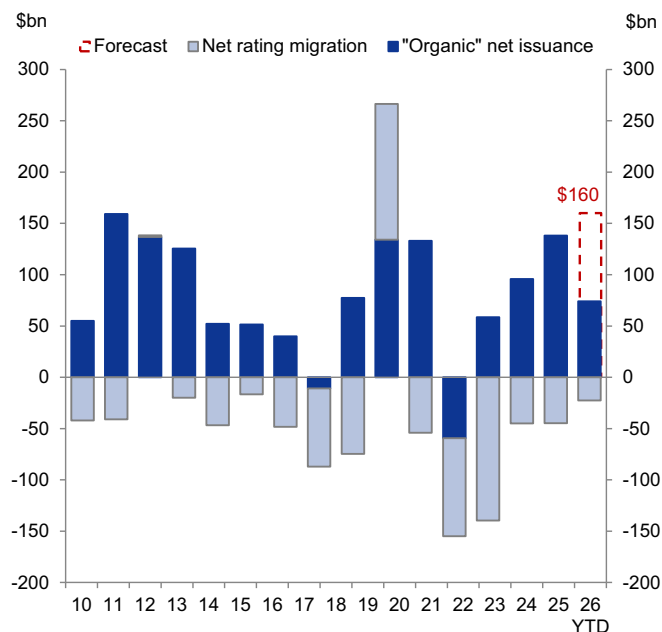
Note: Decomposition of the change in par value outstanding into net ratings migration vs. "organic" net issuance (gross issuance minus redemptions/matured bonds)

Exhibit 98: Annual USD IG net issuance



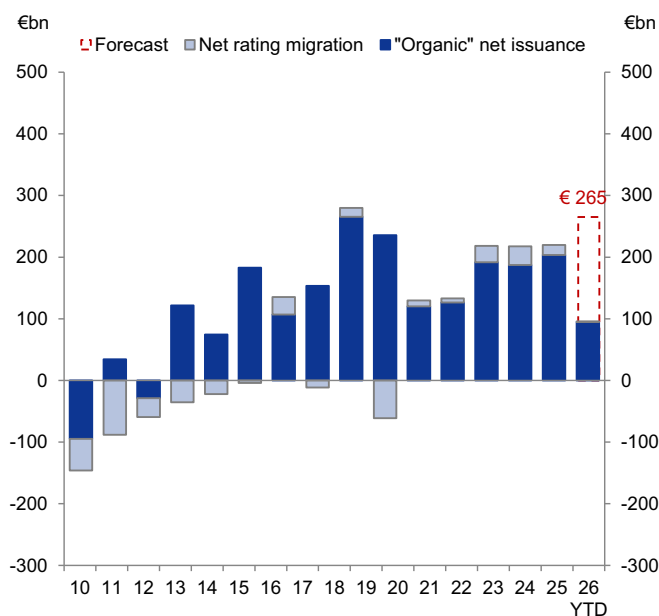
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 99: Annual USD HY net issuance



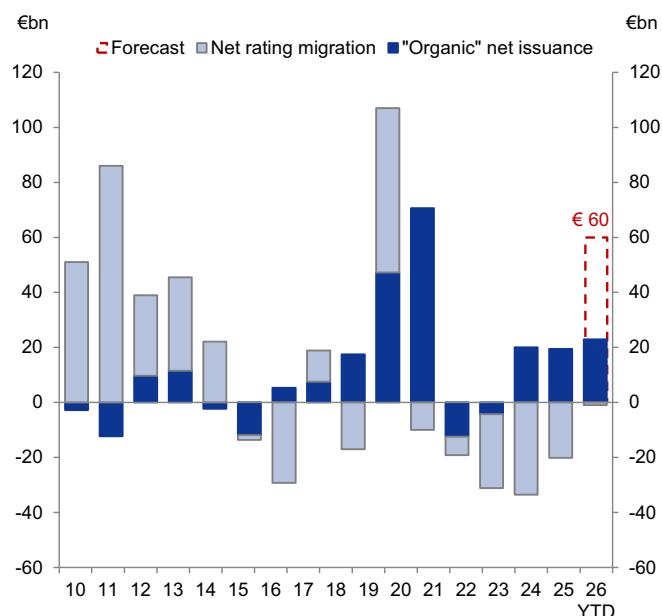
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 100: Annual EUR IG net issuance



Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 101: Annual EUR HY net issuance

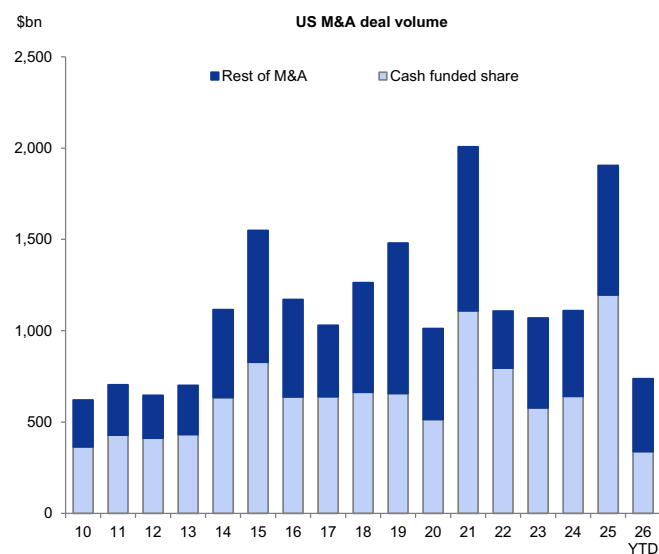


Source: Bloomberg, Goldman Sachs Global Investment Research

M&A volume

Exhibit 102: Annual US M&A deal volume

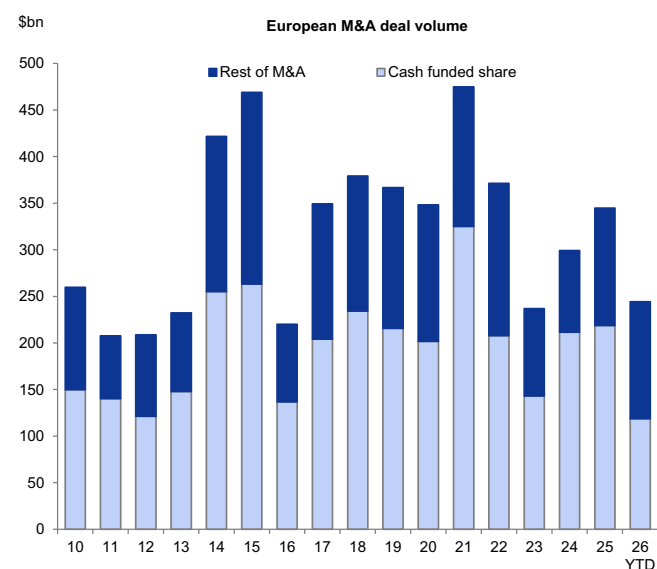
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 103: Annual European M&A deal volume

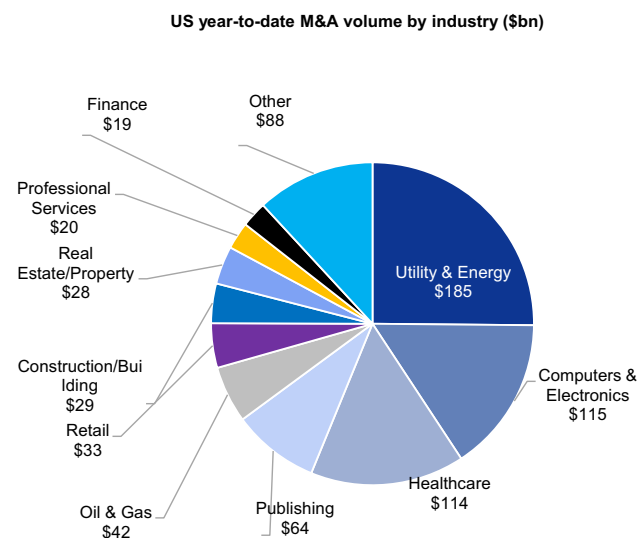
Includes strategic and sponsored deals



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 104: Year-to-date US M&A volume by sector

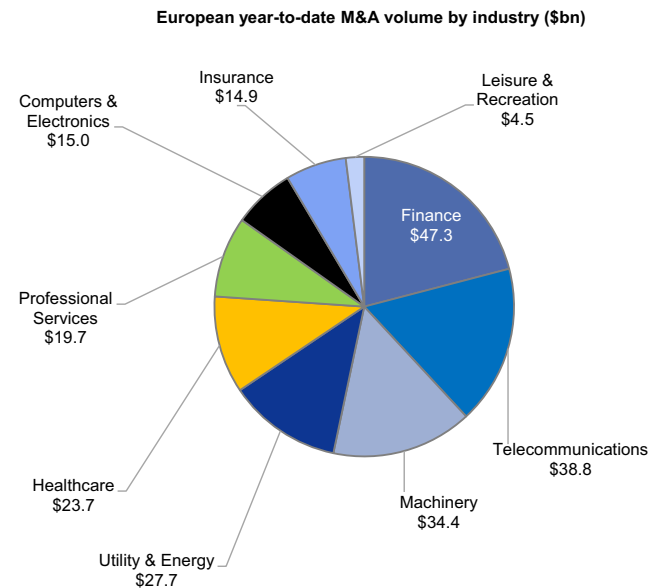
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 105: Year-to-date European M&A volume by sector

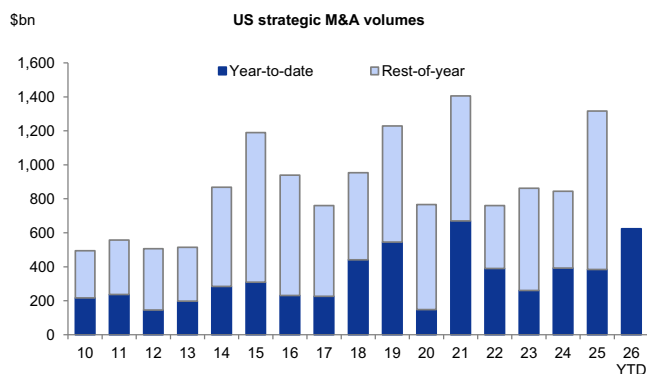
Includes strategic and sponsored deals



Source: Dealogic, Goldman Sachs Global Investment Research

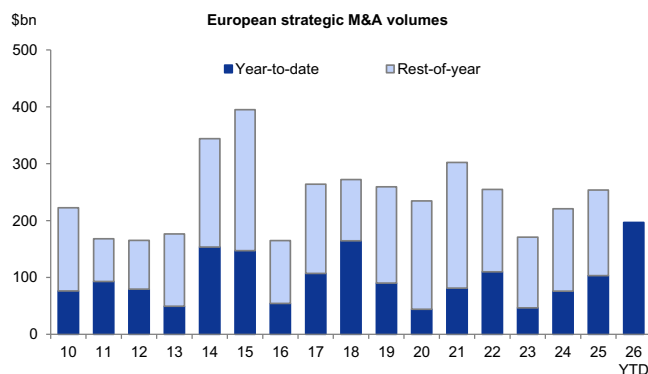
M&A volume (Strategic deals only)

Exhibit 106: Annual US strategic M&A volumes



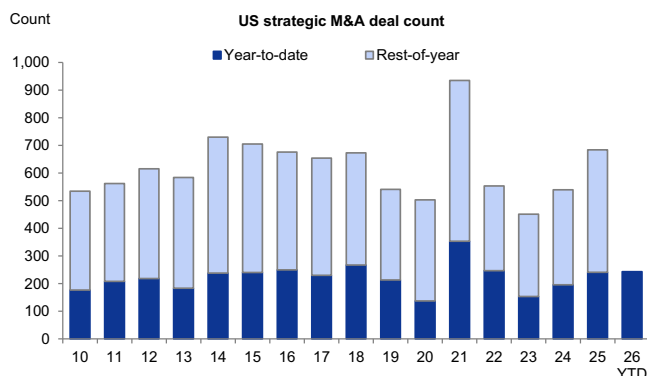
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 107: Annual European strategic M&A volumes



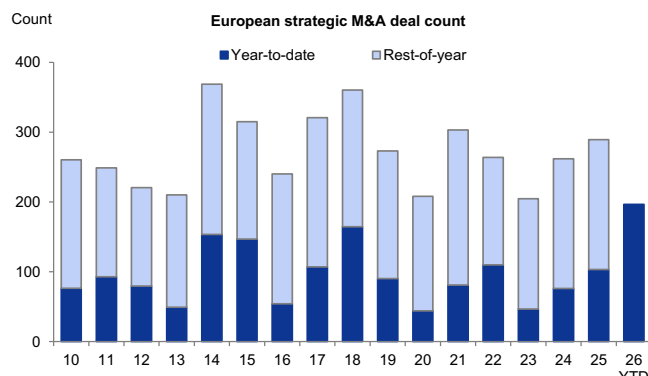
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 108: Annual US strategic M&A deal count



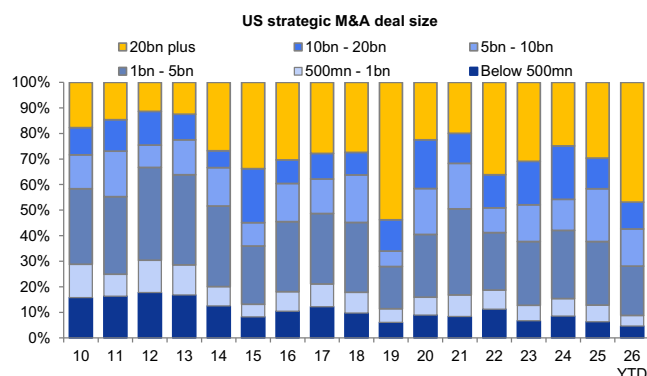
Source: Dealogic, Goldman Sachs Global Investment Research

Exhibit 109: Annual European strategic M&A deal count



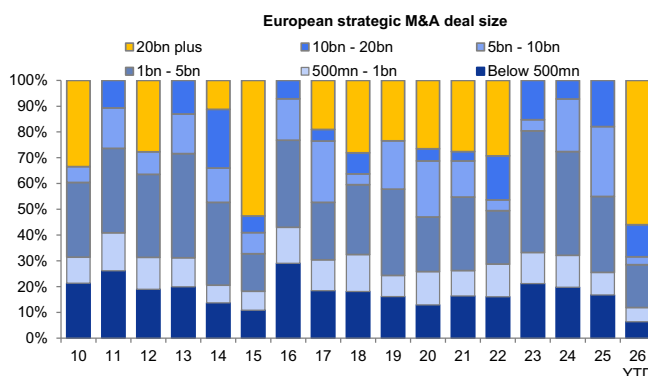
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 110: Annual US strategic M&A deal size



Source: Dealogic, Goldman Sachs Global Investment Research

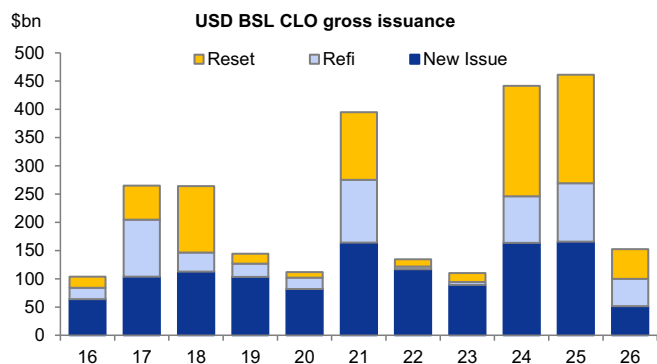
Exhibit 111: Annual European strategic M&A deal size



Source: Dealogic, Goldman Sachs Global Investment Research

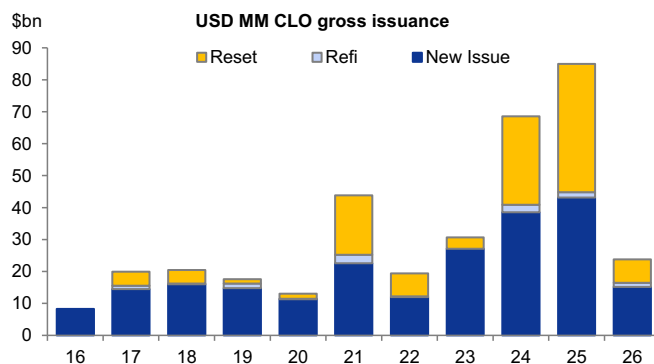
Primary market dashboard: CLO creation and refinancing

Exhibit 112: Annual USD Broadly syndicated loan CLO creation



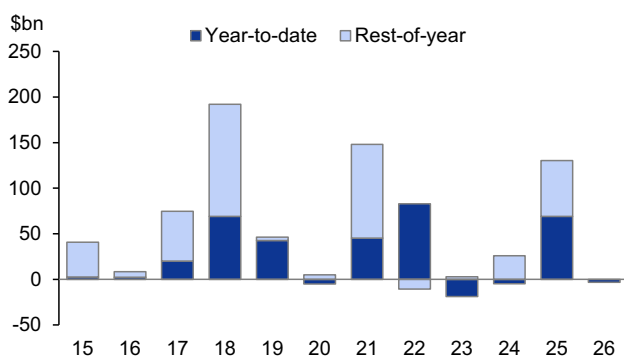
Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 113: Annual USD Middle market CLO creation



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

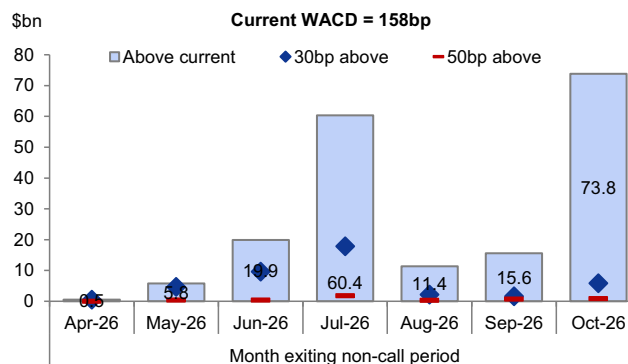
Exhibit 114: Annual USD leveraged loan net issuance



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 115: USD CLO refinance monitor

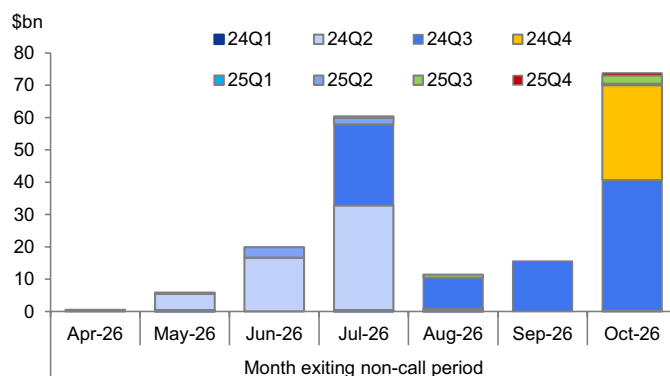
Upcoming notional above prevailing WACD



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 116: USD CLO refinance monitor

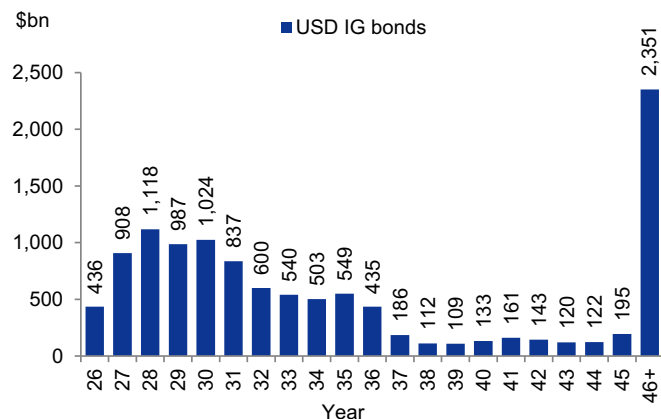
Notional above prevailing WACD by origination quarter



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

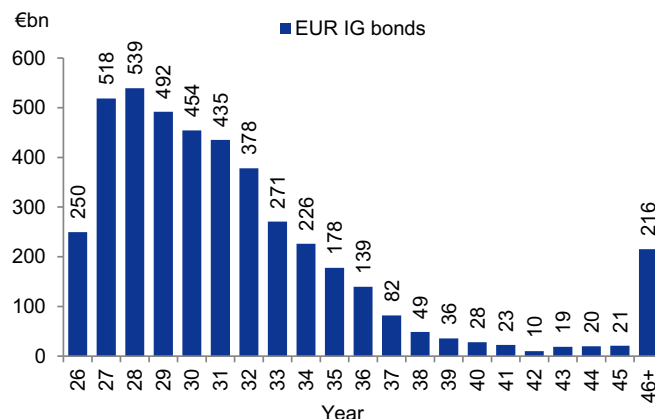
Primary market dashboard: Maturity walls

Exhibit 117: USD IG bond market maturity wall



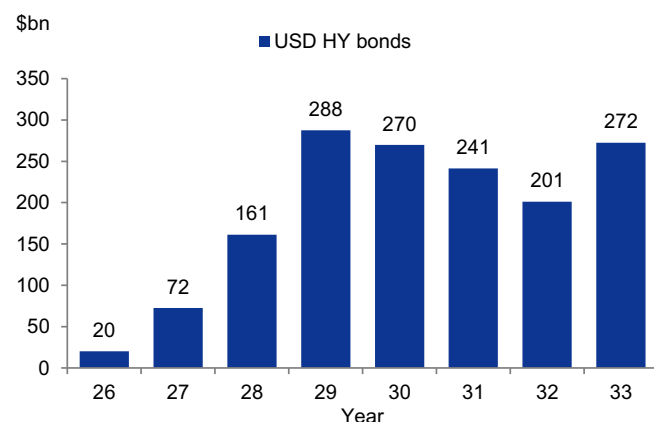
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 118: EUR IG bond market maturity wall



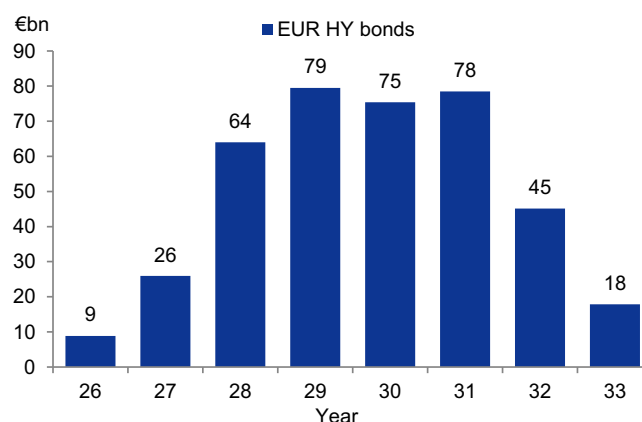
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 119: USD HY bond market maturity wall



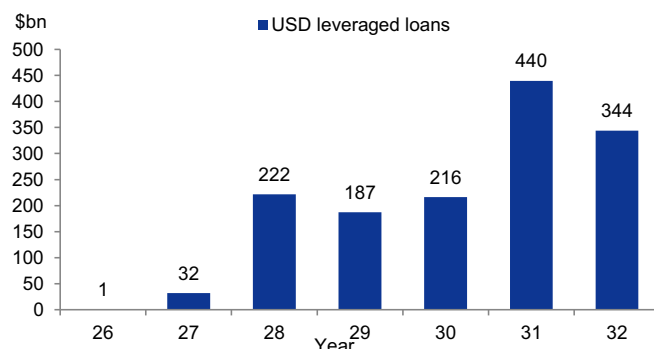
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 120: EUR HY bond market maturity wall



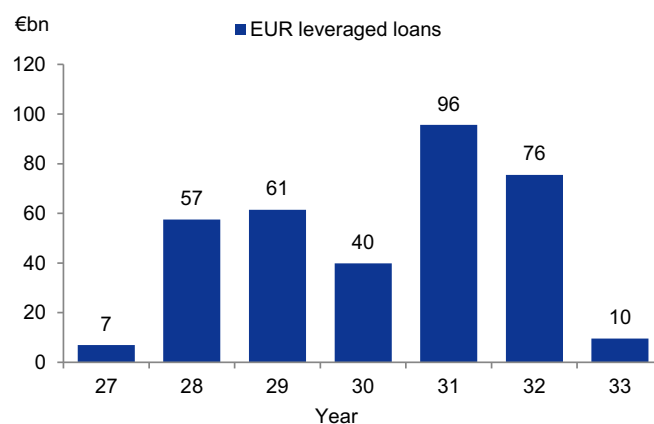
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 121: USD leveraged loan market maturity wall



Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Exhibit 122: EUR leveraged loan market maturity wall

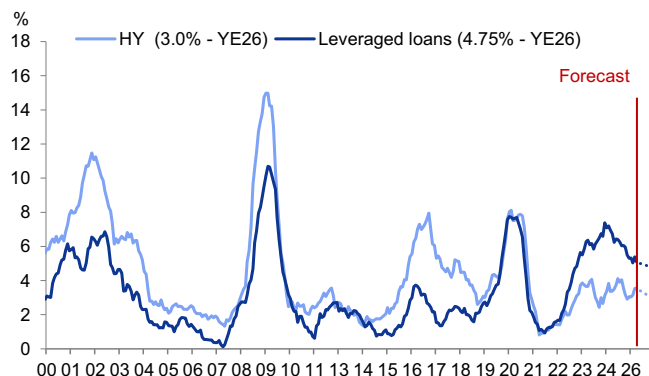


Source: Pitchbook LCD, Goldman Sachs Global Investment Research

Default and ratings migration dashboard: Default trends

Exhibit 123: Default rates for US HY bond and leveraged loan issuers

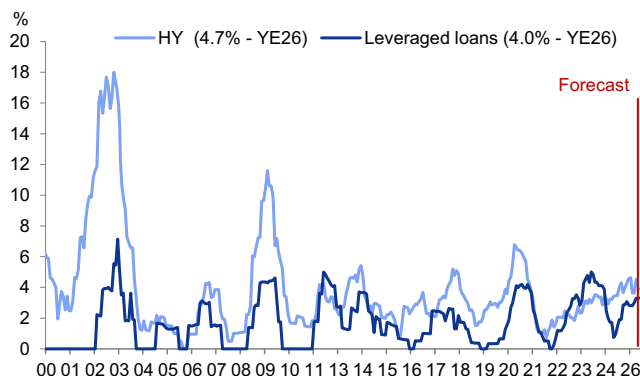
12-month trailing issuer-weighted default rate in the US HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 124: Default rates for European bond and leveraged loan issuers

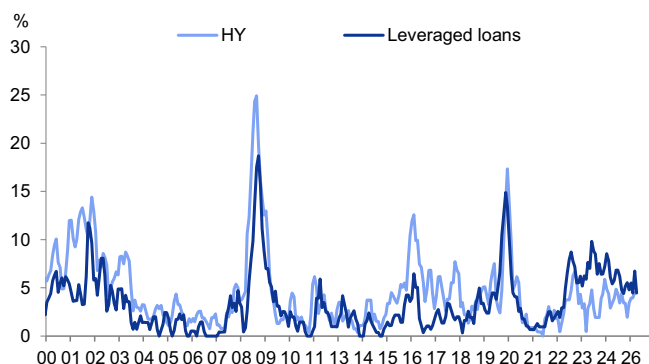
12-month trailing issuer-weighted default rate in the European HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 125: High frequency default rates for US HY bond and leveraged loan issuers

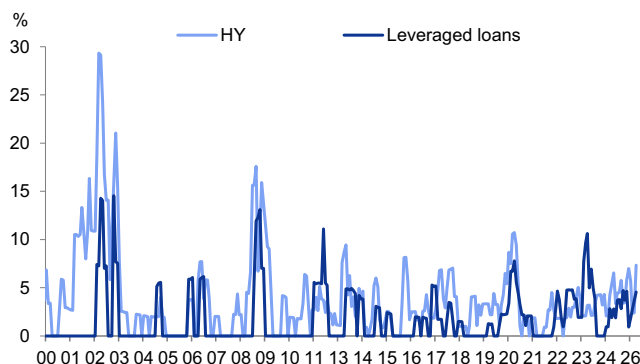
3-month trailing issuer-weighted default rate in the US HY bond and leveraged loan markets



Source: Moody's, Goldman Sachs Global Investment Research

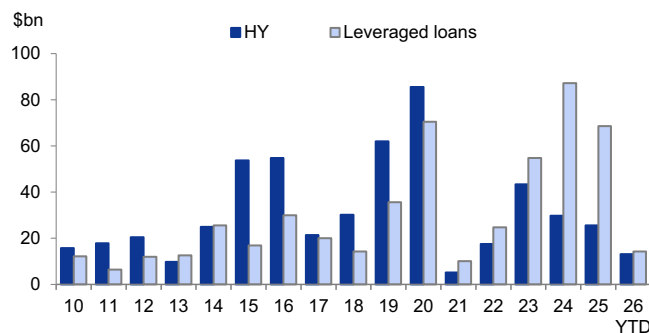
Exhibit 126: High frequency default rates for European HY bond and leveraged loan issuers

3-month trailing issuer-weighted default rate in the European HY bond and leveraged loan markets



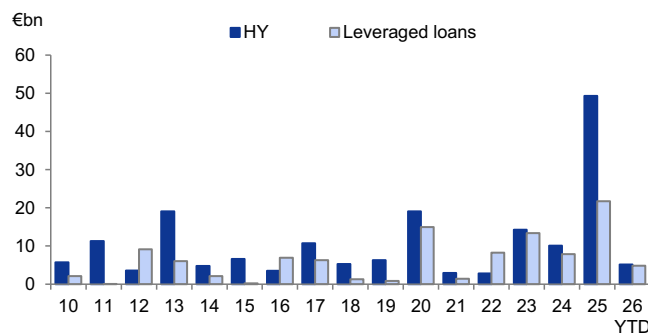
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 127: Annual par value amounts of defaulted USD HY bonds and leveraged loans



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 128: Par value amounts of defaulted EUR HY bonds and leveraged loans



Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 129: Top 5 largest defaults in the USD HY bond and leveraged loan markets in the past 12 months

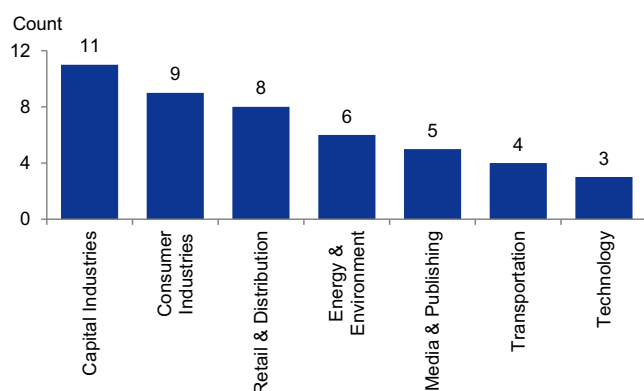
	Default date	Issuer	Industry	Default type	Notional amount (\$bn)
HY	1/29/2026	LABL, Inc.	Containers, Packaging, & Glass	Prepackaged Chapter 11	2.9
	11/20/2025	New Fortress Energy Inc.	Energy: Oil & Gas	Missed interest payment	2.7
	1/14/2026	Saks Global Enterprises LLC	Retail	Chapter 11	2.7
	4/16/2026	QVC, Inc.	Retail	Prepackaged Chapter 11	2.7
	8/8/2025	Saks Global Enterprises LLC	Retail	Distressed exchange	2.1
Loans	9/28/2025	First Brands Group, LLC	Automotive	Chapter 11	4.7
	6/30/2025	Radiate HoldCo, LLC	Telecommunications	Distressed exchange	3.8
	5/30/2025	Quest Identity Intermediate Limited	High Tech Industries	Distressed exchange	3.2
	5/2/2025	Ivanti Software, Inc.	High Tech Industries	Distressed exchange	2.8
	2/17/2026	Vibrantz Technologies Inc.	Chemicals, Plastics, & Rubber	Distressed exchange	2.6

Source: Moody's, data compiled by Goldman Sachs Global Investment Research

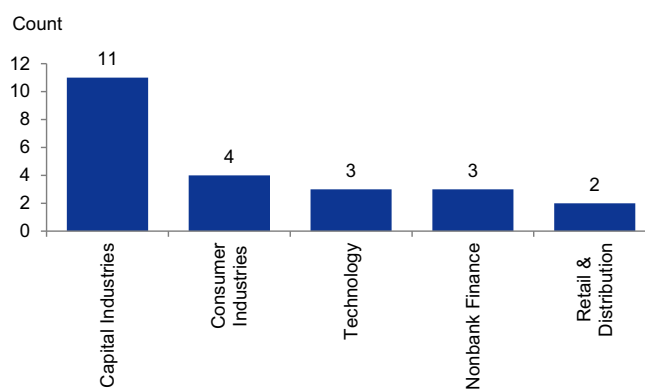
Exhibit 130: Top 5 largest defaults in the EUR HY bond and leveraged loan markets in the past 12 months

	Default date	Issuer	Industry	Default type	Notional amount (€bn)
HY	6/10/2025	Altice France SAS	Telecommunications	Suspension of payments	6.0
	6/16/2025	Altice France Holding S.A.	Telecommunications	Missed interest payment	2.1
	2/19/2026	Garfunkelux Holdco 3 S.A.	Finance	Distressed exchange	1.7
	6/25/2025	Garfunkelux Holdco 3 S.A.	Finance	Distressed exchange	1.7
	11/12/2025	Ardagh Group S.A.	Containers, Packaging, & Glass	Distressed exchange	1.4
Loans	6/10/2025	Altice France SAS	Telecommunications	Suspension of payments	3.5
	6/3/2025	Colisee Group	Healthcare & Pharmaceuticals	Missed interest payment	1.3
	4/30/2026	Poseidon BidCo S.A.S	High Tech Industries	Missed interest payment	1.3
	1/29/2026	Eos Finco S.a.r.l.	Wholesale	Distressed exchange	1.3
	5/9/2025	Eos Finco S.a.r.l.	Wholesale	Distressed exchange	1.0

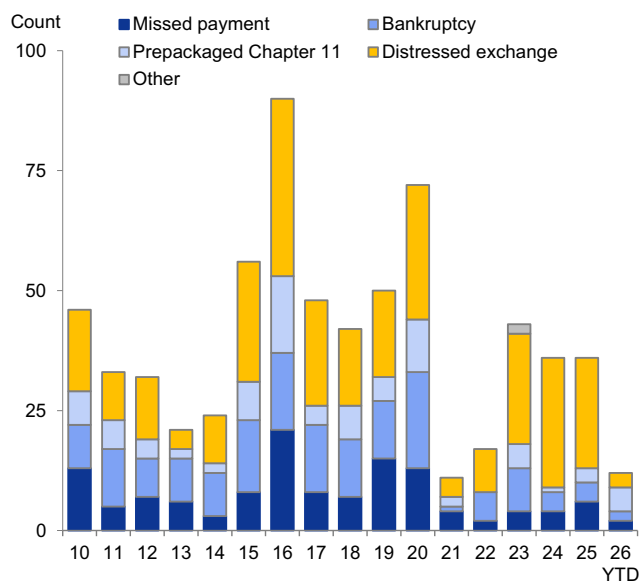
Source: Moody's, data compiled by Goldman Sachs Global Investment Research

Exhibit 131: Industry default count in the US

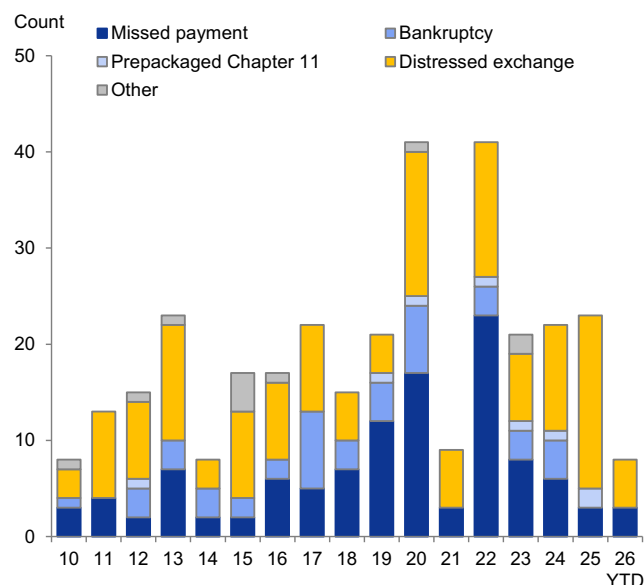
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 132: Industry default count in Europe

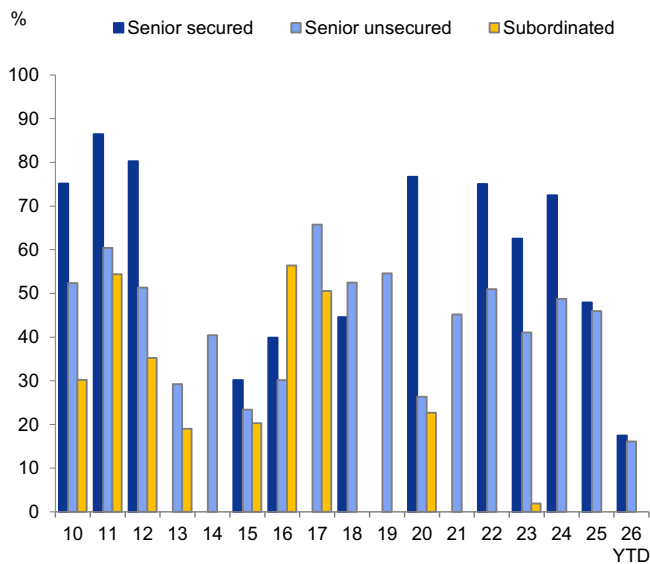
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 133: Composition of defaults in the USD HY bond market by type

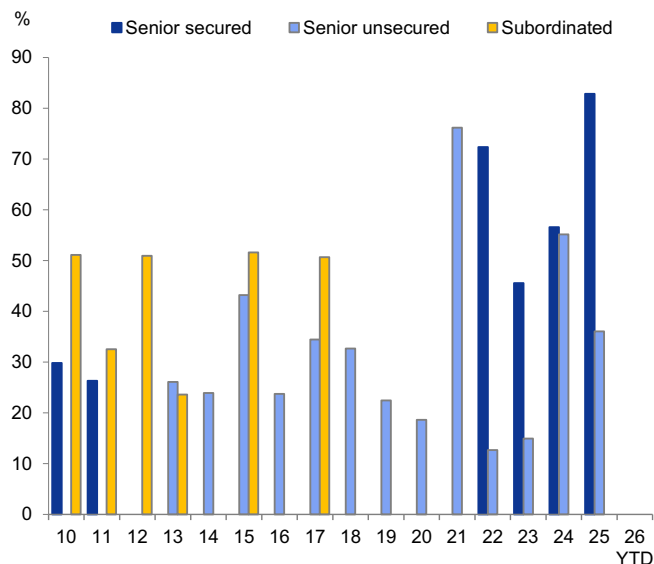
Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 134: Composition of defaults in the EUR HY bond market by type

Source: Moody's, Goldman Sachs Global Investment Research

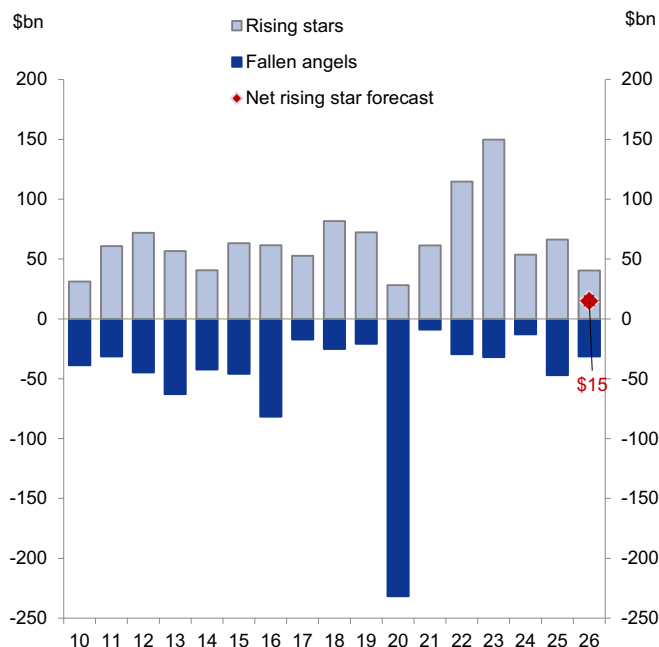
Exhibit 135: Annual average recovery rates by seniority for US issuers

Source: Moody's, Goldman Sachs Global Investment Research

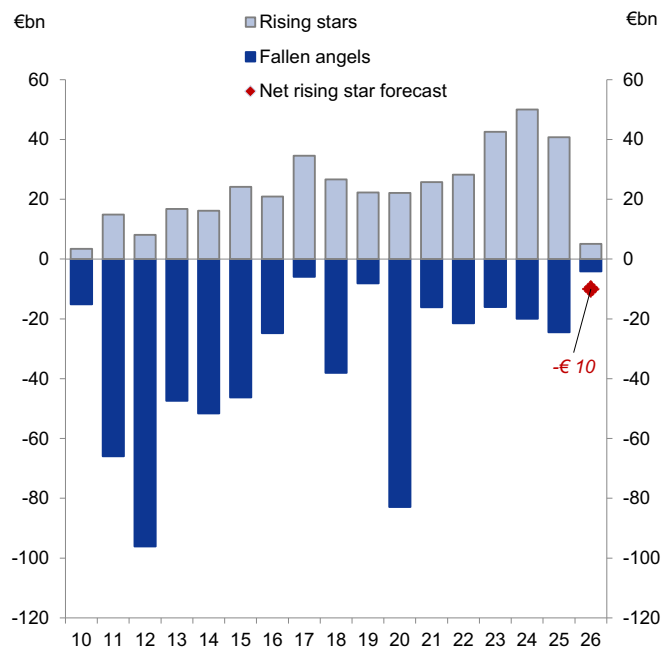
Exhibit 136: Annual average recovery rates by seniority for European issuers

Source: Moody's, Goldman Sachs Global Investment Research

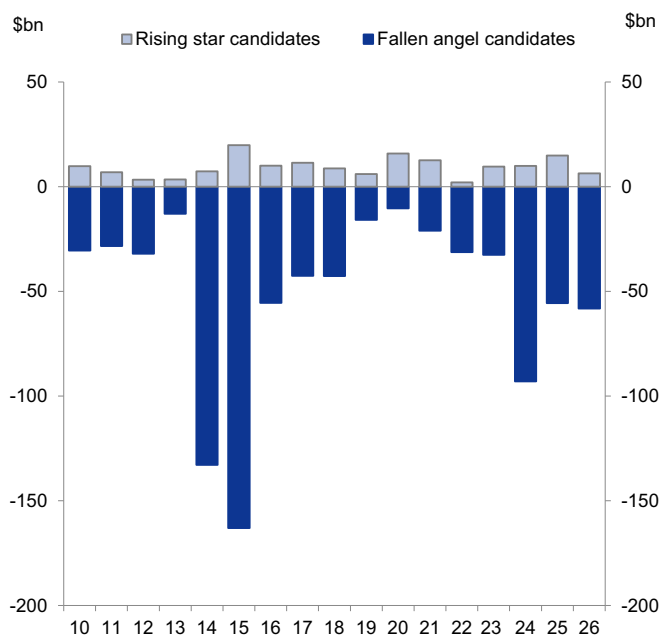
Default and ratings migration dashboard: Upgrades and downgrades

Exhibit 137: Fallen angels vs. rising stars in the USD market


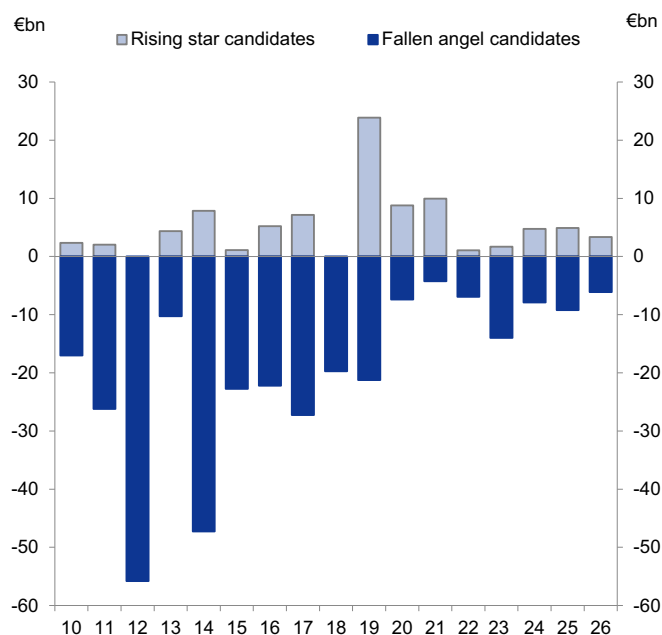
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 138: Fallen angels vs. rising stars in the EUR market


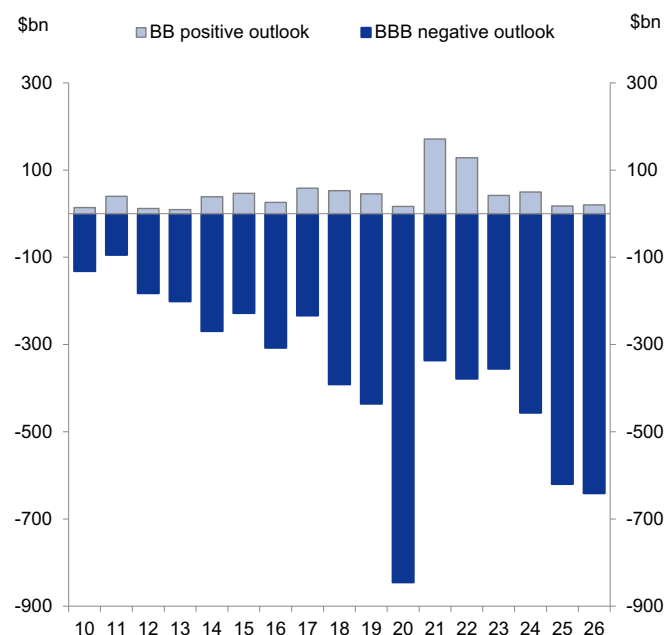
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 139: BB-rated bonds on upgrade watch (upgrade candidates) vs. BBB-rated bonds on downgrade watch (downgrade candidates) in the USD market


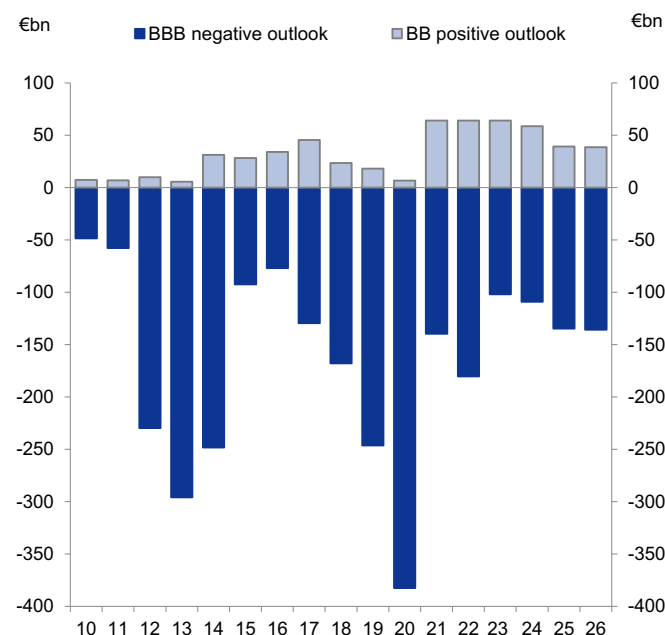
Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 140: BB-rated bonds on upgrade watch (upgrade candidates) vs. BBB-rated bonds on downgrade watch (downgrade candidates) in the EUR market


Source: Bloomberg, Goldman Sachs Global Investment Research

Exhibit 141: BB-rated bonds on outlook positive vs. BBB-rated bonds on outlook negative in the USD market

Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 142: BB-rated bonds on outlook positive vs. BBB-rated bonds on outlook negative in the EUR market

Source: Moody's, Goldman Sachs Global Investment Research

Exhibit 143: Top 10 USD fallen angel issuers in the last 12 months

Company	Notional downgraded (\$bn)
Centene Corp	14.7
Paramount Global	13.0
Fs Kkr Capital Corp	3.8
Fmc Corp	2.5
Huntsman International L	1.5
Banco Santander Mexico	1.3
Pacificorp	0.9
Blackrock Tcp Cap Corp	0.7
Horizon Mutual Holdings	0.6
Ses Americom Inc	0.5

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 144: Top 10 USD rising star issuers in the last 12 months

Company	Notional upgraded (\$bn)
Vistra Operations Co Llc	6.3
Medline Borrower Lp	4.5
1011778 Bc / New Red Fin	4.0
Permian Resourc Optg Llc	3.6
Unicredit Spa	3.5
Frontier Communications	2.8
Frontier Communications	2.8
Constellation En Gen Llc	2.3
Medline Borrower/Medi Co	1.5
Standard Chartered Plc	1.5

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 145: Top 10 EUR Fallen angel issues

Company	Notional downgraded (€bn)
Ses Sa	3.4
Lanxess Ag	2.2
Worldline Sa/France	2.2
Citycon Treasury Bv	1.7
Verallia Sa	1.6
Mediobanca Di Cred Fin	1.5
Unibail-Rodamco-Westfld	1.1
Adecco Int Financial Svs	0.5
Illimity Bank Spa	0.3
Solaben Luxembourg	0.1

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Exhibit 146: Top 10 EUR rising star issuers in the last 12 months

Company	Notional upgraded (€bn)
Banca Monte Dei Paschi S	3.0
Terna Rete Elettrica	1.9
Aib Group Plc	1.7
Asr Nederland Nv	1.5
Unipol Assicurazioni Spa	1.3
Intesa Sanpaolo Assicura	1.1
Bper Banca	1.0
Piraeus Bank Sa	1.0
Czechoslovak Group	1.0
Abanca Corp Bancaria Sa	1.0

Source: Bloomberg, data compiled by Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Amanda Lynam, CPA, Spencer Rogers, CFA, Sara Grut and Shamshad Ali, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

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Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

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